Thermodynamic Dynamics of Observer-Memory Entanglement

Memory, self-frames and continued calibration

Complete Collected Volume

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Introduction

An agent need not acquire a complete image of itself to keep acting. It needs records that improve the decisions available to its continuing body. The three papers develop memory as a resource, a history-selected frame, and continued calibration when that frame drifts or the body loses access. The central claim concerns functional necessity in specified tasks, not an unconditional theorem that every memory generates experience.

I supplies the correlation and thermodynamic account. II constructs the usable frame. III determines when stored history changes detection, maintenance and action decisions. Reading them in this order exposes what each transition requires, including paid references and information that must remain inaccessible in a claimed decision witness.

Reading map

TD-I *Memory as a Usable Resource: Correlations, Records and Complete Endpoint Accounting.* The work value of records depends on their physical readout and retained side information. The endpoint account keeps input purity and the outputs of correction visible.

TD-II *Reference Frames Selected by History: The Capability and Cost of a Bounded Agent's Self-Frame.* An agent's usable frame is selected by its actual history. Oriented records and legal readouts determine what it can calibrate and control.

TD-III *Keeping a Self-Model Calibrated: Drift Detection, the Price of Keeping Up, and Reliability in a Closed Loop.* Drift makes continued self-calibration a task. Retained histories determine reference requests, paid maintenance and action authority on one continuing body.

The papers retain their complete proofs, appendices and references, and their own pagination. The contents list PDF page numbers; PDF bookmarks also provide section-level navigation.

Memory as a Usable Resource: Correlations, Records and Complete Endpoint Accounting

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Abstract

A stored correlation has value only through an operation that can use it. We ask which correlations can fund a free-energy gain, who can read them and what reading, feedback and memory reset cost. The survival functional $\mathcal{S} = \Delta F - W$ obeys an exact balance retaining mutual information, environmental displacement from equilibrium, control energy and interaction energy. Three models distinguish their roles: Lorentz–Drude dephasing has no backflow at any temperature but a nonzero interaction energy; an autonomous two-qubit recurrence saturates the correlation budget without a pointer record; fresh-memory collisions create records available to several readers. A classical memory used for feedback is worth at most $k_B T I(X{:}M)$, with matched quasistatic feedback approaching the value. Every feedback protocol assumes a posterior and loses at least $k_B T$ times its expected divergence from the true posterior. A wrong readout model can therefore make a record worse than no memory. For a tail-controlled estimate from n references per sign, calibration has leading loss $k_B T(1+g^2/(1-b^2))/(4n)$ and pays through reuse. Within a reset menu with no added purity, positive read–harvest–reset work requires accessible correlated side information. Two partially mixed record ports realize this possibility: one drives matched feedback and the other reduces the reset cost through a reversible parity operation. These results establish the memory–self–introspection route's first functional link: relevant memory permits a gain, a model of one's readout realizes its full value, and calibration builds that model at a calculable cost. A separate finite-history classification bound proves that, for each fixed observation window, a longer finite episode can improve the attainable decision risk.

Keywords: open quantum systems, information thermodynamics, system–environment correlations, quantum records, feedback control, self-calibration, interaction energy, non-Markovian dynamics

1 Introduction

1.1 What memory can pay for

A system coupled to a thermal bath can lose the free energy needed for further action. An autonomous detailed-balance semigroup makes that loss precise; a structured environment can instead return correlations at a later time. Persistent structures, from molecular machines to organisms, maintain themselves by importing low-entropy energy and exporting entropy [2]. Within that general picture a narrower question can be answered exactly: when can correlations that a system has already built with its environment fund a later interval, and what does it take for an agent to turn a stored record into usable work?

Two errors are easy to make here. One is to treat any memory effect in the reduced dynamics as a resource. A memory kernel represents retained history; it does not guarantee that the history can be used, and the sign of the free-energy balance is fixed by the resources and the endpoint terms, not by Markovianity as such. The other is to treat information as free once it is stored. A record has value only for an agent that can read it, act on it with a correct model of its own reading, and dispose of the memory afterwards; each step has a price. The endpoint balance and the feedback task below specify which terms must be paid.

1.2 Position within the series

This paper is the first of the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other two lines are HAFF and Q-RAIF.

Line	**Question**		**Role here**
HAFF [25, 26]	How do stable, public records form?	Ocean	Supplies the collision model of records
Q-RAIF [27]	How does a subsystem read and act on its world?	Fish	Reading and control interfaces; costs of persistence
T-DOME I (this work)	What can stored correlations fund, and for whom?	Seed	Ledger, correlation types, value of memory and readout model
T-DOME II [28]	How can a bounded agent organize what it keeps?	Ego	Frame and feature selection under a budget
T-DOME III [29]	How does an agent keep its model of itself calibrated?	Loop	Detection, tracking and body-gated repair

The three T-DOME papers form a conditional sequence of models along what the programme calls the memory–self–introspection route: an agent keeps records of its world, models its own access to that world, and checks that model against references. The present paper supplies the first link and the thermodynamic currency for the other two. It shows

in one explicit task that a memory is informationally necessary for a work gain, that every protocol acts on an implied posterior and obtaining the full value requires that posterior to match the agent's readout, and that calibrating the model from references is an achievable mechanism whose cost and benefit can be computed. Paper II shows that under a reading budget a self-frame formed from records is informationally necessary for the task, prices it and identifies the blind spot it creates; Paper III develops drift detection, priced tracking and history-selected repair of body-gated access.

1.3 Results

- An exact accounting identity for the survival functional with every endpoint term retained (Theorem 8), a budget bound and a charge–discharge interval closure (Corollary 9, Proposition 10), and a restricted autonomous ceiling for a detailed-balance semigroup (Theorem 6).

- Three exact models with different roles (Section 5): no backflow at any temperature and a nonzero interaction-energy term in the Lorentz–Drude pure-dephasing model (Propositions 11 and 12); exact saturation of the correlation budget without any record in an autonomous two-qubit model (Proposition 13); the complete ledger of the collision model of records (Proposition 14).

- The work value of a record held in a classical memory and its information-theoretic ceiling (Theorem 16); the price of acting on a wrong posterior, which every protocol implicitly assumes (Theorem 17); the local metric of that price and the loss after calibration from references (Proposition 18), with the resulting amortization rule (Corollary 20) and a drift law for self-calibration, proved in Paper III (Proposition 21).

- A read–harvest–reset bound for a specified reset menu (Proposition 22), and a two-mixed-port realization with a constructive conditional reset (Proposition 23). The task restores the private memory and the retained record's marginal, not the complete record source.

Sections 8–10 distinguish temporal influence from kernel memory, price specified archives and compare retained history with fixed observation windows.

2 Thermodynamic setting

2.1 Open systems and memory kernels

Consider $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ with total Hamiltonian

$$H(t) = H_S(t) \otimes \mathbb{1}_E + \mathbb{1}_S \otimes H_E + H_{\mathrm{int}}(t), \qquad H_S(t) = H_R + H_{\mathrm{ctrl}}(t), \tag{1}$$

where H_R is the fixed bare system Hamiltonian, H_{ctrl} the control and H_{int} the coupling. Under the Born–Markov and secular approximations the reduced dynamics obeys a GKSL equation [3, 4]. For a fixed environmental reference σ_E, set $\mathcal{P}A = \mathrm{tr}_E(A) \otimes \sigma_E$, $\mathcal{Q} = \mathrm{id} - \mathcal{P}$

and $\mathcal{L}(t)A = -i[H(t), A]$. Eliminating the $\mathcal{Q}$ component gives the exact Nakajima–Zwanzig equation

$$\dot{\rho}_S(t) = -i[H_{\text{eff}}(t), \rho_S(t)] + \int_0^t ds\, \mathcal{K}(t,s)\, \rho_S(s) + \mathcal{I}_0(t), \tag{2}$$

where $H_{\text{eff}}(t) = H_S(t) + \text{tr}_E[H_{\text{int}}(t)(\mathbb{1}_S \otimes \sigma_E)]$ and

$$\mathcal{K}(t,s)A = \text{tr}_E[\mathcal{L}(t)\mathcal{G}_Q(t,s)\mathcal{Q}\mathcal{L}(s)(A \otimes \sigma_E)]\,, \qquad \mathcal{I}_0(t) = \text{tr}_E[\mathcal{L}(t)\mathcal{G}_Q(t,0)\mathcal{Q}\rho_{SE}(0)]\,.$$

Here $\mathcal{G}_Q$ propagates the $\mathcal{Q}$ subspace with generator $\mathcal{Q}\mathcal{L}(t)\mathcal{Q}$ [5, 6]. The inhomogeneous term vanishes for $\rho_{SE}(0) = \rho_S(0) \otimes \sigma_E$; correlated inputs allowed below generally retain it. Non-Markovian dynamics can show information backflow, a temporary increase of the trace distance between two evolving states [14]. Neither the kernel nor backflow is used as a resource below; the accounting identity applies directly to the joint unitary dynamics.

2.2 Conventions

Definition 1 (Thermodynamic setup)**.** *Let $\beta = (k_B T)^{-1}$. The reference state of the system is $\rho_{\text{eq}} = e^{-\beta H_R}/Z_R$, and its nonequilibrium free energy is*

$$F(\rho) = \text{tr}(\rho H_R) - \beta^{-1} S(\rho), \qquad D(\rho\|\rho_{\text{eq}}) = \beta\big(F(\rho) - F_{\text{eq}}\big) \geq 0. \tag{3}$$

The environmental reference is the fixed Gibbs state $\rho_E^{\text{th}} = e^{-\beta H_E}/Z_E$. The inclusive work over $[0,\tau]$ is $W = \int_0^\tau \text{tr}\big(\rho(t)\,\partial_t H(t)\big)dt$, so that $W = \Delta\langle H\rangle$ exactly.

Definition 2 (Standing assumptions)**.** *Unless stated otherwise:*

(A1) *$\mathcal{H}_S \otimes \mathcal{H}_E$ is finite dimensional and the total evolution is unitary, generated by* (1)*. Infinite baths are treated as limits of finite mode sets in which the quantities below have finite limits.*

(A2) *Every energy change is booked: $W = \Delta\langle H_R\rangle + \Delta\langle H_{\text{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\text{int}}\rangle$. Dropping $\Delta\langle H_{\text{int}}\rangle$ is a declared approximation, never a consequence of autonomy.*

(A3) *H_E is time independent; the actual initial state of the environment need not be ρ_E^{th}.*

(A4) *The initial state $\rho_{SE}(0)$ may be correlated.*

(A5) *Relative entropies are finite (full rank, or restriction to supports).*

2.3 The survival functional

Definition 3 (Survival functional)**.** *For an interval $[a,b]$,*

$$\mathcal{S}[a,b] := F(\rho_S(b)) - F(\rho_S(a)) - W[a,b], \qquad \beta\,\mathcal{S} = -\Sigma, \quad \Sigma := \beta(W - \Delta F). \tag{4}$$

The name records the intended use: $\mathcal{S} > 0$ means the system gained more free energy than the external protocol invested, a thermodynamic profit; $\mathcal{S} < 0$ means it paid more than it gained. Mathematically $\mathcal{S}$ is a generalized entropy-balance functional. For an uncorrelated thermal initial environment and zero changes in the endpoint control and interaction energies, $\Sigma \geq 0$ and $\mathcal{S} \leq 0$. When those endpoint terms change, their signs enter the balance of Theorem 8. For correlated initial conditions Σ can be negative [15], and the next sections say what pays for it.

In measurement-based feedback, the generalized second law $\Sigma \geq -I_{\mathrm{fb}}$ allows $\mathcal{S} > 0$ at the expense of the information I_{fb} gained by measurement [7, 8, 9]. Pre-existing correlations enter the balance in an analogous role. A memory kernel is not itself a measurement record, and the two protocols are kept distinct; Section 6 returns to feedback with an explicit record.

3 A restricted autonomous baseline

Definition 4 (Autonomous Davies baseline)**.** *The autonomous baseline has $H_{\mathrm{ctrl}} = 0$, fixed H_R, and a time-independent GKSL generator $\mathcal{L}$ obtained as the weak-coupling limit of coupling to one thermal bath, satisfying quantum detailed balance with respect to the faithful Gibbs state ρ_{eq} [11, 1]. No measurement, feedback or initial system–bath resource is included.*

Lemma 5 (Spohn [11])**.** *For a GKSL semigroup $\Lambda_t = e^{\mathcal{L}t}$ with invariant state ρ_{eq}, $t \mapsto D(\Lambda_t\rho\|\rho_{\mathrm{eq}})$ is nonincreasing.*

Proof. Relative entropy is contractive under CPTP maps and $\Lambda_t\rho_{\mathrm{eq}} = \rho_{\mathrm{eq}}$. □

Theorem 6 (Restricted autonomous ceiling)**.** *Under Definition 4, $\mathcal{S}[0,\tau] = F(\rho(\tau)) - F(\rho(0)) \leq 0$.*

Proof. $W = 0$, and by (3) and Lemma 5, $\beta\Delta F = \Delta D(\rho\|\rho_{\mathrm{eq}}) \leq 0$. □

The theorem is a free-energy monotonicity statement for one autonomous detailed-balance class. It is not a ceiling for driven or feedback-controlled Markovian agents, which can persist under continuous driving; time-dependent protocols need an instantaneous detailed-balance hypothesis and a different theorem. The reduced semigroup omits explicit inter-epoch correlations, which is a modelling loss, not a proof that such correlations are unusable.

4 Initial-correlation accounting

4.1 The identity

The quantum mutual information $I(S{:}E) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}) = D(\rho_{SE}\|\rho_S \otimes \rho_E)$ measures all correlations between system and environment. For any full-rank σ_E,

$$D(\rho_{SE}\|\rho_S \otimes \sigma_E) = I(S{:}E) + D(\rho_E\|\sigma_E), \tag{5}$$

an algebraic identity that holds for correlated ρ_{SE} [16].

Lemma 7 (Information–thermodynamic identity)**.** *For unitary evolution from any initial state,*

$$\Delta I(S{:}E) + \Delta D(\rho_E \| \rho_E^{\mathrm{th}}) = \Delta S_S + \beta\, \Delta\langle H_E \rangle. \tag{6}$$

Proof. By (5) with $\sigma_E = \rho_E^{\mathrm{th}}$ and $\ln \rho_E^{\mathrm{th}} = -\beta H_E - \ln Z_E$, the quantity $I + D(\rho_E \| \rho_E^{\mathrm{th}})$ equals $-S(\rho_{SE}) + S(\rho_S) + \beta\langle H_E \rangle + \ln Z_E$. Unitarity keeps $S(\rho_{SE})$ fixed, and taking differences gives (6). □

Theorem 8 (Survival balance)**.** *Under* (A1)–(A5)*, for any interval and any initial state,*

$$\boxed{\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D(\rho_E \| \rho_E^{\mathrm{th}}) - \beta\, \Delta\langle H_{\mathrm{ctrl}} \rangle - \beta\, \Delta\langle H_{\mathrm{int}} \rangle.} \tag{7}$$

Proof. By (A2) and $\Delta F = \Delta\langle H_R \rangle - \beta^{-1} \Delta S_S$, $\Sigma = \beta(W - \Delta\langle H_R \rangle) + \Delta S_S = (\Delta S_S + \beta \Delta\langle H_E \rangle) + \beta \Delta\langle H_{\mathrm{ctrl}} \rangle + \beta \Delta\langle H_{\mathrm{int}} \rangle$. Lemma 7 replaces the bracket by $\Delta I + \Delta D$, and $\beta\mathcal{S} = -\Sigma$. □

The identity is exact. It is the Esposito–Lindenberg–Van den Broeck decomposition [15] written with arbitrary initial correlations and with both endpoint terms kept. It is dynamics-agnostic: Markovianity does not enter, and a memory kernel matters only if a model shows that it preserves correlations across an interval boundary and later allows their consumption. Autonomy removes the control term but not the interaction term; Section 5.1 gives an autonomous model in which the interaction energy changes at every time.

4.2 Regimes and the budget

Corollary 9 (Regimes and budget)**.** *Suppose* $\Delta\langle H_{\mathrm{ctrl}} \rangle = \Delta\langle H_{\mathrm{int}} \rangle = 0$ *on the interval, as for a switched protocol whose control and interaction Hamiltonians are zero at both endpoints.*

1. *If the interval starts from a product state with a thermal environment,* $\beta\mathcal{S} = -I(S{:}E; b) - D(\rho_E(b) \| \rho_E^{\mathrm{th}}) \le 0$.
2. *If correlations are consumed,* $\Delta I < 0$*, and* $|\Delta I| > \Delta D(\rho_E \| \rho_E^{\mathrm{th}})$*, then* $\mathcal{S} > 0$.
3. *In every case* $\beta\mathcal{S}[a, b] \le I(S{:}E; a) + D(\rho_E(a) \| \rho_E^{\mathrm{th}})$.

Proof. Insert the hypotheses into (7) and use $I \ge 0$, $D \ge 0$ at the end of the interval. □

When the endpoint terms do not vanish they are added to the budget with their signs. The budget in (iii) is the profit ceiling: correlations present at the start of the interval plus the environment's displacement from equilibrium.

Proposition 10 (Charge–discharge interval closure)**.** *Under the hypotheses of Corollary 9 at the times* 0, t^* *and* τ*, and with* $I(S{:}E; 0) = 0$ *and* $\rho_E(0) = \rho_E^{\mathrm{th}}$*,*

$$\begin{aligned} \beta\mathcal{S}[0, t^*] &= -I(t^*) - D(t^*) \le 0, \\ \beta\mathcal{S}[t^*, \tau] &= -[I(\tau) - I(t^*)] - [D(\tau) - D(t^*)], \\ \beta\mathcal{S}[0, \tau] &= -I(\tau) - D(\tau) \le 0, \end{aligned}$$

with $D(t) = D(\rho_E(t) \| \rho_E^{\mathrm{th}})$.

Proof. ΔF and W are additive over concatenated intervals, and Theorem 8 applies to each interval. □

A positive second interval is funded by the correlations built in the first, and their concatenation cannot have positive $\mathcal{S}$. No restoration of the complete joint state is assumed in this accounting statement. This charge–discharge structure is realized exactly by the two-qubit model of Section 5.2. Under the Born approximation every step restarts from a product state, so the resource is removed by construction; a reduced semigroup law by itself does not say that every microscopic dilation is uncorrelated. BLP backflow and correlation consumption can coincide in particular models, but neither implies the other in general [10], so backflow is not a witness for the sign of $\mathcal{S}$ without the full balance.

5 Which correlations can be used: three exact models

5.1 Pure dephasing with a Lorentz–Drude bath: no backflow, nonzero interaction energy

Let $\hbar = 1$ and

$$H = \frac{\omega_0}{2}\sigma_z + \sum_k \omega_k b_k^\dagger b_k + \sigma_z \sum_k (g_k b_k + g_k^* b_k^\dagger), \qquad J(\omega) = \sum_k |g_k|^2 \delta(\omega - \omega_k) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \tag{8}$$

with $\lambda, \gamma > 0$. With this normalization $\int_0^\infty J(\omega)\omega^{-1} d\omega = \pi\lambda$. The initial state is any system state times the thermal state of the bare bath at inverse temperature $\beta \in (0, \infty]$. Removing the free rotation, the coherence is multiplied by a factor of modulus

$$|p(t)| = e^{-G_\beta(t)}, \qquad G_\beta(t) = 4\int_0^\infty \frac{J(\omega)}{\omega^2}(1 - \cos\omega t)\coth\frac{\beta\omega}{2}\, d\omega, \tag{9}$$

with $\coth = 1$ at $\beta = \infty$. This is the standard exact solution: the two eigenvalues of σ_z displace each bath mode in opposite directions, the displacement difference is $2g_k(1-e^{-i\omega_k t})/\omega_k$, and the thermal characteristic function gives the factor $\exp[-|\delta\alpha_k|^2 \coth(\beta\omega_k/2)/2]$ per mode [1, 17]. The integral converges for every finite t, so $|p(t)| > 0$.

Proposition 11 (No backflow at any temperature)**.** *For every $\beta \in (0, \infty]$ and $\lambda, \gamma > 0$, $G'_\beta(t) > 0$ for all $t > 0$. Hence $|p(t)|$ decreases strictly, the trace distance of any two system states does not increase, the reduced dynamics is CP-divisible, and for $\beta < \infty$, $\mathcal{S}[a, b] < 0$ on every interval $0 \le a < b$ when the system starts with nonzero coherence.*

Proof. Differentiating under the integral sign, which is allowed because $|\sin\omega t| \le \min(1, \omega T)$ for $t \le T$ gives an integrable majorant, yields

$$G'_\beta(t) = 8\lambda\gamma \int_0^\infty f(\omega)\sin(\omega t)\, d\omega, \qquad f(\omega) = \frac{\coth(\beta\omega/2)}{\omega^2 + \gamma^2}.$$

The function f is positive and strictly decreasing on $(0, \infty)$ and tends to 0; near $\omega = 0$ it behaves as $2/(\beta\gamma^2\omega)$, and $f(\omega)\sin\omega t$ stays bounded there. Substituting $u = \omega t$ and splitting

$(0,\infty)$ into half-periods,

$$\int_0^\infty f(\omega)\sin(\omega t)\,d\omega = \frac{1}{t}\sum_{j=0}^{\infty}(-1)^j a_j, \qquad a_j = \int_0^\pi f\Big(\frac{j\pi+x}{t}\Big)\sin x\,dx.$$

The a_j are positive, strictly decreasing and tend to 0, so the alternating sum is at least $a_0 - a_1 > 0$. Between times $s < t$ the coherence factor is multiplied by $p(t)/p(s)$ with modulus less than 1, which is a CPTP dephasing map, and qubit trace distances shrink under it. Populations are conserved and $W = 0$, so $\beta\mathcal{S}[a,b] = -[S(\rho_S(b)) - S(\rho_S(a))]$, which is negative because the entropy of a qubit with fixed populations increases as its coherence decreases. □

For $\lambda = 1$ and $\gamma = 0.5$ at zero temperature, $|p| = 0.313$, 0.0360 and 2.2×10^{-4} at $t = 0.5$, 1 and 2; at $\beta = 2$ the values are 0.138, 1.43×10^{-3} and 1.0×10^{-9}. Changing γ/λ or the temperature does not produce a backflow interval in this model.[1]

Proposition 12 (Interaction energy)**.** *For any initial system state and a bath that starts in its bare thermal state with vanishing first moments,*

$$\langle H_I\rangle_t = -2\int_0^\infty \frac{J(\omega)}{\omega}(1-\cos\omega t)\,d\omega = -2\pi\lambda\big(1-e^{-\gamma t}\big), \qquad \Delta E_B[0,t] = 2\pi\lambda\big(1-e^{-\gamma t}\big),$$

independently of the temperature, while $\Delta E_S = 0$ *and* $W = 0$*.*

Proof. The Heisenberg equations give $b_k(t) = e^{-i\omega_k t}b_k(0) + (g_k^*/\omega_k)\sigma_z(e^{-i\omega_k t} - 1)$. With $\sigma_z^2 = \mathbb{1}$ and $\langle b_k(0)\rangle = 0$, $\langle\sigma_z(g_k b_k(t) + g_k^* b_k^\dagger(t))\rangle = -2|g_k|^2(1-\cos\omega_k t)/\omega_k$. Summing with the spectral density and using $\int_0^\infty \cos(\omega t)/(\omega^2+\gamma^2)\,d\omega = \pi e^{-\gamma t}/(2\gamma)$ gives the formula. The total energy is conserved and σ_z commutes with H, so $\Delta E_B = -\Delta\langle H_I\rangle$. □

The balance on $[a,b]$ is therefore $\beta\mathcal{S} = -\Delta I - \Delta D_B - \beta\Delta\langle H_I\rangle = -\Delta S_S$, with $\beta\Delta\langle H_I\rangle = -2\pi\beta\lambda(e^{-\gamma a} - e^{-\gamma b}) \neq 0$. The evolution is autonomous and the interaction-energy term still cannot be dropped. At zero temperature the dynamics and energies are well defined, but the relative-entropy terms of the finite-temperature ledger cannot be used with β^{-1} treated as a finite energy unit.

5.2 An autonomous two-qubit recurrence: correlation without a record

Let a system qubit and an environment qubit evolve under the time-independent, mutually commuting Hamiltonian

$$H = \frac{\omega_0}{2}\sigma_z + \frac{\Omega}{2}\tau_z + g\,\sigma_z\tau_z, \qquad \rho_{SE}(0) = \frac{\mathbb{1} + r\sigma_x}{2}\otimes\tau_E, \quad \tau_E = \frac{e^{-\beta\Omega\tau_z/2}}{Z_E}, \tag{10}$$

with $g, \Omega, \beta > 0$ and $0 < r < 1$. Put $m = \tanh(\beta\Omega/2)$, $t_c = \pi/(4g)$ and $t_r = \pi/(2g)$.

[1] The oscillatory decoherence function, survival-gain table and figure in the July 2026 example were not solutions of its stated pure-dephasing model. Propositions 11 and 12 replace them and restore the interaction-energy term.

Proposition 13 (Exact saturation of the correlation budget)**.** *The coherence of the system is* $\frac{r}{2}e^{-i\omega_0 t}\big(\cos 2gt + im\sin 2gt\big)$*, of modulus* $\frac{r}{2}\sqrt{\cos^2 2gt + m^2\sin^2 2gt}$*. The environment marginal stays* τ_E*, and* $\Delta E_S = \Delta E_E = \Delta\langle H_I\rangle = W = 0$*. On the discharging interval*

$$\beta\mathcal{S}[t_c, t_r] = h\Big(\frac{1+rm}{2}\Big) - h\Big(\frac{1+r}{2}\Big) = I(S{:}E; t_c) > 0,$$

where h *is the binary entropy in nats, so the budget of Corollary 9(iii) is attained with equality. The charging interval* $[0, t_c]$ *has the opposite value and* $\mathcal{S}[0, t_r] = 0$*. The two conditional environment states for the two pointer values coincide, so the environment holds no information about the pointer:* $\chi(\text{pointer}{:}E) = 0$ *at all times.*

Proof. The Hamiltonian is diagonal in the product basis. The off-diagonal element of S acquires the phase $e^{-i\omega_0 t}e^{-2igst}$ when the environment is in the τ_z eigenstate $s = \pm 1$, which has weight $(1 \mp m)/2$; averaging gives the coherence. The diagonal blocks are unchanged, so the energies and the environment marginal are constant and the global entropy is constant; hence $I(S{:}E; t) = S(\rho_S(t)) - S(\rho_S(0))$. At t_r the coupling factor of the evolution operator is $e^{-i(\pi/2)\sigma_z\tau_z} = -i\sigma_z \otimes \tau_z$, so the evolution operator is a product of local unitaries and the correlation vanishes. With the energies constant, $\beta\mathcal{S} = -\Delta S_S$, which gives the displayed values. The environment state conditional on either pointer value is τ_E, because τ_E commutes with τ_z. □

For $(\beta, \Omega, g, r) = (1.2, 1.1, 0.3, 0.7)$, $m = 0.578363$, $t_c = 2.617994$, $t_r = 5.235988$ and $\beta\mathcal{S}[t_c, t_r] = 0.1860846$. The model realizes the charge–discharge cycle of Proposition 10 with an autonomous constant coupling, and it is exact. Its environment remembers in the sense that the correlation returns at a fixed time; it does not remember in the sense of holding a readable record. The reduced dynamics is not CP-divisible, since the coherence revives.

5.3 Records: the collision model

The collision model of the companion papers supplies correlations of a third kind [25, 26]. A pointer qubit with $H_R = \varepsilon P_1$ starts in $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$; memories $E_1, \ldots, E_n$ with degenerate Hamiltonians start in $|0\rangle$, and the k-th collides once through $P_0 \otimes \mathbb{1} + P_1 \otimes R_k$ with $\langle 0|R_k|0\rangle = c_k$. The collision can be generated by $g_k(t)P_1 \otimes \sigma_y$, which commutes with H_R and vanishes at both ends of the pulse, so $W = 0$ and $\Delta\langle H_{\text{int}}\rangle = 0$. With $r = \prod_k c_k$ and $s_n = H_p(r)$, where $H_p(r) = h\big((1 + \sqrt{1 - 4p(1-p)(1-r^2)})/2\big)$, a fragment F holds $\chi(X{:}E_F) = H_p(r_F)$ nats about the pointer label X [25].

Proposition 14 (Ledger of the collision model)**.** *With the environmental reference* $\mathbb{1}/2^n$*:*

1. *from the coherent initial state,* $\Delta I(S{:}E) = 2s_n$*,* $\Delta D = -s_n$ *and* $\beta\mathcal{S} = -s_n$*;*
2. *from the diagonal initial state* $pP_0 + (1-p)P_1$*, every record quantity is the same,* $\Delta I = s_n$*,* $\Delta D = -s_n$ *and* $\beta\mathcal{S} = 0$*;*
3. *the exact inverse protocol, which needs access to* S *and all memories, gives* $\Delta I = -2s_n$*,* $\Delta D = +s_n$ *and* $\beta\mathcal{S} = +s_n$*, and the full forward–backward cycle has* $\mathcal{S} = 0$*.*

Proof. The global state is pure in case (1), so $I(S{:}E) = 2S(\rho_E) = 2s_n$, and $D(\rho_E \| \mathbb{1}/2^n) = n \ln 2 - s_n$. Populations of S are conserved and $W = 0$, so $\beta\mathcal{S} = -\Delta S_S = -s_n$, in agreement with (7). In case (2) the global entropy is $h(p)$ throughout, $S(\rho_S) = h(p)$ is unchanged and $S(\rho_E) = s_n$, so $I = s_n$. Case (3) reverses case (1). □

For $p = 1/2$, $c_k = 0.8$ and $n = 60$, $s_{60} = 0.693147$ nats, equal to $\ln 2$ to nine digits. Records form along a CP-divisible reduced dynamics, and the diagonal case shows that the formation of records does not by itself cost the system free energy; the memories' initial purity deficit supplies the record budget [26].

5.4 Who can use which correlation

	Lorentz–Drude dephasing	Two-qubit recurrence	Collision records
Correlation built	yes	yes	yes
Returns to S by itself	never	at t_r, fixed by g	no
Pointer information in E	yes, spread over a continuum of modes	none ($\chi = 0$)	yes, in many separate copies
Who can use it	an agent with coherent control of the bath	the autonomous dynamics, at one time	an agent with fragment readout and feedback permissions
Endpoint terms	$\Delta\langle H_I \rangle \neq 0$	zero	zero

Table 1: Three forms of system–environment correlation.

In the accounting identity, consuming one nat of mutual information contributes $+k_B T$ to $\mathcal{S}$, with the other terms retained. The models show that the identity does not say who can consume it. Correlation created by a structured bath can be dissipated without ever returning, although the bath holds information about the pointer spread over its modes. Correlation held by a single environmental degree of freedom with a commensurate spectrum can return at a fixed time, without anyone reading anything. Correlation copied into records can be read by many agents at their own times, and the rest of this paper studies what such an agent must do to turn a record into work.

6 The value of a record, and of a model of one's own readout

6.1 The task

The pointer S of Section 5.3 has a degenerate Hamiltonian during the task ($\varepsilon = 0$) and has been decohered by memories the agent does not read, so that its state is diagonal: $\rho_S = pP_0 + (1-p)P_1$, correlated with the memories through X. The agent reads a fragment

F by a measurement N and stores the outcome in a classical memory M. It then runs a feedback protocol on S and a bath B at inverse temperature β, chosen according to the stored value m, in which the system Hamiltonian returns to zero and the coupling vanishes at both ends. The work extracted is $W_{\text{ext}} = -W$. The task is to extract as much work as possible on average.

Lemma 15 (Extraction from a qubit)**.** *For a qubit with degenerate Hamiltonian in a state ρ, every such protocol extracts on average at most $k_B T\, D(\rho\|\mathbb{1}/2) = k_B T[\ln 2 - S(\rho)]$. If the diagonal of ρ is a distribution P and Q is a distribution with full support, the protocol that quenches the levels to $E_x = -k_B T \ln Q(x)$, lets the system thermalize and returns the levels quasistatically to degeneracy extracts*

$$k_B T\big[\ln 2 - H(P) - D(P\|Q)\big].$$

Proof. For the bound: the joint state of system and bath evolves unitarily from $\rho\otimes\tau_B$; since the system Hamiltonian is zero at both ends and the coupling vanishes there, $W = \Delta E_B$. Subadditivity gives $S(\rho') + S(\rho'_B) \geq S(\rho) + S(\tau_B)$, and $D(\rho'_B\|\tau_B) \geq 0$ gives $\beta\Delta E_B \geq S(\rho'_B) - S(\tau_B) \geq S(\rho) - S(\rho') \geq S(\rho) - \ln 2$. For the protocol: the quench costs $\sum_x P(x)E_x = k_B T[H(P) + D(P\|Q)]$; thermalization at fixed levels costs no work; the quasistatic return costs the free-energy difference $-k_B T \ln 2 - (-k_B T \ln \sum_x Q(x)) = -k_B T \ln 2$. The work done on the system is $k_B T[H(P) + D(P\|Q) - \ln 2]$. Off-diagonal elements do not affect it, since the levels are diagonal and the bath relaxes the system to the Gibbs state of the quenched levels. □

Quasistatic values throughout this section are ideal limiting values. A full-support posterior needs finite quenched gaps, but reversible thermalization and return are still limits of physical protocols. A posterior on the boundary additionally needs a divergent level gap; it is treated by a full-support approximation.

6.2 Memory is necessary, and its value is the accessible information

Theorem 16 (Value of a stored record)**.** *In the task above, let $W_0 = k_B T[\ln 2 - h(p)]$ be the largest average work obtainable without the record. For every measurement N, every processing of its outcome into M and every feedback protocol,*

$$\mathbb{E}[W_{\text{ext}}] - W_0 \leq k_B T\, I(X{:}M) \leq k_B T\, I_{\text{acc}}(X{:}E_F) \leq k_B T\, \chi(X{:}E_F) = k_B T\, H_p(r_F).$$

The first inequality is approached by the quasistatic protocol of Lemma 15 with $Q = P(\cdot \mid m)$ when every posterior $P(\cdot \mid m)$ has full support; otherwise it is the supremum over such protocols. For $p = 1/2$ the Helstrom measurement gives $I(X{:}M) = I_{\text{acc}}(X{:}E_F) = \ln 2 - h\big((1 - \sqrt{1 - r_F^2})/2\big)$.

Proof. Given $M = m$ the state of S is $\sum_x P(x \mid m)P_x$, because S is diagonal and correlated with F only through X. Lemma 15 bounds the extracted work by $k_B T[\ln 2 - H(X \mid M = m)]$, with equality in the matched quasistatic limit. Averaging over m and subtracting W_0 gives $k_B T[h(p) - H(X \mid M)] = k_B T\, I(X{:}M)$. Since M is obtained from a measurement on F

followed by processing, $I(X{:}M) \le I_{\mathrm{acc}}(X{:}E_F)$, and Holevo's bound gives $I_{\mathrm{acc}} \le \chi$ [18]. The value of χ and the equal-prior accessible-information formula are the pure-fragment results of Paper E [25]. For mixed record ports the first inequality follows from the same extraction argument; their actual information replaces $H_p(r_F)$. □

The theorem gives the memory link of the route in the strongest form available for this task. A work gain ΔW is possible only if the agent holds a memory with $I(X{:}M) \ge \beta\Delta W$, and only if it reads a fragment with $H_p(r_F) \ge \beta\Delta W$; the fragment lengths of Paper E then say how many environmental memories it must control. Conversely, the gain $k_B T\, I(X{:}M)$ is the matched quasistatic limit. With $p = 1/2$ and $c_k = 0.8$, one memory gives at most $0.1927\, k_B T$ to an agent with a classical memory although its Holevo information is 0.3251; five memories give $0.5668\, k_B T$ against 0.6385.

The gap between I_{acc} and χ has a thermodynamic reading. An agent that does not measure the fragment but keeps it and acts on S and F jointly can, in the asymptotic limit, use F as quantum side information, for which the relevant quantity is the conditional entropy $S(X \mid F) = H(X) - \chi(X{:}E_F)$ [19]; the Holevo value is then the ceiling. The accessible information is the value of the same record to an agent that first writes it into a classical memory.

6.3 A model of one's own readout is necessary

Theorem 17 (Every protocol assumes a posterior)**.** *For every feedback protocol* Π *on* S *and a bath, of the kind allowed in the task, there is a distribution* Q_Π *with full support such that for every diagonal state with populations* P

$$\mathbb{E}_\Pi[W_{\mathrm{ext}} \mid P] \le k_B T\big[\ln 2 - H(P) - D(P\|Q_\Pi)\big],$$

with equality for all P *when* Π *is the protocol of Lemma 15 designed for* Q_Π*. Consequently, if the agent runs* Π_m *after reading* m*,*

$$\mathbb{E}[W_{\mathrm{ext}}] - W_0 \le k_B T\Big[I(X{:}M) - \sum_m P(m)\, D\big(P(\cdot \mid m) \,\|\, Q_{\Pi_m}\big)\Big],$$

with equality for the matched two-step protocols. The record has negative value whenever the expected divergence exceeds $I(X{:}M)$*, and the value* $k_B T\, I(X{:}M)$ *is attained only if* $Q_{\Pi_m} = P(\cdot \mid m)$ *for every* m *with* $P(m) > 0$*.*

Proof. For a fixed protocol the extracted work is an affine function of the initial system state, $E(\rho) = -\operatorname{tr}\big[(U^\dagger H_{\mathrm{end}} U - H_{\mathrm{start}})(\rho \otimes \tau_B)\big]$, and by Lemma 15 it lies below the convex function $f(\rho) = k_B T\, D(\rho\|\mathbb{1}/2)$. On diagonal states $f(P) = k_B T[\ln 2 - H(P)]$, whose gradient $k_B T \ln P$ (up to a constant) takes every value as P ranges over distributions with full support. Choose Q_Π so that the tangent $T(P) = f(Q_\Pi) + \langle \nabla f(Q_\Pi), P - Q_\Pi\rangle = k_B T[\ln 2 - H(P) - D(P\|Q_\Pi)]$ has the same slope as E. Then $E - T$ is constant, and at $P = Q_\Pi$ it equals $E(Q_\Pi) - f(Q_\Pi) \le 0$. So $E \le T$ everywhere. The quasistatic two-step protocol for Q approaches T by Lemma 15. Averaging over m and subtracting W_0 gives the second display, and the divergence vanishes only when the distributions coincide. □

The extra dissipation $k_B T\, D(P\|Q)$ of a protocol tuned to Q and run on P is a special case of the mismatch cost of Kolchinsky and Wolpert, for a process whose output does not depend on its input [20]; the tangent argument adds that every protocol, tuned or not, has such a Q. The distribution Q_Π is what the protocol assumes about the system, whether or not anything in the agent represents it explicitly. An agent that obtains $P(\cdot \mid m)$ from the prior and its readout channel $P(m \mid x)$, that is, from how its own apparatus responds to the two pointer values, acts on a correct assumption only if its model of that channel is correct where the posterior depends on it.

What the agent needs is therefore not a full description of its apparatus but the part on which the posterior depends: given the prior, the likelihood ratio of its readout. Getting that part wrong has a price that no choice of protocol avoids. With one memory of overlap 0.8 and the Helstrom readout, the true error is 0.2 and the record is worth $0.1927\, k_B T$. An agent that believes its error is 0.01 loses $0.4287\, k_B T$ and ends $0.2359\, k_B T$ below an agent with no memory; one that believes 0.1 keeps $0.1483\, k_B T$; one that believes 0.4 keeps $0.1012\, k_B T$. Overconfidence in one's own readout is more expensive than underconfidence of the same size because the quench pushes the level of the excluded value up by $-k_B T \ln Q$.

6.4 Calibration from references

Let the readout be the binary channel of T-DOME III, $P(y = +1 \mid x) = (1 + gx + b)/2$ with pointer values written $x = \pm 1$, gain g and bias b, $|g| + |b| < 1$, and prior $1/2$ [29]. The agent estimates $\theta = (g, b)$ from n reference reads of each known pointer value.

Proposition 18 (Metric and calibrated loss)**.** *Let $P_\theta(m \mid x)$ be a smooth family and $L(\hat\theta) = \sum_m P_\theta(m) D(P_\theta(\cdot \mid m) \| P_{\hat\theta}(\cdot \mid m))$ the expected divergence of Theorem 17. Then*

$$L(\hat\theta) = \tfrac{1}{2}(\hat\theta - \theta)^\top G\,(\hat\theta - \theta) + O(|\hat\theta - \theta|^3), \qquad G = J_{XM} - J_M,$$

where J_{XM} is the Fisher information of the joint distribution of (x, m) with x drawn from the prior and J_M that of the marginal of m. If $\hat\theta$ is estimated from n reference reads per pointer value with second-moment matrix $\mathbb{E}(\hat\theta - \theta)(\hat\theta - \theta)^\top = (2nJ_{XM})^{-1} + o(1/n)$, third absolute moments $O(n^{-3/2})$, and $\mathbb{E}\big[L(\hat\theta)\mathbf{1}\{|\hat\theta - \theta| > \epsilon\}\big] = o(1/n)$ for some $\epsilon > 0$, then

$$\mathbb{E}\, L = \frac{1}{4n}\Big[\dim\theta - \operatorname{tr}\big(J_M J_{XM}^{-1}\big)\Big] + o(1/n).$$

For the gain–bias channel this is $\mathbb{E}\, L = \dfrac{1}{4n}\Big(1 + \dfrac{g^2}{1 - b^2}\Big) + o(1/n)$. The tail condition is needed: the plug-in frequency estimate meets the other conditions but returns a boundary value with positive probability, at which the model posterior excludes an outcome the true posterior allows, so its expected loss is infinite for every n; add-one-half estimates satisfy it.

Proof. For each m, $D(P_\theta(\cdot \mid m) \| P_{\hat\theta}(\cdot \mid m)) = \tfrac{1}{2}\delta^\top J_{X|m}\delta + O(|\delta|^3)$ with $\delta = \hat\theta - \theta$ and $J_{X|m}$ the Fisher information of the posterior family. The chain rule for Fisher information, $J_{XM} = J_M + \sum_m P(m) J_{X|m}$, gives G. On a fixed interior neighborhood, Taylor's remainder has expectation $O(n^{-3/2})$. Outside it, the assumed loss tail is $o(n^{-1})$, and the third-moment bound makes the quadratic tail $O(n^{-3/2})$. Thus $\mathbb{E}\, L = \tfrac{1}{2}\operatorname{tr}(G\Sigma) + o(n^{-1})$, with Σ the error

second-moment matrix including bias. With $\Sigma = (2nJ_{XM})^{-1}$, $\frac{1}{2}\operatorname{tr}(G\Sigma) = \frac{1}{4n}\operatorname{tr}\big((J_{XM} - J_M)J_{XM}^{-1}\big)$. For the gain–bias channel with prior 1/2, the efficient reference estimator of T-DOME III has covariance $\frac{1}{2n}\begin{pmatrix} 1-g^2-b^2 & -2gb \\ -2gb & 1-g^2-b^2 \end{pmatrix} = (2nJ_{XM})^{-1}$, the marginal of m is Bernoulli with parameter $(1+b)/2$, so $J_M = \operatorname{diag}(0, (1-b^2)^{-1})$, and $\operatorname{tr}\big(J_M J_{XM}^{-1}\big) = (1-g^2-b^2)/(1-b^2)$. □

Lemma 19 (Tail control for add-one-half calibration)**.** *For a fixed interior gain–bias channel, let independent $K_\pm \sim \mathrm{Bin}(n, p_\pm)$, $p_\pm = (1+b\pm g)/2$. Set $\hat{p}_\pm = (K_\pm+1/2)/(n+1)$, $\hat{g} = \hat{p}_+ - \hat{p}_-$ and $\hat{b} = \hat{p}_+ + \hat{p}_- - 1$. These estimates meet all moment and tail hypotheses of Proposition 18.*

Proof. The estimates are a linear transform of two independent frequencies. Their bias is $O(n^{-1})$ and their second-moment matrix is $(2nJ_{XM})^{-1} + O(n^{-2})$; the fourth centred binomial moment is $O(n^2)$, so their third absolute error moment is $O(n^{-3/2})$. Every estimated likelihood is at least $1/[2(n+1)]$. For the fair prior every model posterior probability is therefore at least $1/[4(n+1)]$, giving $0 \le L \le \log[4(n+1)]$. For any fixed sufficiently small neighborhood of the interior true parameter, Hoeffding's inequality bounds its complement by Ce^{-cn}, with $c > 0$. Its contribution to expected loss is at most $C\log[4(n+1)]e^{-cn} = o(n^{-1})$. □

The matrix G is the nonnegative posterior Fisher form on the agent's model of itself, a metric on its identifiable directions, and Proposition 18 gives it a thermodynamic meaning: a calibration error δ costs $\frac{1}{2}k_BT\,\delta^\top G\delta$ of work per use. The subtraction of J_M has a simple content. Unlabelled readings already carry information about the bias, which enters the marginal of m, and the posterior is less sensitive to that direction. For $g = 0.55$ and $b = 0$ the leading term is $1.3025/(4n)$; exact enumeration over the reference outcomes, with estimates $(k+\frac{1}{2})/(n+1)$ for the two response probabilities, gives $\mathbb{E}\,L = 0.02064$, 0.005113 and 0.001273 for $n = 16$, 64 and 256, ratios 1.014, 1.005 and 1.001 to the leading term. The same channel carries a record worth $0.1600\,k_BT$, and an agent that assumes $g = 0.8$ without calibrating loses $0.0666\,k_BT$ per use.

Corollary 20 (When calibration pays)**.** *Suppose each reference read costs w_{ref} and the calibrated model is used U times before it is renewed. To leading order the total cost $2nw_{\mathrm{ref}} + Uk_BT\kappa/(4n)$, with $\kappa = 1 + g^2/(1-b^2)$, is minimized at $n^* = \sqrt{U\kappa k_BT/(8w_{\mathrm{ref}})}$, and the overhead per use is $\sqrt{2\kappa k_BT w_{\mathrm{ref}}/U}$. If the known pointer states must be prepared from the thermal state of the degenerate pointer, $w_{\mathrm{ref}} \ge k_BT\ln 2$ by Landauer's bound. Calibration is worthwhile when this overhead is smaller than the loss of the uncalibrated model.*

Proof. Minimize $2nw + Uk_BT\kappa/(4n)$ over $n > 0$. Preparing a pure qubit state from the maximally mixed state with a bath at temperature T costs at least $k_BT\ln 2$ [12, 21]. □

With $g = 0.55$, $b = 0$ and $w_{\mathrm{ref}} = k_BT\ln 2$, the overhead per use is $0.134\,k_BT$ for $U = 100$, $0.0425\,k_BT$ for $U = 1000$ and $0.00425\,k_BT$ for $U = 10^5$, against a record value of $0.1600\,k_BT$. A model of one's own readout is worth building only if it is used many times before it goes stale. How long it stays fresh depends on how fast the apparatus drifts, which is the subject of T-DOME III.

Proposition 21 (Drift law for self-calibration; proved in Paper III)**.** *Let the scalar parameter follow a Gaussian random walk with increment variance ν^2 per use. Suppose each reference has conditional Fisher information at most J, unlabelled observations provide no information about that parameter, and the loss is globally $\frac{1}{2}k_BTG(\hat{\theta}-\theta)^2$ with constant $G > 0$. References cost w_{ref} each. Every predictable calibration policy has long-run average loss plus reference cost at least $c\, w_{\mathrm{ref}}^{1/3}(k_BTG\nu)^{2/3}J^{-1/3} - \frac{1}{2}k_BTG\nu^2$, where $c = 3 \cdot 4^{-2/3}$. With a Gaussian prior and Gaussian observations of controllable real precision at linear cost, a persistent Kalman estimate at rate $\rho^* = \big(k_BTG\nu/(4w_{\mathrm{ref}}\sqrt{J})\big)^{2/3}$ attains $3\rho^* w_{\mathrm{ref}}$ to leading order [29]. For whole reading counts, the continuous-precision optimum is a relaxation bound.*

The rate follows from the steady state of a Kalman filter: with ρ references per use the posterior variance is $\nu/\sqrt{\rho J}$ to leading order, and optimizing $w_{\mathrm{ref}}\rho + \frac{1}{2}k_BTG\nu/\sqrt{\rho J}$ gives the exponent. Paper III proves the lower bound by telescoping the posterior Cramér–Rao recursion, including predictable choices made from earlier records. It also proves the exact predictive-relative-entropy optimum of Consciousness III under a Gaussian prior and controllable precision. Free information changes the optimum: in this static readout experiment, unlabelled readings inform the directions where $J_M \neq 0$; in an acting body, the entire correlated history may inform additional parameters. Such a history needs its joint likelihood. For the gain–bias channel, the work loss is a local divergence, so the quadratic comparison requires Proposition 18's expansion and tail control. A bounded physical gain also needs its actual transition law; clipping a Gaussian walk changes the filtering problem. Smoother drift and vector parameters remain open.

7 Reading, harvesting and resetting

After a harvest the private memory must be retained, reset or replaced. A pure replacement of dimension d_M carries free energy $k_BT \ln d_M$ relative to its degenerate thermal state. Moving the old record to a spent register postpones reset and consumes storage; it does not restore a finite memory bank. We first price reset with a declared side-information menu.

Proposition 22 (Read–harvest–reset task bound)**.** *Let the agent read F into M, extract work from S with feedback on M, and reset M to a standard state by a protocol that couples M and possibly a second fragment G to a bath, leaves the marginal of G unchanged, and vanishes at the endpoints, all registers having degenerate Hamiltonians. The reset bath is thermal and initially independent of MG. No additional purity-bearing workspace or correlated side information is available to the reset. Exact blanking denotes the vanishing-error limit when a finite full-rank thermal bath cannot realize it. Then the net average work satisfies*

$$\beta\big(\mathbb{E}[W_{\mathrm{ext}}] - W_0 - W_{\mathrm{reset}}\big) \leq I(X{:}M) - S(M \mid E_G) = I(X{:}M) - H(M) + \chi(M{:}E_G).$$

Without G, the right side is $-H(M \mid X) \leq 0$, so this task has no positive net gain over the blind harvest. If the records are perfect ($r_F = r_G = 0$), the bound equals $h(p)$ and is approached by reading with a controlled-NOT, extracting with the matched protocol and resetting M by a controlled-NOT from G; with exactly perfect records it is reached only as a limit.

Proof. Theorem 16 bounds the extraction gain by $k_B T\, I(X{:}M)$. For the reset, the joint state of M, G and the bath evolves unitarily from a state in which the bath is thermal and uncorrelated with MG; with degenerate Hamiltonians and vanishing coupling at the endpoints, $W_{\mathrm{reset}} = \Delta E_B$. As in Lemma 15, $\beta \Delta E_B \geq S(\rho_{MG}) - S(\rho'_{MG})$, and with M pure at the end and the marginal of G unchanged, $S(\rho'_{MG}) = S(\rho_G)$, so $\beta W_{\mathrm{reset}} \geq S(M \mid E_G)$. For classical M, $S(M \mid E_G) = H(M) - \chi(M{:}E_G)$. Without G, $I(X{:}M) - H(M) = -H(M \mid X)$. With perfect records, M is a copy of X, $\chi(M{:}E_G) = H(M) = h(p)$, and the controlled-NOT from G maps M to its standard state at no work. □

For pure conditional fragment states with $p = 1/2$, $c_k = 0.8$ and Helstrom readings of five-memory fragments, $I(X{:}M) = 0.5668$, $H(M) = \ln 2$ and $\chi(M{:}E_G) = 0.5231$, so the net task gain is at most $0.3968\, k_B T$ with a second five-memory copy and at most $-0.1263\, k_B T$ without one; with seven-memory fragments the bounds are 0.5511 and −0.0611. Quantum conditional-entropy erasure has asymptotic realizations with the corresponding collective control and side-information permissions [19]. This does not give an arbitrary finite local eraser those permissions. The following classical record pair supplies a direct realization within the present task. If pure blanks are supplied instead, their entropy capacity can lower reset work and can itself fund work extraction; their consumed free energy must enter the input ledger. The harvest's information ceiling alone does not cap harvest minus a work-producing reset.

7.1 Two mixed ports: feedback and an actual conditional reset

Use the fair diagonal pointer and the controlled-X recording protocol of Paper F [26]. The two independent initial ports are $\eta_\alpha = (\mathbb{1} + \alpha Z)/2$, with $0 < \alpha < 1$ and $\epsilon = (1-\alpha)/2$. Their endpoint Hamiltonians are zero. The recording gates act as

$$(x, n_1, n_2) \longmapsto (x, m = x \oplus n_1, g = x \oplus n_2), \qquad X \sim \mathrm{Bern}(1/2), \quad N_1, N_2 \overset{\mathrm{iid}}{\sim} \mathrm{Bern}(\epsilon).$$

Actual Z readout uses the diagonal port M as the feedback control; no extra free copy of its value is introduced. The other physical port G is retained and is accessible to the reset. Preparation, gates and any controller that reads their values remain part of the apparatus.

Proposition 23 (Mixed-record task realization)**.** *In this protocol, put $t = 2\epsilon(1-\epsilon)$. Matched feedback on M has work gain over the blind harvest $k_B T[\ln 2 - h(\epsilon)]$. With G available as a reversible reset control, the infimum reset work is $k_B T h(t)$, and the supremum net task gain is*

$$k_B T\big[\ln 2 - h(\epsilon) - h(t)\big]. \tag{11}$$

Without G, the infimum reset cost is $k_B T \ln 2$ and the net gain is $-k_B T h(\epsilon)$. The supremum in (11) *is approached as a quasistatic, vanishing-reset-error limit and is positive for sufficiently small ϵ. It does not restore the prepared record source or the thermal bath supply.*

Proof. The actual joint record law is $P(m, g \mid x) = P_\epsilon(m \mid x) P_\epsilon(g \mid x)$. Thus M is fair, $H(X \mid M) = h(\epsilon)$ and $I(X{:}M) = \ln 2 - h(\epsilon)$. Lemma 15 with the two full-support posteriors gives the stated harvest. The matched branch protocol ends with a thermal source independent of the records, and leaves the classical joint law of MG unchanged.

The reversible controlled-NOT from G to M changes its physical content to $Z = M \oplus G = N_1 \oplus N_2$. The pair (Z, G) has law $\mathrm{Bern}(t) \otimes \mathrm{Bern}(1/2)$: for each fixed pair of noises the fair X makes G uniform. Hence $S(M \mid G) = h(t)$ before the reset, and after this gate the noisy logical bit is independent of the retained control.

Choose $t = q_0 > q_1 > \cdots > q_\ell = \zeta > 0$. At stage i, swap the degenerate logical memory with a fresh independent Gibbs qubit whose gap is $\Delta_i = k_B T \ln[(1 - q_i)/q_i]$. This is a driven swap, with endpoint interaction off. It resets the logical memory to error ζ, leaves G unchanged and independent of it, and has

$$\beta W_{\mathrm{reset}} = \beta Q_{\mathrm{bath}} = \sum_{i=1}^{\ell} (q_{i-1} - q_i) \ln \frac{1 - q_i}{q_i} \longrightarrow h(t) - h(\zeta).$$

The limit is the thermal-swap construction of Paper F and Reeb–Wolf [26, 21]. Letting $\zeta \downarrow 0$ reaches the entropy lower bound $h(t)$. The post-parity register is reset to the standard zero value, so the parity gate is not undone. Without G the same construction starts from the fair M and approaches cost $\ln 2$. Combining these reset costs with the matched harvest proves the claims. □

At $\epsilon = 0.05$ ($\alpha = 0.9$), the dimensionless harvest is 0.494632, conditional reset costs 0.313956 and net task gain is 0.180676; without the second record the net value is −0.198515. Both ports are partially mixed. Their total initial purity deficit is $2[\ln 2 - h(\epsilon)]$, paid before acquisition; their one-port record values saturate that budget in Paper F. The recording gates themselves have zero net endpoint work. Harvesting is a separate source–bath operation, not work extracted by the recording gate.

The retained G need not be a perfect copy of X: its correlation with the private memory is the useful reset resource. In the present menu it can fund a positive task balance while its marginal is unchanged. Its correlation with the harvested source, the two prepared ports, the driven gaps, clock and fresh Gibbs qubits have not all returned to their input states. A subsequent use must supply a new compatible source–record state and account for that preparation. More generally, an upkeep cost w can fit within the task only if $k_B T[I(X{:}M) - S(M \mid E_G)] \geq w$, before charging any omitted record supply or apparatus costs. This connects readable records to an action budget without identifying it with a sustainable complete-device cycle.

8 Operational temporal relations

The accounting identity does not derive a temporal order. Two notions of influence between times can be defined after a reduced model is specified, and they differ.

Definition 24 (Kernel memory and intervention influence)**.** *For a reduced model with Nakajima–Zwanzig kernel $\mathcal{K}$ and tolerance $\epsilon > 0$, write $s \rightsquigarrow_{\mathcal{K}} t$ if $\|\mathcal{K}(t, s)\| > \epsilon$ for some $s < t$; this is kernel memory. For a record-forming protocol, write $s \rightsquigarrow_{\mathrm{rec}} t$ if an intervention on the system at time s changes the records written at time t, as in the diagnostic of Paper F [26].*

The total linear response of a state at t to a perturbation at s is not a memory diagnostic: a Markov semigroup propagates a perturbation $\delta\rho$ as $e^{(t-s)\mathcal{L}}\delta\rho \neq 0$. The two relations of Definition 24 come apart in both directions. In the collision model the reduced dynamics is a product of dephasing channels with a time-local generator, so there is no kernel memory beyond a single step, while an intervention after the j-th collision changes every later record and none of the earlier ones (the fresh-fragment intervention result of Paper F). In the two-qubit model the coherence revives, so the reduced dynamics is not a semigroup and its Nakajima–Zwanzig kernel is not local in time, while the environment never holds a record. A correspondence between a memory-influence relation and nested record algebras therefore holds in neither direction in general; in fresh-memory collision models the record relation and the intervention order coincide, and kernel memory is absent.

9 The cost of keeping records

Landauer's principle prices logically irreversible resets, not passive storage, sensing, correlation maintenance or reversible computation. If a memory implementation irreversibly erases N_{erase} unbiased bits into a bath at temperature T, its reset work obeys $W_{\text{reset}} \geq k_B T \ln 2\, N_{\text{erase}}$; reversible storage and measurement can postpone this cost until an irreversible operation is performed [12, 13]. No lower bound proportional to stored mutual information follows without an implementation model, and Proposition 22 shows how strongly the reset cost depends on what the eraser can access.

To describe how much of a history is worth keeping, let $\{X_t\}$ be a stationary process describing the environment as seen by the agent. Its entropy rate is $h_\mu = \lim_{k\to\infty} h_k$ with $h_k = H(X_k \mid X_1, \ldots, X_{k-1})$ and $h_1 = H(X_1)$, and its excess entropy, or predictive information, is

$$\mathbf{E} = I(\overleftarrow{X}; \overrightarrow{X}) = \sum_{k=1}^{\infty} \big(h_k - h_\mu\big) \tag{12}$$

whenever the series converges [22, 24, 23]. For a symmetric two-state Markov chain with flip probability $0 < \varepsilon < 1/2$, $h_1 = \ln 2$ and $h_k = h(\varepsilon)$ for $k \geq 2$, so $\mathbf{E} = \ln 2 - h(\varepsilon)$. Finite excess entropy does not prove that a finite physical memory captures every useful feature, and a positive entropy rate does not bound the predictive information from below by h_μ times a memory depth.

Remark 25 (Raw-archive overload)**.** *Consider a specified implementation: an uncompressed archive receives records at r_{raw} bits per unit time, has capacity M bits and evaluates the entire archive once per control cycle. Without deletion it fills after M/r_{raw}. If it irreversibly resets discarded records at rate r_{erase}, Landauer gives $\dot{W}_{\text{reset}} \geq k_B T \ln 2\, r_{\text{erase}}$. For a long-memory or non-stationary source with divergent $\mathbf{E}$, the predictively relevant history may also grow; that is an assumption about the source. The archive can then exceed a finite storage, throughput or reset budget. This is an engineering overload scenario for one implementation.*

An agent using such an archive can respond by enlarging capacity, sampling less, uncomputing reversibly, keeping sufficient statistics, compressing lossily or accepting stale data. Each choice of what to keep needs a criterion that ranks stored correlations by their value.

Section 6 supplies one such value in work units for a single record; Paper II studies what a bounded agent reads, where a preferred frame arises from the agent's history rather than from the task ensemble. Neither compression, nor Clifford coordinates, nor symmetry breaking, nor a "self" is shown here to be thermodynamically necessary in general.

10 A classical filtering illustration

The following hidden Markov model compares finite observation windows with a Bayes filter in an aliased environment. Its reward is classification accuracy, not the survival functional, so it illustrates the task-level value of retained history rather than testing a theorem.

Proposition 26 (A finite-history classification separation)**.** *Let a fair binary hidden state flip with probability ε per step. Conditional on the hidden path, observations are independent, with $\Pr(O = 1 \mid X = 1) = 1/2 + \delta$ and $\Pr(O = 1 \mid X = 0) = 1/2 - \delta$, where $0 < \delta < 1/2$. A decision using only the last k observations, the known model and independent private randomness has classification risk at least*

$$R_k \geq \tfrac{1}{2}(1 - 2\delta)^k.$$

For odd n, the majority of n observations ending at the decision time has risk

$$R_{\mathrm{maj,n}} \leq (n-1)\varepsilon + \exp\bigl(-2n\delta^2\bigr).$$

For every fixed k, some finite n and positive ε make the second bound smaller than the first.

Proof. Every observation pattern of length k, conditional on either terminal hidden value, has probability at least $(1/2-\delta)^k$: each emission factor has that lower bound, and averaging over hidden paths preserves it. For the fair prior the optimal risk is $\frac{1}{2}\sum_o \min\{P(o \mid X = 0), P(o \mid X = 1)\}$. Summing the lower bound over 2^k patterns gives the first inequality; randomization cannot improve the optimal rule.

With probability at least $1-(n-1)\varepsilon$ there is no hidden flip in the n-observation episode. On that event the observations are independent samples with mean $1/2 + \delta$ or $1/2 - \delta$, so Hoeffding's inequality [31] bounds the majority error by $\exp(-2n\delta^2)$. The complement is charged its full probability. Choose odd n so that the exponential is below half the first bound, then choose $\varepsilon > 0$ small enough to make the remaining term smaller than that half. □

For $\delta = 0.05$, $k = 8$, $n = 1001$ and $\varepsilon = 10^{-4}$, these bounds are $R_{\mathrm{maj,n}} \leq 0.106704 < 0.215234 \leq R_8$. The count can be stored in $\lceil \log_2(n+1) \rceil = 10$ bits, in addition to the episode clock, decision and calculation resources; keeping every raw reading would require n bits instead. The comparison excludes additional correlated side information for the window rule. Acquisition, counter updates and any reset of its workspace still require an implementation budget; the classification bounds do not price them. It establishes a finite task separation, not a fixed physical memory that beats all window lengths, and not a bound on thermodynamic profit.

The hidden state $s_t \in \{0, 1\}$ keeps its value with probability $1 - \varepsilon$, $\varepsilon \in [10^{-3}, 10^{-0.7}]$, so the correlation length is $\ell \sim 1/\varepsilon$. Observations are aliased: $\Pr(o_t = A \mid s_t = 0) = 0.5 + \delta$

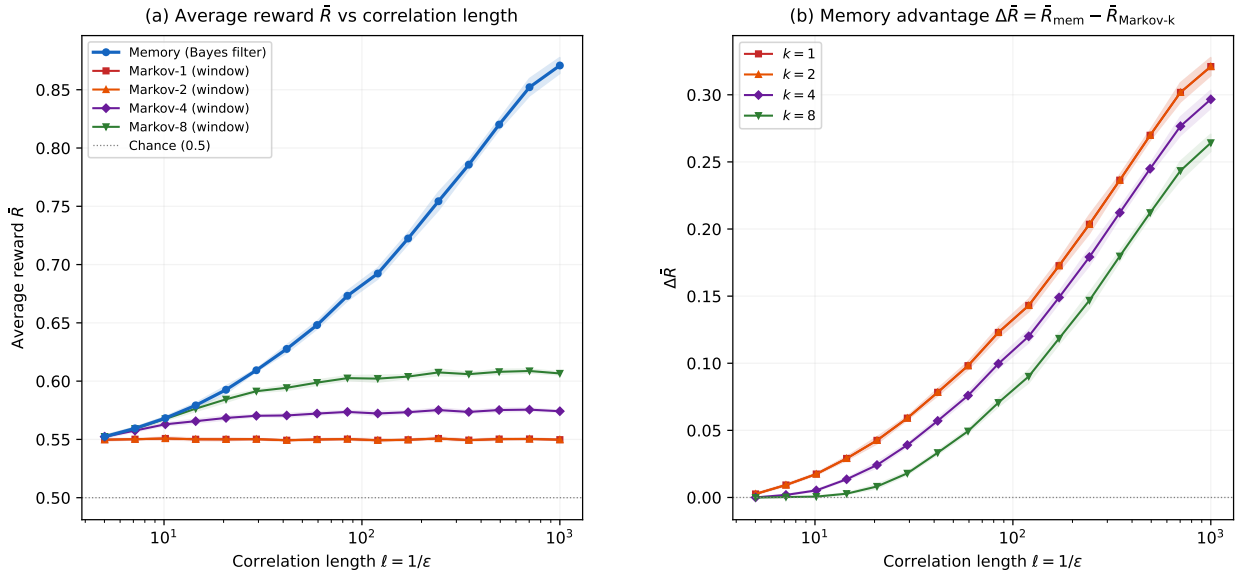


Figure 1: **Finite-window saturation and Bayes-filter advantage.** Historical sweep: $T = 10^5$ total steps, 2000 discarded, 10 seeds. Shading uses 1.96 times the across-seed standard deviation (with NumPy's population convention) divided by $\sqrt{10}$; these are normal-approximation error bands, not rigorous finite-sample confidence intervals. **(a)** Average reward $\bar{R}$ against correlation length $\ell = 1/\varepsilon$. The Bayes filter (blue, bold) rises monotonically; Markov-k agents saturate at k-dependent levels. **(b)** Gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$, which increases with ℓ and is larger for smaller k.

and $\Pr(o_t = A \mid s_t = 1) = 0.5 - \delta$ with $\delta = 0.05$, so a single observation carries 0.0072 bits about the state. The reward is 1 when the action equals s_t. All agents know the model and compute exact posteriors; Markov-k agents ($k \in \{1, 2, 4, 8\}$) use only the last k observations from a uniform prior, and the memory agent runs the full Bayes filter. The supplied script runs $T = 10^5$ total steps, discards the first 2000, and uses 10 seeds at 16 logarithmically spaced values of ε. The supplementary 16-row summary reports this sweep.

The reward of the Bayes filter increases with the correlation length, while each Markov-k agent saturates at a level ordered by k, all below the filter for $\ell \gtrsim 20$ (Figure 1). The Markov-1 and Markov-2 curves nearly coincide at small ℓ because short windows add little in this aliasing regime. The sampled long-correlation regime shows an advantage over the four specified windows. For the given model and observation history, the full posterior Bayes decision minimizes classification risk. The example does not establish tight saturation constants or superiority across different reading and resource menus. Theorem 16 supplies the conversion to work for a different task, in which the retained statistic is used for feedback on a system whose state it predicts. The supplementary source record gives the script, historical sweep and legacy-data distinction; these classification data do not computationally verify a survival-functional theorem.

11 Discussion

11.1 The memory–self–introspection route

In the task of Section 6, a work gain requires a memory carrying the relevant correlation. A protocol matched to its posterior realizes the value $k_B T\, I(X{:}M)$; every error in that posterior has a computable price. Reference calibration builds the required readout model, and reuse pays its preparation cost through the square-root amortization rule. Paper III proves the stated scalar Gaussian drift law. These are the concrete links in the series' memory–self–introspection route. Proposition 22 adds the reset condition for its allowed menu, and Proposition 23 realizes positive task value with two partially mixed records. A useful second record reduces the reset price; a full apparatus cycle still has to renew the source, recording stock and bath supply. The conditions belong to this task and its allowed protocols. Driven Markovian systems can persist without memory or a self-model.

The results concern function. They show what an agent must hold and do to cash a record, and they do not show that holding a model of one's own readout is, or produces, experience. The relation between such functional self-models and first-person organization is treated in the Consciousness series with its own arguments.

The task fixes the record source, pointer basis and feedback device. Paper II addresses how an agent selects what it reads; Paper III addresses detection, the price of tracking, and repair selected by the history of its body-gated access. The three correlation models here remain distinct: the Lorentz–Drude bath carries pointer information without designated fragments, the two-qubit environment carries none, and collision records have no autonomous recurrence. Paper III combines the collision records, a bounded changing instrument, a field-coupled actuator, harvest and reset on one ledger [29]. It exhibits a positive operating margin after body upkeep under supplied food, record purity, drive preparation, phase reference and timing. The calculation does not make those inputs products of the loop. Paper III also constructs a continuing finite joint filter: a bounded changing instrument and controlled body generate the history that revises their model, with a uniform quantization-risk bound and adaptive reference timing. The updating mechanism is thus explicit under its stated transition law and resource permissions. Its processor, reference supply and drive preparation still need one sustainable apparatus ledger.

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Reference Frames Selected by History: The Capability and Cost of a Bounded Agent's Self-Frame

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Abstract

With limited access, a system must learn how a task appears through its own channels before it can choose useful reads. We study this retained relation and the foreground it selects as a self-frame. For a Gaussian task with an unknown, isotropically distributed coupling and $b < D$ linear reads per step, the posterior mean determines the optimal reading subspace and prediction. Beating a history-independent foreground requires a calculable amount of information about the coupling. We distinguish the sharp information price of subspace alignment from the signed information needed to use that subspace: perfect alignment alone can leave every prediction uninformative. In-frame evidence has a further blind spot. It cannot locate the omitted direction or, when signal and noise strengths are unknown, distinguish world noise from its own misalignment. Exact drift curves show when a frozen model becomes worse than no memory and when a retained frame loses its advantage. Off-frame probes revise the frame at a cost in reads and storage; for eight channels and three reads, the specified policy lowers static risk from 0.86 to 0.26. Weighting predictions by their calculated reliability extends the drift range in which this policy beats the fixed-foreground comparator. These task-specific results supply a formation mechanism within the memory–self–introspection route. A learned protective encoding supplies a separate physical example of history changing the relations used by a body. A quantum reference gives a second, finite resource realization of a self-frame. Its advantage in any bounded decision problem is bounded by its asymmetry. For a single qubit phase token we obtain the exact optimal discrimination probability and its loss under dephasing.

1 Introduction

1.1 The question

A system that persists by using its records cannot use all of them at once. It reads some channels and not others, keeps some correlations and lets others go. Whatever rule makes

that choice embodies a view of where the task lies in the system's own coordinates. If the system's coupling to the world is not known in advance, that view has to be learned, and once learned it decides which evidence the system will see next. This paper asks four questions about such a view. When is it informationally necessary? How can it be formed from the agent's own records? What capability does it buy? What does it cost, including what it prevents the agent from seeing?

The series calls this organized view the ego. The word is used here for a definite object: a retained model of how a task-relevant direction sits in the agent's own channels, together with the foreground of reads that the model selects. We call the pair a self-frame. It is a model of the agent's own coupling, because the direction it describes is the image of a world direction under that coupling.

1.2 Position within the series

This paper is the second of the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other lines are HAFF and Q-RAIF.

HAFF-E supplies the actual record protocol [5]; Q-RAIF supplies physical interfaces and their prices [6]. Here we ask how the retained history organizes the reader itself.

Paper I bounds the extra work available from a record by $k_B T\, I(X{:}M)$ in a matched feedback task, explains when that bound is attained, and includes preparation, updates and old correlations in the complete account [3]. It leaves open how an agent comes to select what it keeps. Paper III studies drift detection, the price of tracking a changing instrument, and history-selected repair of body-gated access [4]. The present paper sits between them. Its self-frame is formed from records, it decides which records are gathered next, and its blind spot is the reason a calibration loop is needed at all.

The main results progress from the posterior capability (Theorem 2) to the information needed to select and use a foreground (Theorem 4 and Proposition 5), then to its blind spot and drift. Section 8 gives an explicit revision policy with exact stationary risk. Section 5 treats supplied quantum references separately; Section 9 connects the formation mechanism to learned protection.

1.3 Related work

Bounded rationality [11], rate–distortion theory [12, 13], rational inattention [14] and the information bottleneck [15] treat the choice of what to keep as an optimization under an information constraint. Budgeted learning studies learners that may observe only a few attributes of each example; Cesa-Bianchi, Shalev-Shwartz and Shamir show, among other results, that a budget on the attributes available at prediction time can make arbitrary accuracy unattainable even when a perfect linear predictor exists [16] . The present setting is close to theirs, with a different emphasis: the unknown is the agent's own coupling, the foreground is a function of the agent's memory, and the questions are the information that the foreground must carry and the evidence it excludes. Reference frames as physical resources are reviewed by Bartlett, Rudolph and Spekkens [17]; Skotiniotis and Gour give the accessible information of frame-alignment protocols an operational role for the G-asymmetry [18] .

Theorem 4 is the classical counterpart for a direction on a sphere: an application of the Gibbs variational principle, which we do not claim as new mathematics. Its use here is to price the self-frame of a bounded agent and to connect that price to the agent's capability and blind spot.

2 The budgeted reader

2.1 Channels, task and coupling

At each step t the agent's D channels register a fresh input $x_t \sim \mathcal{N}(0, I_D)$, independent of everything before. After the agent responds, the consequence

$$y_t = \rho\, u_t \cdot x_t + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, s^2), \tag{1}$$

is revealed, with $\rho > 0$ and $|u_t| = 1$. Unless stated otherwise the agent knows ρ and s; Theorem 12(iii) drops this. The agent is scored by the squared error of its anticipation $\hat{y}_t$ of y_t. More generally the consequence may be a vector $y_t = A_t x_t + \xi_t \in \mathbb{R}^p$ with noise covariance of trace ν; we write $W = A^\top A$.

The direction u_t is expressed in the agent's channel coordinates. If e is the task-relevant direction in world coordinates and $R_t \in O(D)$ is the agent's coupling, which maps its channels to the world, then $u_t = R_t^\top e$. A model of u_t is therefore a model of the agent's own coupling as far as this task is concerned. The records cannot separate a change of R_t from a change of e; only their relation enters. Distinguishing the two requires a reference with a known world direction, the role played by reference reads in Paper III, or a known push through an actuator that shares the agent's sensing coupling; Paper III shows that pushes through an independent coupling do not help [4].

Symmetric ensemble. The initial coupling is Haar distributed, so u_0 is uniform on the sphere S^{D-1}, and more generally the law of A_0 is invariant under $A_0 \mapsto A_0 O$ for $O \in O(D)$. Nothing in the task specification prefers any channel.

2.2 The reading budget

At each step the agent may make at most b linear reads $w_1 \cdot x_t, \ldots, w_b \cdot x_t$ with orthonormal w_j chosen from what it retains. The span of the reads used for the response is its *foreground* P_t, an orthogonal projection of rank $k \le b$. Everything the agent retains from past steps is its memory M_t; it may contain past reads, past outcomes and any summaries of them. The foreground and the response are functions of M_t and of the current reads only.

A full probe reads all D channels. With $b < D$ it cannot be made at every step. The resource protocol charges reads per step, start-up probes and retained numbers or bits; it does not assign a thermodynamic price. Section 8.3 relates it to the work currency of Paper I.

Definition 1 (Self-frame)**.** A self-frame is a retained state from which the agent computes, at each step, an estimate of how the task direction sits in its own channels and the foreground that this estimate selects. Its alignment at step t is $c_t = \mathbb{E}|P_t u_t|^2$ for rank-one tasks.

2.3 Comparators

All comparisons in this paper charge the same b reads per step and the same start-up probes.

- *No memory*: $\hat{y} = 0$.
- *No self-frame*: a foreground fixed independently of the history (for definiteness the first b channels), with in-frame coefficients learned from outcomes.
- *Fixed self-frame*: a foreground formed once from the start-up probes, with in-frame coefficients learned afterwards.
- *Frozen self-model*: frame and coefficients formed once and never revised.
- *Calibratable self-frame*: $b - 1$ coordinate probes plus one orthogonal residual read, which together supply the response along the current frame and revise it.

3 What a foreground is worth

Theorem 2 (Capability of a budgeted reader)**.** *Let M be any memory independent of the current input, let P be a rank-k foreground that is a function of M, and let $\hat{y}$ be any function of (Px, M). For the vector task,*

$$\mathbb{E}|y - \hat{y}|^2 \;\geq\; \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\,\mathrm{tr}\big(P\,\widehat{W}_M P\big) \;\geq\; \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\sum_{i\leq k}\lambda_i(\widehat{W}_M), \tag{2}$$

where $\widehat{W}_M = \mathbb{E}[A \mid M]^\top \mathbb{E}[A \mid M]$ and $\lambda_1 \geq \lambda_2 \geq \cdots$ are its eigenvalues. The first bound is attained by $\hat{y} = \mathbb{E}[A \mid M]Px$; the second by taking P to be a top-k eigenspace of $\widehat{W}_M$. For the rank-one task (1)*, with $m = \mathbb{E}[u \mid M]$,*

$$\mathbb{E}(y - \hat{y})^2 \;\geq\; s^2 + \rho^2\big(1 - \mathbb{E}|Pm|^2\big) \;\geq\; s^2 + \rho^2\big(1 - \mathbb{E}|Pu|^2\big). \tag{3}$$

Proof. Write $x = Px + (I - P)x$. Given M, the input x is independent of (A, M), and for fixed P the two Gaussian components are independent of each other. Hence $\mathbb{E}[(A(I-P)x) \mid Px, M] = \mathbb{E}[A \mid M]\,\mathbb{E}[(I-P)x] = 0$ and $\mathbb{E}[APx \mid Px, M] = \mathbb{E}[A \mid M]Px$. The Bayes response is therefore $\mathbb{E}[A \mid M]Px$, and its risk is $\mathbb{E}|y|^2 - \mathbb{E}|\mathbb{E}[A \mid M]Px|^2 = \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\,\mathrm{tr}(P\widehat{W}_M P)$, using $\mathbb{E}[xx^\top] = I$. No other response has smaller risk. Ky Fan's maximum principle gives $\mathrm{tr}(P\widehat{W}P) \leq \sum_{i\leq k}\lambda_i(\widehat{W})$ for every rank-k projection, with equality on a top-k eigenspace. For the rank-one task $\widehat{W} = \rho^2 mm^\top$, which gives the first inequality in (3). For the second, $\mathbb{E}|Pu|^2 = \mathbb{E}\,\mathbb{E}[|Pu|^2 \mid M] \geq \mathbb{E}|P\,\mathbb{E}[u \mid M]|^2$ by Jensen's inequality, since P is fixed given M. □

Corollary 3 (Three limits)**.** *(a)* No history. *If M is independent of the coupling, then $\mathbb{E}[A \mid M] = 0$ in the symmetric ensemble and every response has risk at least $\nu + \mathbb{E}\,\mathrm{tr}\,W$, the risk of $\hat{y} = 0$.*

(b) Known task. *If M determines A, the best rank-k risk is $\nu + \sum_{i>k} \lambda_i(W)$, attained by the top-$k$ eigenspace of W. Each added dimension strictly helps while $\lambda_{k+1} > 0$, and the optimal subspace is unique when $\lambda_k > \lambda_{k+1}$. If $W = \lambda I$, every rank-k subspace is optimal.*

(c) Frame-free level. *If the foreground is independent of the coupling, then $\mathbb{E}|Pu|^2 = k/D$ and every policy has risk at least $s^2 + \rho^2(1 - k/D)$ in the rank-one task.*

Proof. (a) Invariance gives $\mathbb{E}[A] = \mathbb{E}[A]O$ for all $O \in O(D)$, hence $\mathbb{E}[A] = 0$ for $D \geq 2$; use (2). (b) Here $\widehat{W} = W$. (c) For P independent of the uniform u, $\mathbb{E}|Pu|^2 = \operatorname{tr} P/D$. □

Theorem 2 has the same form as Paper I's theorem that every feedback protocol assumes a posterior [3]: whatever a budgeted policy represents, its performance is bounded by the alignment of the foreground it actually uses. The agent need not store an explicit direction. It does need some retained state from which its foreground is computed, and no response computed from that state can do better than the alignment of its foreground allows.

Part (b) is the known-task case of rank selection, a spectral variational fact. It locates the source of any preferred subspace. When W is anisotropic with a known eigenbasis, the selection is fixed by the task, an explicit asymmetry. When the ensemble is symmetric, as here, no subspace is preferred before records arrive. A preferred foreground can then come only from the agent's history, which is the subject of the next section.

4 A frame is a record of the coupling

For uniform u and a fixed rank-k projection P, the overlap $t = |Pu|^2$ has the Beta law $\beta_{k,D} = \mathrm{Beta}(k/2, (D-k)/2)$ with mean k/D and moment generating function $\mathbb{E}e^{\kappa t} = {}_1F_1(k/2; D/2; \kappa)$. Define

$$\Lambda^*_{D,k}(c) = \sup_{\kappa \geq 0} \Big[\kappa c - \ln {}_1F_1\big(\tfrac{k}{2}; \tfrac{D}{2}; \kappa\big) \Big] \tag{4}$$

for $0 \leq c < 1$. It is the upper half of the Legendre transform of the log moment generating function of $\beta_{k,D}$: convex, zero for $c \leq k/D$ and increasing above.

Theorem 4 (The information a self-frame must carry)**.** *In the symmetric rank-one ensemble, let P be a rank-k foreground that is a function of the agent's memory. Then*

$$I(U; P) \;\geq\; \Lambda^*_{D,k}\big(\mathbb{E}|PU|^2\big). \tag{5}$$

For $k/D \leq c < 1$ the bound is attained: if P is Haar distributed and, given P, U has density proportional to $e^{\kappa|PU|^2}$ with respect to the uniform measure, with $\kappa \geq 0$ chosen so that $\mathbb{E}|PU|^2 = c$, then U is uniform and $I(U; P) = \Lambda^*_{D,k}(c)$. (For $\kappa < 0$ the same construction gives the lower branch of the Legendre transform, which (5) does not use.) Consequently, a policy whose risk is at most $r < s^2 + \rho^2(1 - k/D)$ has a foreground with

$$I(U; P) \;\geq\; \Lambda^*_{D,k}\Big(1 - \frac{r - s^2}{\rho^2}\Big) > 0. \tag{6}$$

Proof. Since U is uniform, $I(U;P) = \mathbb{E}_P\, D(\mathcal{L}(U \mid P)\|\mathrm{Unif})$. Fix a value of P and map $u \mapsto |Pu|^2$. Under the uniform law the image is $\beta_{k,D}$; let Q_P be the image of $\mathcal{L}(U \mid P)$. The data-processing inequality gives $D(\mathcal{L}(U \mid P)\|\mathrm{Unif}) \geq D(Q_P\|\beta_{k,D})$. The Gibbs variational inequality $D(Q\|\beta) \geq \kappa\mathbb{E}_Q t - \ln \mathbb{E}_\beta e^{\kappa t}$ for every κ gives $D(Q_P\|\beta_{k,D}) \geq \Lambda^*_{D,k}(\mathbb{E}_{Q_P} t)$. Averaging over P and using convexity of Λ^* (Jensen) proves (5). In the attaining construction, rotation invariance of the uniform measure and of the Haar law of P makes the marginal of U uniform. Given P, the law of U is the exponential tilt of the uniform law by $\kappa|Pu|^2$; its image is the corresponding tilt of $\beta_{k,D}$, both inequalities are equalities, and the relative entropy is $\kappa c - \ln {}_1F_1(k/2; D/2; \kappa)$ with κ chosen so that the tilted mean is c. For (6), Theorem 2 gives $\mathbb{E}|PU|^2 \geq 1 - (r - s^2)/\rho^2$, and Λ^* is increasing on $[k/D, 1)$. □

Proposition 5 (Information needed to use an aligned frame)**.** *Let U be uniform on S^{D-1}, $D \geq 2$, and let $m_M = \mathbb{E}[U \mid M]$. Set*

$$\Psi_D(a) = \sup_{t \geq 0}\{ta - \log \mathbb{E}e^{tU_1}\}, \qquad 0 \leq a \leq 1,$$

with $\Psi_D(1) = +\infty$; the inequalities use extended relative entropy. For any memory and any allowed rank-k foreground,

$$I(U;M) \geq \mathbb{E}\Psi_D(|m_M|) \geq \frac{D}{2}\mathbb{E}|m_M|^2 \geq \frac{D}{2}\mathbb{E}|Pm_M|^2. \tag{7}$$

Consequently a predictor with risk at most $r < s^2 + \rho^2$ needs $I(U;M) \geq \frac{D}{2}[1 - (r - s^2)/\rho^2]$ nats. This bound concerns the signed conditional mean needed for prediction; it is not asserted to be sharp.

Proof. For each posterior, the relative-entropy variational inequality applied to $tv \cdot U$, with $v = m_M/|m_M|$, gives $D(\mathcal{L}(U \mid M)\|\mathrm{Unif}) \geq \Psi_D(|m_M|)$. At $m_M = 0$ take $t = 0$. The uniform sphere has odd moments zero and

$$\mathbb{E}U_1^{2j} = \frac{(2j-1)!!}{D(D+2)\cdots(D+2j-2)} \leq \frac{(2j-1)!!}{D^j}.$$

Termwise comparison of the moment-generating series gives $\log \mathbb{E}e^{tU_1} \leq t^2/(2D)$. Choosing $t = D|m_M|$ gives the second inequality in Eq. (7); averaging gives the mutual information. Projection contracts the norm. Finally Theorem 2 implies $\mathbb{E}|Pm_M|^2 \geq 1 - (r - s^2)/\rho^2$ for any such predictor. □

The attaining law in Theorem 4 is invariant under $U \mapsto -U$ conditional on P. Its posterior mean is therefore zero: a memory consisting only of that P has no predictive advantage, despite its high alignment. At the limit, knowing the unoriented line $UU^\top$ gives alignment one and still leaves the sign undetermined. Outcome-bearing records, not merely an axis label, supply the signed relation required by the response. The alignment table below is an information bound for choosing a subspace, not an achievable risk–information curve. For a finite memory its entropy bounds both the alignment price and the signed price. The step-capacity argument below likewise applies to either necessary information bound.

D	$c = 0.5$	$c = 0.9$	$c = 0.99$	ratio to $\frac{D}{2}\log_2\frac{1}{1-c}$ at $c = 0.9$
2	0	1.18	2.92	0.354
3	0.21	2.80	6.19	0.561
8	2.46	11.07	22.76	0.833
16	6.45	24.35	49.33	0.916
64	30.49	104.07	208.78	0.979
256	126.50	422.97	846.58	0.995

Table 1: Minimum frame information $\Lambda^*_{D,1}(c)$ in bits for a one-dimensional foreground (Theorem 4). For $D = 2$, $c = 1/2$ is the frame-free alignment.

Corollary 6 (Price of forming a frame)**.** *(a) A memory from which a foreground of alignment c is computed holds at least $\Lambda^*_{D,k}(c)/\ln 2$ bits.*

(b) Each step, whatever reads are made, conveys at most $C_1 = \frac{1}{2}\ln(1 + \rho^2/s^2)$ nats about a static coupling. Forming a foreground of alignment c from scratch therefore takes at least $\Lambda^*_{D,k}(c)/C_1$ outcome-bearing steps.

Proof. (a) $I(U;P) \le I(U;M) \le H(M)$. (b) Let H_{t-1} be the history. The reads at step t are functions of x_t and H_{t-1}, so $I(U;\text{reads}_t, y_t \mid H_{t-1}) \le I(U; y_t \mid x_t, H_{t-1})$. Given (x_t, H_{t-1}) the outcome has variance at most $\rho^2 x_t^\top \Sigma x_t + s^2$, with Σ the posterior covariance of u, and given U as well its entropy is that of $\mathcal{N}(\cdot, s^2)$. The Gaussian maximum-entropy bound and Jensen's inequality give at most $\frac{1}{2}\ln(1 + \rho^2 \operatorname{tr}\Sigma/s^2) \le C_1$, since $\mathbb{E}x_t^\top \Sigma x_t = \operatorname{tr}\Sigma \le 1$. The chain rule sums the steps, and $I(U;P) \le I(U;H_n)$. □

Table 1 lists $\Lambda^*_{D,1}$. For $D = 2$ the transform is $\sup_\kappa[\kappa(2c-1) - \ln I_0(\kappa)]$, the von Mises form on the doubled angle; the two expressions agree to 10^{-15} in the check. A Laplace evaluation of the moment generating function for large D gives $\Lambda^*_{D,1}(c) = \frac{D}{2}\ln\frac{1}{1-c}\,(1 + o(1))$ at fixed c, and the last column shows the approach. A self-frame thus costs about half a natural logarithm of the inverse misalignment per channel, the same per-direction precision cost that appears in finite self-models generally [7].

Remark 7 (Symmetry broken by the record)**.** If the agent's update rule is equivariant, so that rotating all channel coordinates rotates its records, memory and foreground alike, then the marginal law of the foreground over histories is Haar: the ensemble prefers no subspace. Conditional on one history the foreground is concentrated, and $I(U;P)$ measures how far. This is the precise sense in which a reference frame is selected spontaneously here. The symmetry of the task ensemble is intact; each history breaks it, and the frame is the trace that the history leaves. The reading budget makes a selection valuable (Corollary 3(c)); the records decide which frame is selected.

Remark 8 (Group-theoretic and quantum settings)**.** Nothing above uses the full orthogonal group beyond its transitivity on directions. If the channels carry a representation of a smaller group acting transitively on the sphere, for example $SO(3)$ acting on the pure quaternions of $Cl(0,2) \cong \mathbb{H}$, the overlap law and Theorem 4 are unchanged. In quantum settings an agent without a frame is restricted to group-averaged operations, and alignment resources

are quantified by asymmetry measures [17, 18]. The next section gives a quantum task bound and an exact finite reference. The spherical information price is not identified with quantum asymmetry.

5 A quantum reference changes the available reading

The classical foreground is a policy selected by memory. A retained quantum reference changes what a symmetry-restricted policy can measure. This is an operational bridge to a fixed-point algebra, rather than an identification of that algebra with a classical subspace.

Proposition 9 (Bounded task value of a retained reference)**.** *Let a compact group G act on a finite reference system, with twirling $\mathcal{G}$. In a specified task the reference starts in τ, independent of the hidden task label; target states and permissions are unchanged when the reference is replaced by $\mathcal{G}(\tau)$. The same class of physical protocols is allowed in both experiments. If losses lie in $[0, L]$, their optimal Bayes risks satisfy*

$$|R^*(\tau) - R^*(\mathcal{G}(\tau))| \leq \frac{L}{2}\|\tau - \mathcal{G}(\tau)\|_1 \leq L\sqrt{\frac{\mathcal{A}_G(\tau)}{2}}, \tag{8}$$

where $\mathcal{A}_G(\tau) = D(\tau\|\mathcal{G}(\tau))$ uses natural logarithms. For jointly G-invariant measurement permissions, an invariant reference induces only invariant effects on the target.

Proof. For each hidden label and each allowed protocol, the input trace distance is $\|\tau - \mathcal{G}(\tau)\|_1/2$. Contractivity bounds the total variation of its outcome laws by the same number. A loss in $[0, L]$ changes by at most L times that variation. Averaging and taking the infimum over the unchanged protocol class gives the first bound. Quantum Pinsker gives the second. The twirled-state relative entropy equals the entropy increase under twirling [2, Proposition 2 and Theorem 3]; the task bound is its combination with distinguishability. If a joint effect commutes with $U_S(g) \otimes U_R(g)$ and τ is invariant, partial expectation over R gives an effect commuting with $U_S(g)$. □

Proposition 10 (One finite phase reference and its passive lifetime)**.** *Two equiprobable target states are $\rho_\pm = (I \pm X)/2$. The retained reference $\tau_r = (I + rX)/2$, $0 \leq r \leq 1$, is prepared relative to their source phase. Allowed joint effects commute with $Z_S + Z_R$. The exact optimal success probability is*

$$P^*(r) = \frac{1}{2} + \frac{r}{4}. \tag{9}$$

With independent phase-dephasing of the unused reference, $r(t) = r_0 e^{-\kappa t}$. A required advantage $\Delta > 0$ above chance can persist without renewal only until $t \leq \kappa^{-1} \ln[r_0/(4\Delta)]$, for $4\Delta \leq r_0$.

Proof. The optimum invariant measurement is the Helstrom measurement of the two jointly twirled input states. Their difference is

$$\mathcal{G}_{SR}(\rho_+ \otimes \tau_r - \rho_- \otimes \tau_r) = \frac{r}{4}(X \otimes X + Y \otimes Y).$$

Its nonzero eigenvalues are $r/2$ and $-r/2$, in the common-charge space spanned by $|01\rangle, |10\rangle$. The trace norm is r, giving Eq. (9); its eigenprojections are permitted. Dephasing multiplies the reference's transverse Bloch vector by $e^{-\kappa t}$, and substitution proves the lifetime. □

Corollary 11 (One noisy carrier as a record and a frame)**.** *Use $X_S + X_R$ as the conserved charge and $(I \pm Z)/2$ as the equiprobable target states. A retained reference with Bloch vector $(0, y, z)$ gives optimal invariant-measurement success*

$$P^*_{\text{frame}} = \frac{1}{2} + \frac{\sqrt{y^2 + z^2}}{4}.$$

Take the reference to be the initially positive record in HAFF-B's transverse-qubit model [1], with known elapsed time and known generator. If $E^*_{\text{record}}(t)$ is the optimal error for reading its two possible initial signs directly, then

$$P^*_{\text{frame}}(t) = \frac{3}{4} - \frac{1}{2} E^*_{\text{record}}(t). \tag{10}$$

Thus the same physical carrier loses its record distinguishability and its one-use reference advantage on the same slow decay scale, even though its exact fixed algebra is scalar.

Proof. A rotation about the charge axis brings the transverse reference vector to one axis without changing its length. It commutes with the symmetry and maps allowed effects to allowed effects. The calculation of Proposition 10, with the Pauli axes relabelled, therefore gives the first equation. The two noisy record states have opposite Bloch vectors of length $\sqrt{y^2 + z^2}$, so their Helstrom error is $(1 - \sqrt{y^2 + z^2})/2$, proving Eq. (10). □

The baseline reference $r = 0$ permits only diagonal target effects and cannot distinguish ρ_+ from ρ_-. Retaining $r > 0$ makes a phase-sensitive reading possible without a non-invariant joint measurement. It does not create an absolute phase: the entire input may be jointly twirled with no change of the reading probabilities. This is the established relational-reference mechanism [17], here used to connect task capability and passive noise. A fresh symmetric history cannot create $r > 0$ by covariant operations alone; each conditional output of an invariant-outcome covariant instrument is still invariant. The source-relative preparation is therefore a priced input, not a frame created from an uninformative record.

The lifetime concerns an unused token followed by one reading. Repeated measurement can disturb it further and requires a reuse account. If a protocol must keep its advantage in this drifting task, it must obtain new phase information, protect that information or refresh the reference; renaming a stale token a self-model does not change its capability. This is a finite quantum counterpart of the self-frame's calibration problem, not a sufficient condition for experience or a theorem that all tasks need asymmetry. Charge-label tasks, for example, can be solved with invariant effects.

6 The frame shapes its own evidence

Once a foreground is chosen, the agent's next records are read through it. The following theorem says what those records can and cannot reveal.

Theorem 12 (In-frame identifiability and its blind spot)**.** *Fix a rank-k foreground P and consider in-frame records (Px_t, y_t) from a static coupling.*

(i) Their joint law depends on (ρ, s, u) only through $b = \rho Pu \in \operatorname{ran} P$ and $V = \rho^2 + s^2$.

(ii) With ρ and s known, the likelihood is constant on each fibre $\{u : Pu = w\}$, a sphere of dimension $D - k - 1$ in the background. The posterior law of the background direction $(I - P)u/|(I - P)u|$ is not updated by any number of in-frame records.

(iii) With ρ and s unknown, every triple (ρ', s', u') with $\rho' Pu' = b$, $\rho'^2 + s'^2 = V$ and $|b| \le \rho' \le \sqrt{V}$ produces the same law. The risk reduction available by reorienting the frame, $\rho'^2 - |b|^2$, ranges over $[0, V - |b|^2]$ within this family: the unexplained variance may be entirely world noise or entirely the agent's own misalignment.

(iv) No rule using only in-frame records and the frame achieves risk below $V - |b|^2 = s^2 + \rho^2(1 - |Pu|^2)$, and in-frame regression attains it asymptotically.

Proof. (Px, y) is Gaussian with mean zero, $\operatorname{Cov}(Px) = P$, $\operatorname{Cov}(Px, y) = \rho Pu$ and $\operatorname{Var} y = \rho^2 + s^2$, which proves (i). Parts (ii) and (iii) follow, since the likelihood of any sample is a function of (b, V); with ρ, s known, b fixes Pu and nothing else, and the conditional law of the background direction given Pu is therefore unchanged by the data. For (iv), given u the part $\rho(I - P)u \cdot x + \xi_t$ of y_t has variance $V - |b|^2$ and is independent of Px_t and of all earlier records, so no rule based on them removes it; regression of y on Px estimates b consistently and attains $V - |b|^2$. □

Theorem 12 states the cost side of a self-frame. A frame that serves the agent well also decides which evidence it will receive, and that evidence cannot show where the missing part of the task has gone. Part (iii) adds a second limitation: from its own in-frame records the agent cannot know whether anything is missing. A large residual is compatible with a well-aligned frame in a noisy world and with a misaligned frame in a quiet one. Separating the two requires either a known noise level or reads outside the frame. This is the agent-level form of the distinction between a change of the world and a change of the agent's own coupling.

Corollary 13 (The two-channel case: an even residual)**.** *Let $D = 2$, $k = 1$, foreground v and $u = \cos\theta\, v + \sin\theta\, v_\perp$. The in-frame law depends on θ only through $\cos\theta$, so θ and $-\theta$ are indistinguishable, and the in-frame residual variance $s^2 + \rho^2 \sin^2\theta$ is even in θ. The Fisher information of one in-frame record about θ is*

$$F(\theta) = \frac{\rho^2 \sin^2\theta \left(\rho^2 + s^2 + \rho^2 \cos^2\theta\right)}{\left(s^2 + \rho^2 \sin^2\theta\right)^2} = \frac{\rho^2(2\rho^2 + s^2)}{s^4}\,\theta^2 + O(\theta^4). \tag{11}$$

One off-frame read along $v_\perp$ gives $\operatorname{Cov}(v_\perp \cdot x, y) = \rho \sin\theta$, which is odd in θ and resolves the sign.

Proof. The covariance of $(v \cdot x, y)$ is $\Sigma = \begin{pmatrix} 1 & a \\ a & V \end{pmatrix}$ with $a = \rho\cos\theta$. For a Gaussian family with parameter-dependent covariance, $F = \frac{1}{2}\operatorname{tr}(\Sigma^{-1}\Sigma'\Sigma^{-1}\Sigma')$; with $a' = -\rho\sin\theta$ and $\det\Sigma = V - a^2$ this equals $a'^2(a^2 + V)/(V - a^2)^2$, which is (11). □

Paper III takes this even residual law as its detection model, proves that it cannot identify the sign of the mismatch, and shows that a regular even record law with a nonzero quadratic leading perturbation has Fisher information quadratic and relative entropy quartic in the mismatch near calibration, which fixes the time needed to detect drift [4]. Corollary 13 derives the structure from a budgeted self-frame: the residual is even because the frame reads only one of two channels, and the sign is the unobservable fibre of Theorem 12. A single off-frame read is the minimal probe that breaks the symmetry. The derivation fixes the structure (evenness, a two-point fibre, quadratic information), not the Gaussian mean model used in Paper III's computations.

7 Drift: fixed and frozen frames

Let the coupling drift. In continuous time we take u_t to be Brownian motion on S^{D-1} with generator $\sigma\Delta_{S^{D-1}}$. A time-homogeneous discrete Markov step equivariant under $O(D)$, including reflections, has the corresponding scalar structure: $\mathbb{E}[u_{t+1} \mid u_t] = \lambda_1 u_t$, and the degree-two harmonics have multiplier λ_2. Equivariance under $SO(D)$ suffices for $D \geq 3$, but not for a directed drift on the circle.

Theorem 14 (Fixed and frozen frames under drift)**.** *For spherical Brownian motion with generator $\sigma\Delta$,*

$$\mathbb{E}[u_t \cdot v] = e^{-(D-1)\sigma t}\, u_0 \cdot v, \qquad \mathbb{E}[(u_t \cdot v)^2] - \tfrac{1}{D} = e^{-2D\sigma t}\big((u_0 \cdot v)^2 - \tfrac{1}{D}\big). \tag{12}$$

(a) A frozen self-model with response $\hat{y} = g{\cdot}x$, $g = \rho u_0$ fixed, has risk $s^2 + 2\rho^2\big(1 - e^{-(D-1)\sigma t}\big)$. It is worse than having no memory after $t^* = \ln 2/((D-1)\sigma)$, and worse than the frame-free level once $2(1 - e^{-(D-1)\sigma t}) > 1 - b/D$. If g is estimated from n_0 full probes, the risk is $s^2 + 2\rho^2 + \epsilon_0 - 2\rho^2 e^{-(D-1)\sigma t}$ with $\epsilon_0 = (D(\rho^2 + s^2) + \rho^2)/n_0$.

(b) A fixed foreground spanned by u_0 and $b - 1$ orthonormal directions orthogonal to it, with exact in-frame coefficients, has risk $s^2 + \rho^2(1 - b/D)\big(1 - e^{-2D\sigma t}\big)$. It never rises above the frame-free level and approaches it at rate $2D\sigma$.

For an $O(D)$-equivariant discrete step the same formulas hold with $e^{-(D-1)\sigma}$ and $e^{-2D\sigma}$ replaced by λ_1 and λ_2 per step. The discrete risks need not be monotone when these multipliers are negative.

Proof. The functions $u \mapsto u \cdot v$ and $u \mapsto (u \cdot v)^2 - 1/D$ are spherical harmonics of degrees one and two, with Laplace–Beltrami eigenvalues $-(D-1)$ and $-2D$; their expectations decay accordingly. For an $O(D)$-equivariant step the stabilizer of u forces $\mathbb{E}[u' \mid u] = \lambda_1 u$ and $\mathbb{E}[u'u'^{\top} \mid u] = aI + \lambda_2 uu^{\top}$, with $Da + \lambda_2 = 1$. Iteration gives the same first- and second-moment formulas with the corresponding powers of λ_1 and λ_2. (a) The risk is $s^2 + \mathbb{E}|\rho u_t - g|^2 = s^2 + \rho^2 + |g|^2 - 2\rho\, g \cdot \mathbb{E}u_t$. With g the mean of $y_i x_i$ over n_0 probes, $\mathbb{E}[g \mid u_0] = \rho u_0$ and $\mathbb{E}|g - \rho u_0|^2 = \epsilon_0$ by the Isserlis identity $\mathbb{E}[(u \cdot x)^2 x_i^2] = 1 + 2u_i^2$. (b) The risk is $s^2 + \rho^2(1 - \mathbb{E}|Pu_t|^2)$, and (12) gives $\mathbb{E}(u_t \cdot u_0)^2 = \frac{1}{D} + (1 - \frac{1}{D})e^{-2D\sigma t}$ and $\mathbb{E}(u_t \cdot w)^2 = \frac{1}{D}(1 - e^{-2D\sigma t})$ for $w \perp u_0$; summing over the b directions gives the stated curve. □

The two Brownian curves separate two failure modes that a single visibility angle cannot tell apart. A self-model that keeps both its frame and its coefficients becomes confidently wrong: its predictions remain as large as before while their direction decorrelates, and after t^* it does worse than predicting nothing. A frame that keeps learning its in-frame coefficients cannot become worse than the frame-free reader, because in-frame regression shrinks the coefficient on a stale direction, but it loses its whole advantage. Neither can recover the advantage without information from outside the frame, by Theorem 12. Paper III compares these curves with the time needed to detect drift and finds that a residual-variance test detects slow drift long before t^* [4].

Proposition 15 (Directed drift of a planar frame)**.** *Let $D = 2$ and $u_t = (\cos\varphi_t, \sin\varphi_t)$, with $\varphi_{t+1} = \varphi_t + \Delta_t$ modulo 2π, where the increments are independent with one common law. Write $\mu_j = \mathbb{E}e^{ij\Delta_t}$. For a known initial direction, a frozen predictor $g = \rho u_0$ has risk*

$$R_{\text{frozen}}(t) = s^2 + 2\rho^2[1 - \operatorname{Re}(\mu_1^t)].$$

A fixed one-dimensional foreground along u_0, using its exact current regression coefficient, has risk

$$R_{\text{axis}}(t) = s^2 + \frac{\rho^2}{2}[1 - \operatorname{Re}(\mu_2^t)].$$

These laws are $SO(2)$-equivariant. Reflection symmetry makes both multipliers real and recovers the scalar discrete formulas.

Proof. Independence gives $\mathbb{E}e^{ij(\varphi_t - \varphi_0)} = \mu_j^t$. The two risks are respectively $s^2 + 2\rho^2\mathbb{E}[1 - \cos(\varphi_t - \varphi_0)]$ and $s^2 + \rho^2\mathbb{E}\sin^2(\varphi_t - \varphi_0)$. Taking real parts proves both identities. For deterministic increments $\Delta_t = \phi$, they involve $\cos(t\phi)$ and $\cos(2t\phi)$, not $(\cos\phi)^t$ and $(\cos 2\phi)^t$. □

A directed rotation can repeatedly return a stale frame to alignment. The Brownian crossing time is therefore a property of that diffusion, not a universal lifetime inferred from rotational symmetry.

8 A calibratable frame and its price

8.1 The policy and its exact stationary risk

The calibratable agent stores a vector $g_t \in \mathbb{R}^D$, its estimate of $\theta_t = \rho u_t$. Before the input arrives it chooses a set S_t of $b - 1$ channel indices uniformly without replacement, independently of the world. It reads those coordinate directions and, as its last read, the normalized component of g_t orthogonal to their span. If this component vanishes it fills the slot with any unused orthogonal direction. These are exactly b orthonormal reads. From them it computes $g_t \cdot x_t$ and responds $\hat{y}_t = \gamma g_t \cdot x_t$, with $\gamma \in [0, 1]$. No repeated or additional read is required: coordinate values and the orthogonal residual reconstruct the noiseless linear functional algebraically. Each coordinate is probed with probability $p = (b-1)/D$. For each probed channel i it updates $g_{t+1,i} = g_{t,i} + \alpha(y_t x_{t,i} - g_{t,i})$; unprobed coordinates are kept. The frame-free comparator stores coefficients β_t on the fixed channels $S = \{1, \ldots, b\}$, reads them every step, updates them by the same rule with $p = 1$, and responds $\gamma\,\beta_t \cdot x_{t,S}$.

Proposition 16 (Exact stationary risk)**.** *Assume $1 < b < D$, $0 < \alpha < 2$, u_0 uniform, a drift step with $\mathbb{E}[u_{t+1} \mid u_t] = \lambda_1 u_t$ and a uniform invariant marginal, with factor $\lambda_1 = 1 - \kappa \in [0, 1]$, fresh inputs and noise, and probe choices independent of the world (correlations among the probe indicators of different channels do not matter). Let m be the number of tracked coordinates ($m = D$ for the calibratable frame, $m = b$ for the frame-free reader), p the probe probability ($p = 1$ for the frame-free reader), $T = \rho^2 m/D$ and $a = p\alpha(2 - \alpha)$. The tracking error e_t of the tracked coordinates has stationary moments*

$$\begin{aligned} c_\infty &= \mathbb{E}[e \cdot \theta] = -\frac{\kappa T}{1 - \lambda_1(1 - p\alpha)}, \\ A_\infty &= \mathbb{E}|e|^2 = \frac{p\alpha^2 m\big(\rho^2 + s^2 + \rho^2/D\big) + 2\kappa T + 2\kappa(1 - p\alpha)c_\infty}{a}, \end{aligned} \tag{13}$$

and the stationary risk with weight γ is

$$R(\gamma) = s^2 + \rho^2 - 2\gamma\,(T + c_\infty) + \gamma^2\,(T + 2c_\infty + A_\infty). \tag{14}$$

In particular $R(1) = s^2 + (\rho^2 - T) + A_\infty$, and the reliability weight $\gamma^* = (T + c_\infty)/(T + 2c_\infty + A_\infty)$ gives $R(\gamma^*) = s^2 + \rho^2 - (T + c_\infty)^2/(T + 2c_\infty + A_\infty) \le s^2 + \rho^2$; the weight lies in $[0, 1]$ under the stated nonnegative first-harmonic assumption.

Proof. Coordinate i is probed with indicator χ_i, $\mathbb{E}\chi_i = p$. The probe value yx_i has conditional mean θ_i given the past and the current coupling, and conditional variance $\rho^2 + s^2 + \rho^2 u_i^2$ by the Isserlis identity; its average over the uniform marginal of u_t, which holds at every t, is $\rho^2 + s^2 + \rho^2/D$. After the update the error is $e_i' = (1 - \alpha\chi_i)e_i + \alpha\chi_i n_i$ with n_i conditionally centred, so $\mathbb{E}|e'|^2 = (1 - a)\mathbb{E}|e|^2 + p\alpha^2 m(\rho^2 + s^2 + \rho^2/D)$ and $\mathbb{E}[e' \cdot \theta] = (1 - p\alpha)\mathbb{E}[e \cdot \theta]$. The drift step then replaces θ by $\theta + \Delta$ and e' by $e' - \Delta$, where Δ is independent of e' given u_t, $\mathbb{E}[\Delta \mid u_t] = -\kappa\theta$ on the tracked coordinates, $\mathbb{E}|\Delta|^2 = 2\kappa T$ and $\mathbb{E}[\Delta \cdot \theta] = -\kappa T$. Hence

$$\mathbb{E}|e_{t+1}|^2 = \mathbb{E}|e'|^2 + 2\kappa(1 - p\alpha)\mathbb{E}[e_t \cdot \theta_t] + 2\kappa T, \qquad \mathbb{E}[e_{t+1} \cdot \theta_{t+1}] = \lambda_1(1 - p\alpha)\mathbb{E}[e_t \cdot \theta_t] - \kappa T.$$

Both recursions are affine with contraction factors $1 - a$ and $\lambda_1(1 - p\alpha)$ of modulus less than one, so the moments converge to the fixed point (13) from any start. The response $\gamma\, g \cdot x$ has risk $s^2 + \mathbb{E}|\theta - \gamma g|^2$ with g supported on the tracked coordinates; expanding with $g = \theta_m + e$ gives (14). The minimizing weight follows by completing the square, and $R(\gamma^*) \le R(0) = s^2 + \rho^2$. □

Only the one-step correlation $\lambda_1 = \mathbb{E}[u_{t+1} \cdot u_t]$ of the drift enters. For spherical Brownian motion sampled at unit time, $\kappa = 1 - e^{-(D-1)\sigma}$. The weight γ^* is a property of the agent's own tracking error. An agent that knows its drift factor and noise level can compute it from (13); an agent that knows only its residuals cannot, because by Theorem 12(iii) its in-frame residual mixes its tracking error with world noise. The reliability weight is thus a small self-assessment, and its effect is measurable.

8.2 Capability and benefit threshold

Table 2 compares the five policies of Section 2 under one protocol. The exact stationary risk agrees with the simulation to within the Monte Carlo error at every drift level, including

κ	Frozen	No frame	Fixed	Calib.	Calib. (th.)	Calib. γ^*	No frame γ^*	Alignment
0	0.414	0.863	0.340	0.2608	0.2610	0.2607	0.863	0.991
1.40×10^{-5}	0.812	0.872	0.559	0.2856	0.2848	0.2852	0.872	0.967
8.75×10^{-5}	1.934	0.897	0.856	0.3367	0.3361	0.3350	0.897	0.921
3.50×10^{-4}	2.448	0.915	0.920	0.4188	0.4190	0.4116	0.914	0.849
1.40×10^{-3}	2.426	0.938	0.941	0.5743	0.5742	0.5477	0.935	0.718
5.54×10^{-3}	2.464	0.994	0.995	0.8411	0.8396	0.7568	0.985	0.506
1.23×10^{-2}	2.449	1.049	1.049	1.0383	1.0402	0.8969	1.030	0.353
2.15×10^{-2}	2.430	1.095	1.096	1.1740	1.1736	0.9944	1.064	0.238
4.61×10^{-2}	2.423	1.169	1.169	1.2516	1.2518	1.1043	1.116	0.128

Table 2: Time-averaged risk in the closed loop ($D = 8$, $b = 3$, $\rho = 1$, $s = 0.5$, start-up $n_0 = 64$ full probes, 24 000 steps, first 8 000 discarded, 200 trajectories with common random numbers). No memory gives 1.25. "Calib. (th.)" is $R(1)$ from Proposition 16; learning rates are set from the formulas, not tuned on the simulation. Alignment is $\mathbb{E}(u \cdot g/|g|)^2$ of the calibratable frame. Standard errors are at most 0.016 for the frame-free and fixed columns (dominated by the sample of couplings), at most 0.05 for the frozen column, and at most 0.0015 for the calibratable columns; the paired difference between the calibratable and frame-free readers has standard error at most 0.016.

levels where the calibratable frame loses. With a static coupling the calibratable frame reaches alignment 0.991 and risk 0.261, against 0.863 for the frame-free reader and 0.340 for the frame fixed after start-up. The frozen self-model is worse than no memory from $\kappa = 8.75 \times 10^{-5}$ on. The fixed frame becomes indistinguishable from the frame-free reader at moderate drift, as Theorem 14(b) predicts.

Without the reliability weight the calibratable frame loses to the frame-free reader between $\kappa = 1.23 \times 10^{-2}$ and 2.15×10^{-2}: it keeps responding at full strength along a direction it can no longer track. With γ^* for both readers it remains better through $\kappa = 4.61 \times 10^{-2}$. Table 3 gives the crossing drift κ_c from the exact formulas for other dimensions and budgets. Weighting by self-assessed reliability raises κ_c by factors from 2.1 ($D = 4$, $b = 3$) to 13.8 ($D = 32$, $b = 3$). Without weighting, $\kappa_c D^2/(b-1)$ stays between 0.14 and 0.94 over the computed cases: the sampled scaling is $\kappa_c \propto (b-1)/D^2$, or D^{-2} at fixed b, rather than $[(b-1)/D]^2$.

Corollary 17 (Static limit)**.** *With $\kappa = 0$ the stationary risk of the calibratable frame is $s^2 + \alpha D(\rho^2 + s^2 + \rho^2/D)/(2-\alpha)$ at weight 1, which tends to s^2 as $\alpha \to 0$, while the frame-free reader stays at or above $s^2 + \rho^2(1 - b/D)$. Under a reading budget $2 \le b < D$, a calibratable self-frame is therefore strictly better than every frame-free policy for all sufficiently slow drift.*

Proof. Set $\kappa = 0$ in (13). The frame-free bound is Corollary 3(c). Continuity of (13) in κ at fixed α extends the strict inequality to small κ. □

8.3 What the agent pays

The exact curves use noiseless linear reads and real-valued coefficients. Orthonormalizing the probe span is a computation on retained g and chosen indices before the input arrives. It

D	b	κ_c, weight 1	κ_c, weight γ^*
4	2	1.47×10^{-2}	3.82×10^{-2}
4	3	1.73×10^{-2}	3.68×10^{-2}
8	2	7.66×10^{-3}	3.49×10^{-2}
8	3	1.29×10^{-2}	5.78×10^{-2}
8	5	1.85×10^{-2}	7.19×10^{-2}
16	2	2.85×10^{-3}	2.23×10^{-2}
16	3	5.07×10^{-3}	4.00×10^{-2}
16	5	8.45×10^{-3}	6.44×10^{-2}
32	2	9.16×10^{-4}	1.26×10^{-2}
32	3	1.68×10^{-3}	2.31×10^{-2}
32	5	2.93×10^{-3}	3.93×10^{-2}

Table 3: Drift $\kappa_c = 1 - \lambda_1$ at which the calibratable frame and the frame-free reader have equal stationary risk, each with learning rate optimized from Proposition 16 ($\rho = 1$, $s = 0.5$). Below κ_c the calibratable frame is better.

needs arithmetic and working storage, not another observation. These resources are counted separately from the b reads. If the implemented response coefficient differs by a vector with mean squared norm at most ϵ^2, the squared-risk increase is bounded by $2\epsilon\sqrt{R - s^2} + \epsilon^2$, by Cauchy–Schwarz. Thus a strict benefit margin survives sufficiently precise finite arithmetic; precision is an additional paid input. Independent noisy hardware reads require their own covariance and calibration account.

All readers above use b reads per step and the same $n_0 D$ start-up reads. They differ in retained state. The frame-free reader keeps b coefficients. The fixed frame keeps b coefficients and its b directions. The calibratable frame keeps the D numbers of g, and Theorem 4 says that some such growth is unavoidable: alignment 0.991 in eight channels requires at least $\Lambda^*_{8,1}(0.991)$ nats in its foreground, 23.3 bits. In time the calibratable frame spends $b - 1$ of its b reads off its own frame. Paper I prices these resources when they are implemented thermodynamically. In Paper I's matched task, a record adds at most $k_B T$ times its task mutual information. Isothermal erasure of independent unbiased degenerate bits without usable side information costs at least $k_B T \ln 2$ per bit; general updates instead require the conditional-entropy and free-energy ledger (a known bit need not cost that amount). Reference preparation requires the free-energy difference for its specified Hamiltonian and initial state [3]. A pure degenerate one-bit reference has price $k_B T \ln 2$; an arbitrary reference does not have that universal price. The present paper keeps reads, probes and stored numbers as separate entries and does not convert them into a single price, because the conversion depends on the implementation.

9 Connections within the theory

9.1 A frame can change the body's protective relations

Foundation III gives a finite body implementation [10]. A diagnostic record of an unknown noise sign selects and later decodes an odd or even protective code on two body bits holding an unknown logical qubit and its inaccessible old reference. The controller sees the record, never the sign. In that protocol the learned code beats both fixed codes when $p > .51430317693$; at $p = .9$ its ideal loss is .0527213975, with early damage included. The full gate program, paid ports and source/return construction belong to Foundation III. This is a separate implementation, not an identification of Gaussian subspace alignment with quantum protection. The available encoders and diagnostic coupling remain inputs; Paper III studies history-selected maintenance when body access becomes unreliable. The positive subject-formation criterion proposed here is that learned relations organize the same system's future access, protection and revision, rather than merely naming a preferred basis. Paper III's body-gated repair gives this criterion an acting interpretation: the acquired model selects which future access is worth restoring. This is an interpretive connection between the models, not a theorem of general subject formation.

Paper I. Paper I establishes the task value of correlations and the conditions under which their preparation and maintenance pay [3]. Its feedback protocols imply a posterior; a record is not valuable independently of its task and implementation. Theorem 2 is the reading-side counterpart: every budgeted policy implies a frame, and its capability is the alignment of that frame. Theorem 4 prices the frame in information, as Paper I prices the record in work.

Paper III. Corollary 13 derives the even residual model that Paper III assumes, including the sign ambiguity and the quadratic Fisher signal. Paper III's reference reads play the role of the known world direction in Section 2: they separate a change of the agent's coupling from a change of the world, which in-frame records cannot do (Theorem 12(iii)). Its memory-under-drift proposition and Proposition 16 have the same structure, one for a scalar instrument parameter and one for a frame in D channels; the second shows that the cost of tracking a frame grows with the number of channels the frame excludes.

Consciousness III. Theorem 12 is an instance of a self-shaped unobservable fibre: what the agent cannot learn about itself is set by its own current frame and moves when the frame moves. Consciousness III develops the general theory of such fibres for finite self-models, including a floor for self-knowledge under drift [7].

Imprint and Brain Anchor. The reader of *The World as Imprint* learns which of two instrument axes it has and retains the corresponding information [8]. The self-frame here is its continuous, budgeted generalization, with the added feature that the learned model controls which records arrive. Brain Anchor's learned relation weights are another case of a frame formed from records and selecting later use [9].

10 Discussion

10.1 What is established

Within the budgeted reader, a self-frame is informationally necessary for any risk below $s^2+\rho^2(1-b/D)$ (Corollary 3 and Theorem 4); it can be formed from the agent's own records, at a price of at least $\Lambda^*_{D,k}(c)/C_1$ outcome-bearing steps and $\Lambda^*_{D,k}(c)/\ln 2$ bits (Corollary 6); it buys the capability computed in Theorem 2 and Proposition 16; and it has a blind spot that only reads outside the frame can repair (Theorem 12). The necessity is created by the reading budget. With $b = D$ the frame-free level falls to s^2 and selecting a foreground gains nothing, although the agent still needs memory of its coefficients.

For the memory–self–introspection route, this gives the self link a definite content. Task-relevant memory becomes a self-frame here when it selects reads through a learned model of the system's own coupling. The blind spot then gives a reason for acquiring new relations outside that frame. A system may implement this calibration function without representing a human-like self. Each statement concerns this task class. It is not claimed that every persistent system needs a self-frame; a system that can read everything it needs, or whose coupling is known, does not.

The self-frame is a functional object. Nothing here shows that holding one is, or produces, experience. The Consciousness series treats the relation between such functional self-models and first-person organization with its own arguments.

10.2 Drift boundaries and two conjectures

Proposition 18 (A drift factor does not determine the tracking problem)**.** *Let $U_t = S_tV$ with V initially uniform on the sphere and fixed, while $S_t \in \{-1, 1\}$ flips independently with probability $\kappa/2$ each step. This rotation-invariant Markov law has the same $\lambda_1 = 1-\kappa$ as a spherical diffusion with that first harmonic. Given a retained exact axis V, a legal reader of that axis has excess risk at most*

$$R - s^2 \le 4\rho^2 \left(\frac{w\kappa}{2} + e^{-wC_1}\right), \qquad C_1 = \frac{1}{2}\ln(1+\rho^2/s^2), \tag{15}$$

using only the last w outcome-bearing reads. In particular it has $O(\kappa\log(1/\kappa))$ excess risk at fixed signal-to-noise ratio.

Proof. Read $z_j = V \cdot x_j$ and estimate the current sign by the sign of $\sum_{j=t-w}^{t-1} z_j y_j$. Predict $\rho\widehat{S}_t V\cdot x_t$. A flip in the w transitions joining these outcomes to the current step has probability at most $w\kappa/2$. On the no-flip event, relabel the common sign positive. For one sample, $y = \rho z + s\eta$, with z, η independent standard normals. At $a = \rho/s^2$,

$$\mathbb{E}e^{-azy} = \mathbb{E}e^{-a\rho z^2 + a^2 s^2 z^2/2} = (1+\rho^2/s^2)^{-1/2}.$$

Chernoff's inequality bounds a wrong sign by e^{-wC_1}. A sign error costs $4\rho^2$ in squared prediction loss; the current Gaussian input is independent of the sign estimate. Taking $w = \lceil \log(1/\kappa)/C_1 \rceil$ proves the order. □

An exact axis is a supplied resource in this statement. A paid initialization with sufficiently small axis error adds a corresponding prediction error; finite precision is not free. The axis can also be learned without access to S_t: for $i \neq j$, $\mathbb{E}[y_t^2 x_{t,i} x_{t,j}] = 2\rho^2 V_i V_j$, while $\mathbb{E}[y_t^2 x_{t,i}^2] = \rho^2 + s^2 + 2\rho^2 V_i^2$. Cycling through coordinate pairs uses two legal reads per training step. These expectations are unchanged by sign flips and identify $VV^\top$ as training grows. Training time, stored moments and the sign window are resources. This law preserves an axis rather than diffusing $D-1$ independent tangent directions. Thus λ_1 suffices for the moment formula of Proposition 16, but cannot justify a universal diffusion tracking lower bound.

Conjecture 19 (Rate optimality of probe calibration)**.** *For the rank-one task with $2 \leq b < D$ noiseless linear reads per step, let U_t be spherical Brownian motion sampled at unit intervals, with generator $\sigma\Delta$ and $\kappa = 1 - e^{-(D-1)\sigma}$. Policies may retain their paid observed history and randomize, but receive no external coupling truth or additional drift reference. Their long-run average risk is at least $s^2 + c_0\rho^2 \min\{1, D\sqrt{\kappa(\rho^2+s^2)/(\rho^2(b-1))}\}$ for a constant $c_0 > 0$ independent of D, b, κ, in the local diffusion regime. Initialization is charged and does not supply future increments.*

The mechanism is that each off-frame read informs one background direction, and $D-1$ directions drift independently. For a single coordinate observed with probability p per step under a random walk with increment variance q, the posterior Cramér–Rao recursion gives a stationary error of order $\sqrt{q/(pF)}$, with F the Fisher information per observation; summing over $D-1$ coordinates with $p = (b-1)/D$, $q \simeq 2\rho^2\kappa/D$ and $F \simeq (\rho^2+s^2)^{-1}$ gives the stated scaling, which Proposition 16 attains with optimized α for small κ. The strongest objection is that the coordinates are not independent: an adaptive policy can probe the directions in which its uncertainty is largest, and the response read itself carries information about the magnitude of the misalignment (Corollary 13). These could improve the constant; the conjecture would fail if they changed the exponent of κ or the power of D.

Conjecture 20 (Sharp quantum counterpart of the frame price)**.** *For a specified quantum direction-estimation task and a compact group acting transitively on its directions, the minimum retained reference resource needed for a given capability admits a sharp task-dependent variational bound whose classical direction limit is Theorem 4.*

Proposition 9 supplies a general bounded-risk inequality, and Proposition 10 supplies an exact quantum task. What remains is sharpness and the classical limit, not the existence of any risk bound. Equal asymmetry need not give equal capability in different tasks, so the task family cannot be omitted. The noise-selected algebra of Consciousness I is a symmetry-fixed algebra; the present reference enlarges effective measurement permissions through a state, not by claiming that every fixed algebra becomes a larger abstract algebra.

10.3 Open problems

The task class is linear and Gaussian with a rank-one or fixed-rank target. Several competing frames and structured non-Gaussian inputs remain open. Paper III joins a binary self-frame to a world-changing loop: acting cannot identify parameters moved by its record-law symmetry, while explicit experiments in its stationary nondegenerate model identify

persistence, squared gain and flip reliability [4]. Symmetry invariance is therefore a necessary condition, not a general identification theorem. A continuous frame with matrix-valued loop gains, and a thermodynamic implementation of the reading and retention budgets, are next steps.

A Simulation protocol for Table 2

Parameters: $D = 8$, $b = 3$, $\rho = 1$, $s = 0.5$. The coupling starts uniform and moves by the rotation-invariant step $u \mapsto (u + \eta g_\perp)/|u + \eta g_\perp|$ with $g_\perp$ the projection of a standard Gaussian vector orthogonal to u, for $\eta \in \{0, 0.002, 0.005, 0.01, 0.02, 0.04, 0.06, 0.08, 0.12\}$; $\lambda_1 = \mathbb{E}(1 + \eta^2 \chi^2_{D-1})^{-1/2}$ is computed by quadrature. Each agent receives the same $n_0 = 64$ start-up probes, from which g_0 is the mean of $y_i x_i$. The frozen agent responds $g_0 \cdot x$. The frame-free agent reads channels 1–3; the fixed agent reads $g_0/|g_0|$ and two fixed random orthonormal directions orthogonal to it; both learn their three coefficients by the update of Section 8 with $p = 1$. The calibratable agent chooses two coordinate probes uniformly without replacement and reads those axes plus the normalized component of g orthogonal to them. It reconstructs $g \cdot x$ from these three orthonormal reads and updates only the probed coordinates. The recorded noiseless simulation used the algebraically equivalent nonorthogonal functionals; this equivalence need not hold for hardware with independent read noise. Learning rates and weights are set by minimizing the risks of Proposition 16 over $\alpha \in [10^{-4}, 0.999]$ (floor 0.002). Risks are evaluated exactly given the state, $s^2 + |\rho u - c|^2$ with c the agent's effective coefficient vector, and averaged over steps 8 000–23 999 and over 200 trajectories that share the world's random numbers across agents. The code is `td2_loop.py`; the thresholds of Table 3 are computed by `td2_threshold.py`, and the analytic checks of Theorems 2–14 by `td2_checks.py`.

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Keeping a Self-Model Calibrated: Drift Detection, the Price of Keeping Up, and Reliability in a Closed Loop

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Abstract

A learned instrument model helps an observer decide which parts of its own access are worth maintaining. We study this decision together with drift detection and paid calibration. Smooth even record laws delay linear-drift detection to order $\varepsilon^{-4/5}$, compared with $\varepsilon^{-2/3}$ for first-order probes. For scalar Gaussian transitions, no free background information and quadratic loss, the all-policy long-run cost lies between $C - \frac{1}{2}k_BTG\nu^2$ and C, with $C \propto \nu^{2/3}$. The bounds have the same slow-drift leading term; the upper bound uses a Gaussian prior and reads with controllable continuous precision, while integer reads inherit the relaxation lower bound. In an acting body, stationary correlations identify gain and actuator success, and labelled standards supply orientation. Adaptive reference timing has a strict advantage and an exact optimum in a finite-rate example. When the body's working state gates reading and processing, latched repair changes future access and the kernel between acquired records. Retained body history raises task value and, under positive gain and sign drift, can reverse optimal repair at equal sign confidence. Two posterior moments determine terminal access value; continuing enquiry can require a third moment and the joint belief. In a two-observer quadratic task, estimation-optimal and task-optimal authority weights differ. These constructions use one acquired model for task permission, future enquiry and repair under declared physical supplies.

Keywords: self-calibration, self-models, Fisher information, change detection, posterior Cramér–Rao bound, drift tracking, information thermodynamics, feedback control

1 Introduction

1.1 The question

Paper I bounds the usable value of a stored record by $k_BT\,I(X{:}M)$ and associates every feedback protocol with an implicit matching distribution [1]. This distribution need not be

represented inside the agent. When an agent does represent its own readout channel and decodes with a wrong model, it loses part of the record's value and can fall below an agent without memory. Paper II studies a retained self-frame that selects reads under a channel budget; in-frame records cannot locate its missing task direction [2]. Here such a represented model is checked against an external reference or the pointer value selected by the body's working state. The later body constructions show how acquired records form that model and change its next operations.

We ask how quickly drift is detected, what calibration costs, and how the retained model changes action when the body's function determines future access.

1.2 Position within the series

This paper completes the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other lines are HAFF and Q-RAIF.

The three T-DOME papers follow a route that the programme calls memory–self–introspection. An agent keeps records of its world (Paper I), organizes its access to them through a model of itself (Paper II), and checks that model against references while it acts (this paper). The links share their objects. The gain–bias channel whose calibrated loss Paper I computes is the instrument of Section B; the even residual law that Paper II derives from a budgeted frame is the detection model of Section 3; the conditional drift law cited in Paper I is proved in Section 5. Each link is established in a stated model.

1.3 Main results

The analytical chain runs from detection (Section 3) through all-policy Gaussian maintenance costs (Section 5) to tracking (Section 6) and implementation (Appendix F). The joint body model of Section 7 supplies the main acting example. Section 8 then makes working access conditional on the observer's body: actual history changes harvest, permission to act, and paid repair. Sign confidence is not a sufficient repair statistic. Terminal access has a two-moment value, while continued enquiry generally needs the joint belief. The older numerical checks and gain–bias bank models are retained in Appendices A–B; the finite-filter implementation certificates are in Appendix H.

1.4 Relation to internal selection

Paper G of HAFF gives a typed model of internal selection: a program register holds a choice, a fixed executor turns program states into couplings or readouts, records are outcomes of instruments, and a fixed rule rewrites the program from the records [8]. A finite system can select its own readouts and revise that selection. Within this target/program architecture, a selected target operation is not a complete implementation of the larger executor acting on both registers; a stored description or an additional selector is not excluded. The calibration loop uses this architecture. The self-model is the program, decoding and command rules are the executor, and calibration is the revision rule. History changes the program and the operations it selects. The executor and the labelled standard specify how those changes are used.

1.5 Related work

The Fisher information metric was introduced by Rao [13] and characterized by Čencov [14]; Amari developed the natural gradient and information geometry [15, 16]. The Bayesian Cramér–Rao (van Trees) inequality [17] and its recursive form for filtering, the posterior Cramér–Rao bound of Tichavský, Muravchik and Nehorai [18], are the estimation tools of Sections 4–5; the Kalman filter [19] attains the latter in the linear Gaussian case. The trade-off between noise and lag in adaptive estimation, and the persistent-excitation condition, are standard in adaptive control [21]. Thermodynamic length and minimum-dissipation protocols are due to Crooks and Sivak [22, 23], and Fisher information enters entropy production through thermodynamic uncertainty relations and information geometry [24, 25]. Second-order cybernetics, the control of the observing system itself, goes back to von Foerster, building on Ashby's cybernetics [26, 27]. The data-processing bound for hypothesis tests used in Section 3 is standard [28]. We do not claim that the telescoping inequality behind Theorem 11 is new to filtering theory; its use here is to settle the drift law of Paper I for all policies and to give it a thermodynamic reading.

2 Setting

2.1 Inputs from Papers I and II

Paper I supplies the record's matched work value, the posterior mismatch loss and its calibrated expansion [1]. Its gain–bias example has leading loss $k_B T(1 + g^2/(1 - b^2))/(4n)$ after n add-half reference reads per sign. Paper II supplies a retained foreground, an in-frame blind spot and, for its two-channel reader, the even information law [2]

$$F(\theta) = \frac{\rho^2 \sin^2\theta\,(\rho^2 + s^2 + \rho^2\cos^2\theta)}{(s^2 + \rho^2\sin^2\theta)^2}. \tag{1}$$

An off-frame read resolves its sign fibre. The Gaussian drift law cited by Paper I is proved here; its assumptions do not identify either reader with the bounded acting body below.

2.2 Fisher information and the Cramér–Rao bounds

Definition 1 (Fisher information)**.** Let $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ be a family of densities for which differentiation and integration may be interchanged. Its Fisher information matrix is

$$g_{ij}(\theta) := \mathbb{E}_\theta\big[\partial_i \log p(x \mid \theta)\, \partial_j \log p(x \mid \theta)\big] = -\mathbb{E}_\theta\big[\partial_i\partial_j \log p(x \mid \theta)\big]. \tag{2}$$

Where it is positive definite it is a Riemannian metric on Θ.

By Čencov's theorem, on finite probability simplices the Fisher metric is, up to a constant factor, the unique Riemannian metric invariant under statistically sufficient embeddings [14]. This makes it the canonical local measure of distinguishability; it does not make a Fisher-based detector physically mandatory, and it does not choose the agent's likelihood or objective. The metric is the Hessian of relative entropy,

$$D\big(p_\theta \,\|\, p_{\theta+d\theta}\big) = \tfrac{1}{2}\, g_{ij}(\theta)\, d\theta^i\, d\theta^j + O(|d\theta|^3), \tag{3}$$

and it bounds estimation: for an unbiased estimator from n independent observations, $\mathrm{Cov}(\hat{\theta}) \succeq g(\theta)^{-1}/n$ [13, 28]. For a correlated stream use the Fisher information of its joint likelihood. A decorrelation time alone does not justify replacing the sample count by T/τ_E.

2.3 Standing assumptions

Definition 2 (Standing assumptions)**.** Sections 3–6 assume the following. Section B states its own inputs.

(C1) **Self-model.** The agent's model of itself is a parameter ϑ in a finite-dimensional manifold Θ: a frame of Paper II, the gain and bias of an instrument, or a scalar mismatch coordinate. The family of records $p(e \mid \vartheta)$ that the agent uses to assess it is specified.

(C2) **Drift.** The parameter that best describes the instrument, $\vartheta^*(t)$, moves. Each result states its drift law: linear, spherical Brownian, a Gaussian random walk, or bounded speed.

(C3) **Regularity.** Fisher informations are finite and differentiation under the integral sign is allowed to the order used. Uniform lower bounds on information are assumed separately where they are needed.

3 Detecting drift from one's own records

A self-model goes stale when the instrument it describes changes. The agent's first question is whether it can notice this from the records it already gathers. Paper II shows that records read through a frame depend on the frame's misalignment only through an even function of it. This section works out what evenness implies.

3.1 Even models

Proposition 3 (Sign non-identifiability)**.** *If a record family satisfies $p(e \mid \theta) = p(e \mid -\theta)$ for all θ, then the joint law of any number of conditionally independent records is invariant under $\theta \mapsto -\theta$, and no statistic of the records distinguishes θ from $-\theta$.*

Proof. The joint likelihood is a product of factors that are each invariant under the sign change. □

Paper II derives this structure instead of assuming it: a frame that reads one of two channels sees an in-frame law that depends on the angle θ to the coupling only through $\cos\theta$, and the sign is the unobservable fibre of its blind spot [2]. Evenness has a second consequence, which governs how fast drift can be noticed at all.

Proposition 4 (Information and relative entropy of an even model)**.** *Let $p(e \mid \theta)$ be even in θ and smooth enough near $\theta = 0$ that the expansions below may be integrated term by term (this holds for the Gaussian families of Corollary 5). Write $\ell(\theta) = \log p(e \mid \theta)$ and*

$$\kappa := \mathbb{E}_0\big[\ell''(0)^2\big].$$

Assume $\kappa > 0$ for the sharp detection exponents below. Then the score vanishes at $\theta = 0$, so the Fisher information $\mathcal{I}_1(0)$ of one record is zero, and

$$\mathcal{I}_1(\theta) = \kappa\,\theta^2 + O(\theta^4), \qquad D\big(p_\theta \,\|\, p_0\big) = \frac{\kappa}{8}\,\theta^4 + O(\theta^6). \tag{4}$$

Proof. For each e the function ℓ is even, so $\ell'(0) = \ell'''(0) = 0$ and $\ell'(\theta) = \theta\,\ell''(0) + O(\theta^3)$. Since $p_\theta - p_0 = O(\theta^2)$, $\mathcal{I}_1(\theta) = \mathbb{E}_\theta[\ell'(\theta)^2] = \theta^2\mathbb{E}_0[\ell''(0)^2] + O(\theta^4)$. For the relative entropy write $p_\theta/p_0 = 1+\eta$ with $\eta = \frac{1}{2}\theta^2 H + O(\theta^4)$ and $H = p_0''/p_0 = \ell''(0) + \ell'(0)^2 = \ell''(0)$. Normalization gives $\mathbb{E}_0\eta = 0$, and $(1+\eta)\ln(1+\eta) = \eta + \frac{1}{2}\eta^2 + O(\eta^3)$, so $D(p_\theta\|p_0) = \frac{1}{2}\mathbb{E}_0\eta^2 + O(\theta^6) = \frac{1}{8}\theta^4\mathbb{E}_0 H^2 + O(\theta^6)$. □

Corollary 5 (Three record families)**.** *(a)* Gaussian mean model, $e \mid \theta \sim \mathcal{N}(a\theta^2, \sigma_e^2)$*:* $\mathcal{I}_1(\theta) = 4a^2\theta^2/\sigma_e^2$ *and* $D(p_\theta\|p_0) = a^2\theta^4/(2\sigma_e^2)$ *exactly, so* $\kappa = 4a^2/\sigma_e^2$*.*

(b) Paper II's in-frame record $(v\cdot x, y)$ *with* $D = 2$*:* $\mathcal{I}_1 = F(\theta)$ *of* (1)*, and*

$$D(p_\theta\|p_0) = \frac{\rho^2(1-\cos\theta)}{s^2} - \frac{1}{2}\ln\Big(1 + \frac{\rho^2\sin^2\theta}{s^2}\Big), \qquad \kappa = \frac{\rho^2(2\rho^2+s^2)}{s^4}. \tag{5}$$

(c) An off-frame probe record $(v_\perp\cdot x, y)$*:* $D(p_\theta\|p_0) = -\frac{1}{2}\ln\big(1 - \rho^2\sin^2\theta/(\rho^2+s^2)\big)$*, and its Fisher information*

$$F_\perp(\theta) = \frac{\rho^2\cos^2\theta\,(\rho^2\sin^2\theta + \rho^2 + s^2)}{(\rho^2 + s^2 - \rho^2\sin^2\theta)^2} \tag{6}$$

is positive at calibration, $F_\perp(0) = \rho^2/(\rho^2+s^2)$*.*

Proof. (a) is the normal location family with mean $a\theta^2$. For (b) and (c) the record is a zero-mean Gaussian pair with covariance $\Sigma_\theta = \left(\begin{smallmatrix} 1 & c(\theta) \\ c(\theta) & V \end{smallmatrix}\right)$, $V = \rho^2 + s^2$, where $c = \rho\cos\theta$ in (b) and $c = \rho\sin\theta$ in (c). Then $D(p_\theta\|p_0) = \frac{1}{2}[\mathrm{tr}\big(\Sigma_0^{-1}\Sigma_\theta\big) - 2 - \ln(\det\Sigma_\theta/\det\Sigma_0)]$ gives the two relative entropies, and the Fisher information of a Gaussian family with parameter-dependent covariance, $\frac{1}{2}\,\mathrm{tr}(\Sigma^{-1}\Sigma'\Sigma^{-1}\Sigma') = c'^2(c^2+V)/(V-c^2)^2$, gives (1) and (6). Expanding (5) to fourth order gives $\kappa/8 = \rho^2(2\rho^2+s^2)/(8s^4)$, in agreement with $F(\theta)/\theta^2 \to \rho^2(2\rho^2+s^2)/s^4$. □

Model (a) is a convenient assumption; model (b) is what a budgeted frame actually produces. Proposition 4 shows that the quadratic onset of information does not depend on the choice: evenness gives the expansion; a nonzero κ gives its quadratic onset. The constant κ is the only feature of the model that enters the leading order. For records at rate r under linear drift $\theta(t) = \theta_0 + \varepsilon t$, the accumulated local information $J_T := \sum_{t_m \le T}\mathcal{I}_1(\theta(t_m)) \simeq r\int_0^T \mathcal{I}_1(\theta_0 + \varepsilon t)\,dt$ is

$$J_T = \frac{\kappa r\varepsilon^2 T^3}{3} + O(\varepsilon^4 T^5) \quad (\theta_0 = 0), \qquad J_T = \frac{4a^2 r}{3\sigma_e^2\varepsilon}\big[(\theta_0+\varepsilon T)^3 - \theta_0^3\big] \quad \text{in model (a).} \tag{7}$$

It enters the estimation bound of Section 4. Detection is governed by a different quantity, the accumulated relative entropy.

3.2 How long detection takes

Fix a horizon T and let records arrive at rate r, conditionally independent given the mismatch path $\theta_m = \theta(m/r)$. A test is a function of the records up to time T with the values "calibrated" and "drifted". It has errors (α, β) if it says "drifted" with probability at most α when $\theta_m = 0$ for all m, and "calibrated" with probability at most β under the drifted law; for a random drift path the drifted law is the mixture over paths. Write $d(p\|q) = p \ln \frac{p}{q} + (1-p) \ln \frac{1-p}{1-q}$.

Proposition 6 (Detection time)**.** *Every test with errors (α, β), $\alpha + \beta < 1$, satisfies*

$$\sum_{m \le rT} \mathbb{E}\, D\big(p_{\theta_m} \,\|\, p_0\big) \;\ge\; d(1-\beta \,\|\, \alpha), \tag{8}$$

with the expectation over the drift path if it is random. Consequently, as the drift rate tends to zero:

(a) for an even model with $D(p_\theta\|p_0) \le \kappa\theta^4/8 + C\theta^6$ for all θ and linear drift $\theta(t) = \varepsilon t$, $T \ge \big(40\, d/(\kappa r \varepsilon^4)\big)^{1/5}(1 + o(1))$;

(b) if probe positions are fixed in advance with counting function $N_\perp(t) = prt + o(t)$ for $p > 0$, their record law has $D(p_\theta\|p_0) \le \frac{1}{2}F_1\theta^2 + C\theta^4$ with $F_1 > 0$ its Fisher information at calibration, as off-frame probes do, and the others are as in (a), $T \ge \big(6\, d/(pF_1 r\varepsilon^2)\big)^{1/3}(1 + o(1))$;

(c) for Paper II's reader with $D = 2$ under spherical Brownian drift with generator $\sigma\Delta$, $T \ge \big(2\, d/(\kappa r \sigma^2)\big)^{1/3}(1 + o(1))$,

where $d = d(1-\beta\|\alpha)$.

Proof. Let P_0 and P_1 be the laws of the records up to T under the two hypotheses. The test's decision is a function of the records, so by the data-processing inequality $D(P_1\|P_0)$ is at least the relative entropy between the two laws of the decision, which are Bernoulli with parameters $p_1 \ge 1-\beta$ and $p_0 \le \alpha$; since $d(p_1\|p_0)$ increases in p_1 and decreases in p_0 for $p_1 > p_0$, it is at least $d(1-\beta\|\alpha)$ [28]. For a fixed path the chain rule gives $D(P_1\|P_0) = \sum_m D(p_{\theta_m}\|p_0)$; for a random path P_1 is a mixture and joint convexity gives $D(P_1\|P_0) \le \mathbb{E}\sum_m D(p_{\theta_m}\|p_0)$. This proves (8). For (a), summing the hypothesis over $m \le rT$ gives $\kappa r \varepsilon^4 T^5/40\,(1 + O(\varepsilon^2 T^2))$, and the solution $T \propto \varepsilon^{-4/5}$ has $\varepsilon T \to 0$. For (b), summation by parts gives $\sum_{t_m \le T,\, m \text{ probe}} t_m^2 = prT^3/3 + o(T^3)$. The probe records therefore give $pF_1 r\varepsilon^2 T^3/6\,(1 + o(1) + O(\varepsilon^2 T^2))$ and the in-frame records only $O(\varepsilon^4 T^5)$, which vanishes at $T \propto \varepsilon^{-2/3}$. For (c), the generator $\sigma\Delta$ on the circle makes the unwrapped angle a Brownian motion with variance $2\sigma t$, and the relative entropy (5) is periodic and bounded, so $D(p_\theta\|p_0) \le \kappa\theta^4/8 + C\theta^6$ for all real θ. With $\mathbb{E}\theta_t^4 = 12\sigma^2 t^2$ and $\mathbb{E}\theta_t^6 = 120\sigma^3 t^3$ the sum is $\kappa r \sigma^2 T^3/2\,(1 + O(\sigma T))$, and $\sigma T \to 0$ at $T \propto \sigma^{-2/3}$. The families of Corollary 5 satisfy the hypotheses of (a) and (b): in (a) with $C = 0$, in (b) and (c) because the relative entropies are bounded and smooth. □

The bound holds for any test, whatever statistic it uses. For a known drift it is also of the right order: the log-likelihood ratio has mean equal to the accumulated relative entropy and, for small drift, variance about twice that mean, so the likelihood-ratio test succeeds at a constant multiple of the bound. In the example of Section A it reaches equal errors of 0.05 at about 1.3 times the bound. Evenness therefore costs time. A frame that sees only its own channel needs a time of order $\varepsilon^{-4/5}$ to notice a drift of rate ε; a stream that includes probes outside the frame needs a time of order $\varepsilon^{-2/3}$, which is shorter for slow enough drift at any fixed fraction p of probes.

Corollary 7 (Detection before failure)**.** *For Paper II's reader with $D = 2$ under spherical Brownian drift, a frozen self-model becomes worse than no memory at $t^* = \ln 2/\sigma$. Let the agent compare the mean S_T of the squared in-frame residuals $e_m = y_m - \rho\, v \cdot x_m$ over a window of length T with the threshold that gives type I error α under calibration. For every $\beta > 0$ there is a constant K, depending on α, β, ρ, s and r, such that at $T = K\sigma^{-2/3}$ the type II error is at most β for all small enough σ. Together with Proposition 6(c), the detection time is of order $\sigma^{-2/3}$, and its ratio to t^* tends to zero: slow drift can be detected from in-frame records long before the frozen model fails.*

Proof. Paper II's frozen-model curve with $D = 2$ gives t^*. Given the path, $e_m \sim \mathcal{N}(0, s^2 + 2\rho^2(1 - \cos\theta_m))$ independently, and under calibration nS_T/s^2 is χ^2_n with $n = rT$, so the threshold is $s^2(1 + z_\alpha\sqrt{2/n}\,(1 + o(1)))$. Write $\theta_{uT} = \sqrt{2\sigma T}\,B_u$ with B a standard Brownian motion. Then the conditional mean of $S_T - s^2$ is $2\sigma T\rho^2 Z_n(1 + O(\sigma T))$ with $Z_n \to Z = \int_0^1 B_u^2\,du$, and its conditional fluctuation is $s^2\sqrt{2/n}\,(1 + o(1))$ times a standard normal variable. At $T = K\sigma^{-2/3}$ the ratio of the shift to the null scale is $\sqrt{2}\,\rho^2\sqrt{r}\,K^{3/2}Z/s^2$, independent of σ, so the type II error tends to $P\big(N + \sqrt{2}\rho^2\sqrt{r}K^{3/2}Z/s^2 \le z_\alpha\big)$ with N standard normal and independent of Z. Since $Z > 0$ almost surely, this tends to zero as $K \to \infty$. □

For a general predetermined probe schedule, let χ_m indicate a probe at $t_m = m/r$. Under the linear drift of Proposition 6(b), the leading term of the probe entropy upper bound is

$$\frac{F_1\varepsilon^2}{2}\sum_{m\le rT}\chi_m t_m^2,$$

with remainder $O(\varepsilon^4 rT^5)$ under its global entropy bounds. The prefix-density assumption makes this sum $prT^3/3 + o(T^3)$. If only the horizon count $\sum\chi_m = prT + o(T)$ is specified, placing all probes first gives $p^3rT^3/3$, while placing them last gives $[1 - (1-p)^3]rT^3/3$, to leading order. These are the sharp extremes because the weights t_m^2 increase. A probe fraction alone therefore does not determine the detection constant. The placement is a genuine use of the access budget, even though all three schedules have the same $\varepsilon^{-2/3}$ scaling of the necessary detection time at fixed $p > 0$.

Detection is not correction. An even record can signal that the frame has moved but not in which direction (Proposition 3). One off-frame read or one reference read supplies the sign and a first-order signal at the same time. This is the division of labour in Paper II's calibratable frame and in Section B: most reads serve the task through the frame, and a small budget of reads outside it keeps the frame correct.

4 Estimating the mismatch with a prior self-model

For an agent that retains a model of its instrument, earlier calibration records supply a prior π on the drift parameter $\delta\vartheta$. Its Fisher information measures how firmly that estimate is held. Paper II's foreground uses such a retained model; Paper I's implicit protocol distribution alone does not require one.

Definition 8 (Rigidity of a self-model)**.** The rigidity of a prior self-model π is its Fisher information

$$\mathcal{I}_{\text{self}} := \int \big(\partial_{\delta\vartheta} \log \pi(\delta\vartheta)\big)^2 \, \pi(\delta\vartheta)\, d(\delta\vartheta). \tag{9}$$

Theorem 9 (Van Trees bound with a prior self-model)**.** *Let $\delta\vartheta$ lie in a compact interval $[a, b]$, let the prior π be C^1 with finite rigidity and vanish at a and b, and let the record consist of n_{eff} conditionally independent samples (or a joint experiment with that exact information), each with Fisher information $\mathcal{I}_1(\delta\vartheta)$ under a regular likelihood. Then every estimator $\delta\hat{\vartheta}$ based on the record satisfies*

$$\mathbb{E}\big[(\delta\hat{\vartheta} - \delta\vartheta)^2\big] \;\geq\; \frac{1}{n_{\text{eff}}\, \mathbb{E}_\pi[\mathcal{I}_1(\delta\vartheta)] + \mathcal{I}_{\text{self}}}. \tag{10}$$

Proof. The van Trees inequality [17] bounds the Bayesian mean squared error of any estimator by the inverse of the prior-averaged Fisher information of the record plus the Fisher information of the prior; the boundary conditions remove the boundary term of its integration by parts. Conditional independence makes the record's information $n_{\text{eff}}\mathcal{I}_1$. □

The prior helps when it is well placed and hinders when it is not. In the conjugate Gaussian location model, with constant per-sample precision $\bar{\mathcal{I}}_1$, the posterior mean is the precision-weighted average

$$\hat{\vartheta} = \frac{n_{\text{eff}}\bar{\mathcal{I}}_1\, \hat{\vartheta}_{\text{MLE}} + \mathcal{I}_{\text{self}}\, \vartheta_0}{n_{\text{eff}}\bar{\mathcal{I}}_1 + \mathcal{I}_{\text{self}}} \tag{11}$$

of the maximum-likelihood estimate and the prior centre ϑ_0. A rigid self-model centred on the wrong value pulls every estimate toward it and resists recalibration; a soft one follows the noise. Without a model of the drift one can only say that an optimal rigidity exists when the loss is lower semicontinuous on a compact set of rigidities. Under a random-walk drift the optimum is exact (Corollary 15).

5 The price of keeping up

5.1 The drift law

Let a scalar self-parameter, for instance the gain of the agent's instrument, drift as a Gaussian random walk $\theta_k = \theta_{k-1} + \omega_k$, with independent $\omega_k \sim \mathcal{N}(0, \nu^2)$ per use and θ_0 drawn from a prior with finite Fisher information Q_0 ($Q_0 = 1/P_0$ for a Gaussian prior of variance P_0). In use k the agent makes $\rho_k \geq 0$ reference reads, each costing w and carrying Fisher information at most J about θ_k under the pointwise or Bayesian conditional-information

budget of Lemma 10, together with any unlabelled reads, which carry expected information at most J_0 per use at no extra cost. It then forms an estimate $\hat{\theta}_k$ from all its records so far and incurs a loss. The number ρ_k may be a real reading intensity, and it may depend on the records of earlier uses. Define $\bar{\rho}_k = \mathbb{E}\rho_k$, $\psi(Q) = (Q^{-1} + \nu^2)^{-1}$ and

$$Q_k = \psi(Q_{k-1}) + J_0 + J\bar{\rho}_k, \qquad P_k = Q_k^{-1} \qquad (k \geq 1). \tag{12}$$

Lemma 10 (Predictable design and stopped reading scores)**.** *Fix a use k with θ_k unchanged during its reads. Let $\mathcal{H}_{j-1}$ contain the preceding complete acquired history, selected designs and private coins, including earlier reads of this use, and set $\mathcal{G}_{j-1} = \mathcal{H}_{j-1} \vee \sigma(\theta_0, \dots, \theta_k)$. Assume a parameter-independent, nonanticipating design rule and regular dominated likelihoods. The next reading score s_j has conditional mean zero given $\mathcal{G}_{j-1}$, and put $I_j = \mathbb{E}[s_j^2 \mid \mathcal{G}_{j-1}]$. Either the pointwise bound $I_j \leq J$, or the weaker Bayesian bound $\mathbb{E}[I_j \mid \mathcal{H}_{j-1}] \leq J$ under the actual joint prior and experiment, suffices. For a stopping count ρ_k with $\mathbb{E}\rho_k < \infty$ and $\{\rho_k \geq j\} \in \mathcal{H}_{j-1}$, the Bayesian expected reading information is at most $J\mathbb{E}\rho_k$. A separately declared free record channel contributes its joint conditional-information budget J_0.*

Proof. The joint likelihood factors as $p(\theta_0)\prod_k p(\theta_k \mid \theta_{k-1})\prod_j p(z_{kj} \mid \theta_k, d_{kj})$ times design probabilities. At the realized record the selected designs are fixed under differentiation, so design contributes no parameter score. The scores are martingale differences in the enlarged filtration $\mathcal{G}_j$, not necessarily in $\mathcal{H}_j$. Writing $A_j = \mathbf{1}\{\rho_k \geq j\}$, orthogonality and the tower property give

$$\mathbb{E}S^2_{\rho_k \wedge m} = \sum_{j \leq m} \mathbb{E}[A_j I_j] = \sum_{j \leq m} \mathbb{E}[A_j \mathbb{E}(I_j \mid \mathcal{H}_{j-1})] \leq J\mathbb{E}(\rho_k \wedge m).$$

The same argument bounds the squared L^2 tail by $J\mathbb{E}(\rho_k - m)_+ \to 0$. The transition information blocks of the posterior Cramér–Rao recursion are unchanged by this factorization. The conditional-average alternative proves a Bayesian information bound, not a separate bound at every θ_k. Parameter-dependent design or informative status records need their own scores; an unconditional per-read average alone does not justify stopping on the record. □

Theorem 11 (Drift law)**.** *Let $\ell : (0, \infty) \to [0, \infty)$ be a continuous function such that the expected loss in use k is at least $\ell(P_k)$ for every estimator. Then every reading schedule and every estimator satisfy*

$$\liminf_{K \to \infty} \frac{1}{K} \sum_{k=1}^{K} \big(w\bar{\rho}_k + \mathbb{E}[\mathrm{loss}_k]\big) \;\geq\; \inf_{0 < P \leq P_{\mathrm{free}}} h(P), \qquad h(P) = \frac{w}{J}\Big(\frac{\nu^2}{P(P+\nu^2)} - J_0\Big) + \ell(P), \tag{13}$$

where $P_{\mathrm{free}} = 1/Q_{\mathrm{free}}$ and Q_{free} is the positive fixed point of $Q = \psi(Q) + J_0$ ($P_{\mathrm{free}} = \infty$ if $J_0 = 0$). For the quadratic loss $\mathbb{E}(\hat{\theta}_k - \theta_k)^2$ times a constant c the hypothesis holds with $\ell(P) = cP$. If the reads are Gaussian, the loss of the posterior-optimal decoder is exactly $\ell(P_k)$, and arbitrary nonnegative observation precision is an allowed control, and the initial prior is Gaussian, a constant precision attaining a minimizing P^* attains the bound. When reads must be counted in whole units, the same expression is a lower bound from a continuous-precision relaxation; equality requires an admissible schedule.

Proof. Information bound. For a random walk with increment variance ν^2 the posterior Cramér–Rao recursion of Tichavský, Muravchik and Nehorai [18] has $D_{11} = -D_{12} = -D_{21} = \nu^{-2}$ and $D_{22} = \nu^{-2} + I_k$, where I_k is the expected Fisher information of the records of use k about θ_k, and it reads $\mathcal{J}_k = D_{22} - D_{21}(\mathcal{J}_{k-1} + D_{11})^{-1}D_{12} = \psi(\mathcal{J}_{k-1}) + I_k$, with $\mathbb{E}(\hat{\theta}_k - \theta_k)^2 \geq \mathcal{J}_k^{-1}$. Lemma 10 supplies $I_k \leq J_0 + J\bar{\rho}_k$ for predictable designs, including finite-mean within-use stopping counts. Whole-use Gaussian precision has its declared conditional information, without a discrete stopping argument. Since ψ is increasing, induction gives $\mathcal{J}_k \leq Q_k$, so the mean squared error is at least P_k, and the quadratic loss is at least cP_k.

Telescoping. By (12), $J\bar{\rho}_k = Q_k - \psi(Q_{k-1}) - J_0$. Summing,

$$\sum_{k=1}^{K}\big(Q_k - \psi(Q_{k-1})\big) = \sum_{k=1}^{K-1}\big(Q_k - \psi(Q_k)\big) + Q_K - \psi(Q_0) \;\geq\; \sum_{k=1}^{K-1}\big(Q_k - \psi(Q_k)\big) - \psi(Q_0),$$

and $Q - \psi(Q) = \nu^2/(P(P+\nu^2))$ for $P = 1/Q$. Hence

$$\sum_{k=1}^{K}\big(w\bar{\rho}_k + \mathbb{E}[\mathrm{loss}_k]\big) \;\geq\; \sum_{k=1}^{K-1} h(P_k) + \ell(P_K) - \frac{w}{J}\big(\psi(Q_0) + J_0\big) \;\geq\; \sum_{k=1}^{K-1} h(P_k) - \frac{w}{J}\big(\psi(Q_0) + J_0\big).$$

Each term of the sum is paired with a single P_k: no convexity is needed.

Range. Since $\bar{\rho}_k \geq 0$, $Q_k \geq f(Q_{k-1})$ with $f(Q) = \psi(Q) + J_0$ increasing, so $Q_k \geq f^k(0)$, and $f^k(0)$ increases to Q_{free}. For every $\eta > 0$, $P_k \leq (1+\eta)P_{\text{free}}$ for all large k. Dividing by K, letting $K \to \infty$ and then $\eta \to 0$, which is allowed because h is continuous, gives (13).

Attainment. With a Gaussian prior and Gaussian measurements, the allowed constant precision makes Q_k converge to $1/P^*$. A posterior-optimal decoder with loss exactly $\ell(P_k)$ then has average cost converging to $h(P^*)$. □

Corollary 12 (Paper I's drift law)**.** *Take Paper I's quadratic loss $\frac{1}{2}k_BTG(\hat{\theta} - \theta)^2$ per use and $J_0 = 0$, and let*

$$C = 3 \cdot 4^{-2/3}\, w^{1/3}\, (k_BTG\nu)^{2/3}\, J^{-1/3}. \tag{14}$$

Every schedule and estimator has long-run average reference cost plus loss at least $C - \frac{1}{2}k_BTG\nu^2$ per use. With a Gaussian prior and reads of controllable real precision, the best stationary policy achieves at most C, so the least achievable average L^* in that class satisfies $C - \frac{1}{2}k_BTG\nu^2 \leq L^* \leq C$. The optimal reference rate is $\rho^* = \big(k_BTG\nu/(4w\sqrt{J})\big)^{2/3}(1 + o(1))$ as $\nu \to 0$, and $C = 3w\rho^*$ to leading order. The optimal estimator is a persistent memory: the steady-state Kalman filter updates its estimate with the constant weight $\alpha^* = P^*J\rho^* = \nu\sqrt{\rho^* J}\,(1 + o(1))$, which is the memory of Proposition 43 with that weight.

Proof. With $c = \frac{1}{2}k_BTG$ and $Q = P + \nu^2$, the inequalities $\nu^2/Q^2 \leq \nu^2/(P(P+\nu^2)) \leq \nu^2/P^2$ give $h(P) \geq (w/J)\nu^2/Q^2 + cQ - c\nu^2$ and $h(P) \leq (w/J)\nu^2/P^2 + cP$. Both $\min_{Q>0}[(w/J)\nu^2/Q^2 + cQ]$ and $\min_{P>0}[(w/J)\nu^2/P^2 + cP]$ equal $3 \cdot 2^{-2/3}c^{2/3}(w\nu^2/J)^{1/3} = C$. Theorem 11 gives the lower bound and, with Gaussian reads, the upper bound. At the stationary point $\nu^2/(P^*(P^*+\nu^2)) = J\rho^*$ and $P^* = \nu/\sqrt{\rho^* J}\,(1 + o(1))$; minimizing $w\rho + c\nu/\sqrt{\rho J}$ gives ρ^* and the value $3w\rho^*$. The stationary Kalman gain is the posterior variance times the precision of the use's reads, $P^*J\rho^*$. □

Paper I conjectured this law with an unspecified universal constant, the rate ρ^* and the value $3w\rho^*$ [1]. Corollary 12 proves it, with the constant $3 \cdot 4^{-2/3} \approx 1.191$, for every schedule including time-varying and adaptive ones in its scalar Gaussian-transition model, for any read family whose Fisher information is bounded by J, and for the quadratic loss defined by Paper I's metric G. For Paper I's exact divergence loss the statement holds to leading order in the calibration error; a global version holds with G replaced by any $G' > 0$ such that the loss is at least $\frac{1}{2}k_B T G'(\hat{\theta}-\theta)^2$ on the explored range, under a uniform quadratic separation of posteriors on that range. Injectivity alone does not guarantee a positive quadratic minorant.

Corollary 13 (The relative-entropy floor of Consciousness III for all policies)**.** *In the Gaussian self-model of Consciousness III [3], let the initial prior be Gaussian and the self-state $\mathcal{N}(\theta_t, v)$ drift with increment variance q, with readings $\theta_t + \mathcal{N}(0, r)$ costing w each. Every predictable observation-precision schedule and decoder has long-run average relative-entropy loss plus reading cost at least*

$$\min_{P>0}\Big[\frac{wrq}{P(P+q)} + \tfrac{1}{2}\ln\Big(1+\frac{P}{v}\Big)\Big],$$

For a Gaussian initial prior and controllable real observation precision at linear cost, the constant-rate Kalman schedule attains it. Integer reading counts give the corresponding relaxation bound. The stationary constant-rate optimum computed in Consciousness III is therefore optimal among all policies.

Proof. Let $\mathcal{H}_k$ contain the full readings, chosen precisions, and private randomness through use k. A precision chosen from $\mathcal{H}_{k-1}$ does not contribute an extra parameter-dependent likelihood factor. Gaussian conjugacy therefore gives, path by path,

$$P_k^{-1} = (P_{k-1}+q)^{-1} + \rho_k/r, \qquad \theta_k \mid \mathcal{H}_k \sim \mathcal{N}(m_k, P_k).$$

The posterior predictive law is $\mathcal{N}(m_k, v + P_k)$, with conditional optimal relative-entropy loss $\ell(P_k) = \frac{1}{2}\ln(1 + P_k/v)$. Put $Q_k = P_k^{-1}$ and $\psi(Q) = (Q^{-1}+q)^{-1}$. For every realized observation schedule,

$$\sum_{k=1}^{K}[w\rho_k + \ell(P_k)] = \sum_{k=1}^{K-1}\left[\frac{wrq}{P_k(P_k+q)} + \ell(P_k)\right] + \ell(P_K) + wr[Q_K - \psi(Q_0)].$$

The terminal terms are at least $-wr\psi(Q_0)$. Take expectations and divide by K. A constant allowed precision at the minimizer attains the resulting bound. □

5.2 What the floor means

Remark 14 (Local thermal comparison)**.** At a fixed interior gain g, the Bernoulli reference information and Paper I's local loss metric are $J = G = 1/(1-g^2)$. Substitution into the exact Gaussian comparison yields

$$3 \cdot 4^{-2/3}(\ln 2)^{1/3}\frac{k_B T \nu^{2/3}}{(1-g^2)^{1/3}} - \frac{k_B T \nu^2}{2(1-g^2)}. \tag{15}$$

This expression gives the local quadratic comparison at thermally prepared reference cost $k_B T \ln 2$. A Gaussian random walk leaves the physical gain interval almost surely, so this substitution alone does not prove an infinite-time lower bound for the Bernoulli instrument. A bounded-drift theorem or a uniform finite-window limit is needed for that conclusion. The Gaussian theorem itself retains its all-policy strength. An alternative coordinate is $p = \sin^2 \phi$, whose Bernoulli Fisher information is 4 where $0 < p < 1$. This alone cannot supply a global quadratic KL minorant on the circle: ϕ and $\pi - \phi$ give identical laws while their circle distance can be nonzero. A bounded-instrument extension must use an identifiable quotient or chart, establish its loss separation and treat its actual transition law; the Gaussian information recursion is not coordinate-invariant under that change.

Paper I's amortization rule says that the overhead of a static calibration falls as $U^{-1/2}$ with the number U of uses, and so can be made as small as desired. The exact Gaussian comparison removes that freedom: with no free information its price per use has order $k_B T \nu^{2/3}$. For $g = 0.55$ the local Gaussian comparison value is $0.012\, k_B T$ per use at $\nu = 10^{-3}$, where the calibration error at the optimum is about 0.1 in g and the leading-order statement is accurate to a few per cent; the quadratic-loss value at $\nu = 10^{-2}$ is $0.055\, k_B T$, but there the error is about 0.23 and the divergence loss is about 7% below its quadratic approximation (Section A). A record read through the same channel is worth $0.160\, k_B T$ per use in Paper I's example. For a bounded physical gain, a uniform information bound and a quadratic loss minorant must be paired with the actual transition law; clipping the Gaussian walk changes its information recursion. With the quadratic loss the two would be equal near $\nu = 0.05$, where calibration errors are no longer small, so that crossing is only indicative. A physical persistence calculation must combine the actual record reset and the cost of its admissible calibration policy; Section G.1 supplies such a bounded example.

Corollary 15 (Optimal rigidity under drift)**.** *In the optimal stationary policy of Corollary 12, the information that the self-model carries into each use from earlier uses is* $\mathcal{I}^*_{\text{self}} = 1/(P^* + \nu^2)$*, the information gathered in the use is* $J\rho^*$*, and their ratio is* $(1 - \alpha^*)/\alpha^* = (\nu\sqrt{\rho^* J})^{-1}(1 + o(1))$.

Proof. At stationarity $Q^* = \psi(Q^*) + J\rho^*$ with $\psi(Q^*) = 1/(P^* + \nu^2)$, and the Kalman weight is $\alpha^* = J\rho^*/Q^*$. □

A more rigid self-model would ignore real drift; a softer one would spend references on noise. The trade-off that Section 4 could only assert exists has, under drift, an exact optimum set by the drift speed and the price of a reference.

Remark 16 (What the drift law does not cover)**.** (i) *Free information.* If unlabelled reads inform the parameter ($J_0 > 0$), the floor is lower, and when $\nu/\sqrt{J_0}$ is small enough the agent can track without references at an error of about $\nu/\sqrt{J_0}$; the $\nu^{2/3}$ law then no longer describes the optimum. In Paper I's setting with prior 1/2, unlabelled reads inform the bias and not the gain. In a world that the agent changes, an unlabelled read has mean $gx + b$ with the world state x unknown, so world reads alone do not separate the instrument from the world, which is Paper II's blind spot in another form. Section B uses references for this reason; Section 7 shows that a closed loop supplies free information about some invariant self-parameters in its stationary nondegenerate model, while the symmetry obstructs absolute sign identification.

(ii) *Linear drift.* Under the linear drift of Proposition 43, a memory with a constant weight has mean squared error $\nu^2(1-\alpha)^2/\alpha^2 + \alpha\sigma^2/(2-\alpha)$; optimizing the weight and the number of reads gives a cost proportional to $\nu^{2/5}$, worse than $\nu^{2/3}$. A memory that also estimates the drift rate drives the error to zero under exactly linear drift, so linear drift has no positive floor. (iii) *Vector parameters.* For a self-model with several drifting parameters the same telescoping, applied to traces, gives a lower bound by relaxation; whether it is attained when one reference informs several parameters at once is open.

Conjecture 17 (Smoother drift)**.** *Let the self-parameter be an m-fold integrated random walk, so that its m-th differences are independent $\mathcal{N}(0,\nu^2)$ variables, and let the other inputs be as in Theorem 11 with quadratic loss. Then every policy pays at least $c_m\, w^{(2m-1)/(4m-1)}\, \nu^{2/(4m-1)}$ per use, where $c_m > 0$ depends only on m, J and the loss constant, and the stationary Kalman filter with a constant reference rate attains this order.*

The stationary part is a steady-state computation: for an m-fold integrator observed with noise variance R the stationary position error scales as $\nu^{1/m} R^{(2m-1)/(2m)}$, and optimizing $R = 1/(\rho J)$ gives the exponent $2/(4m-1)$, which is $2/3$, $2/7$ and $2/11$ for $m = 1, 2, 3$; a numerical solution of the Riccati equation reproduces these slopes. The open part is the lower bound for time-varying and adaptive schedules. The telescoping of Theorem 11 does not extend directly, because the information recursion is matrix-valued and the reads inform only the first component. The conjecture would fail if a schedule that alternates bursts of reads with silence could exploit the smoothness to beat the stationary exponent.

6 The calibration loop

6.1 The natural-gradient update

A loop that keeps the self-model current can move ϑ along the steepest descent of a frame loss in the geometry of its own records,

$$\dot{\vartheta} = -\eta\, g(\vartheta)^{-1} \nabla_\vartheta L(\vartheta, t), \tag{16}$$

where $\eta > 0$ is the adaptation rate, g is the Fisher metric of the records the loop uses, and $L(\vartheta, t)$ is the risk of acting at time t with self-model ϑ. For Paper II's rank-one frame v, $L(v,t) = s^2 + \rho^2\big(1 - (v \cdot u_t)^2\big)$, which is smallest for the frame aligned with u_t. The update is invariant under reparametrization of Θ; this matters because no coordinate system on a space of frames or instrument models is preferred. It is a definite algorithm. An implementation must supply the likelihood, evaluate or approximate its score and realize the update law; accumulating an unlabelled second moment of residuals is not in general a Fisher computation.

The metric must be nondegenerate where the loop operates, and records read through the frame fail exactly there.

Proposition 18 (The in-frame metric degenerates at calibration)**.** *For Paper II's two-channel reader with the frame at angle θ to the coupling, the metric of in-frame records*

is $F(\theta)$ of (1), and the natural gradient of $L = s^2 + \rho^2 \sin^2\theta$ in this metric is

$$\frac{\partial_\theta L}{F(\theta)} = \frac{\rho^2 \sin 2\theta}{F(\theta)} = \frac{2\rho^2}{\kappa\,\theta}\big(1 + O(\theta^2)\big),$$

which diverges at $\theta = 0$. If a fraction $p \in (0, 1]$ of the records are off-frame probes, the metric $(1-p)F(\theta) + pF_\perp(\theta)$ is at least $p\rho^2 \cos^2\theta/(\rho^2 + s^2)$, and for $|\theta| < \pi/2$ its absolute natural gradient is bounded by $2(\rho^2 + s^2)|\tan\theta|/p$.

Proof. The first statement follows from $F(\theta) = \kappa\theta^2 + O(\theta^4)$ (Corollary 5). In (6) the factor $\rho^2 \sin^2\theta + \rho^2 + s^2$ is at least the denominator's base $\rho^2 + s^2 - \rho^2 \sin^2\theta$, so $F_\perp(\theta) \geq \rho^2 \cos^2\theta/(\rho^2 + s^2 - \rho^2 \sin^2\theta) \geq \rho^2 \cos^2\theta/(\rho^2 + s^2)$; divide $\rho^2 \sin 2\theta$ by p times this. □

A loop that uses only in-frame records therefore takes ever larger natural-gradient steps as it approaches calibration, and by Proposition 3 its records do not even fix the sign of the step. This is the geometric form of Paper II's blind spot, and the remedy is the same: a fixed fraction of reads outside the frame makes the metric positive at calibration, with value $pF_\perp(0) = p\rho^2/(\rho^2 + s^2)$.

The local strong-convexity tracking theorem and proof are retained in Appendix E.1. Its assumptions require a nondegenerate metric and a smooth geodesic neighborhood; they do not repair the in-frame degeneracy. Persistent excitation is a separate input.

Definition 19 (Persistent excitation)**.** The records are persistently exciting with constant $\alpha_{\mathrm{PE}} > 0$ if for some window T

$$\frac{1}{T}\int_t^{t+T} g(\vartheta(s))\,ds \;\succeq\; \alpha_{\mathrm{PE}}\, I \qquad \text{for all } t. \tag{17}$$

Pointwise regularity and a positive entropy rate of the environment do not by themselves give a uniform lower bound of this kind, and no equality between α_{PE} and m_L is assumed. Proposition 18 is a concrete failure of persistent excitation: in-frame records near calibration have $\alpha_{\mathrm{PE}} \to 0$, and probes restore it.

6.2 The four links of the loop

The series uses the following model architecture. It is a template, not a necessity theorem.

Definition 20 (Four links)**.** (I) **Records.** A world leaves readable, stable records, which an agent can store and use (HAFF; Paper I gives their value).

(II) **Reading and control interface.** Instruments read the world and actuators change it, each with a cost and a legal form (Q-RAIF; Foundation III's legal feedback channels [11]). The programme's running example of an internal algebra, a Clifford representation, can serve as the channel space; nothing here depends on it.

(III) **Self-frame.** A retained model of how the task sits in the agent's own channels, formed from records and selecting later reads (Paper II).

(IV) **Calibration.** References, memory and reliability weighting that keep (III) correct while the instrument drifts, and that decide how much authority over the world each reading carries (this paper).

Each link has an exact result in a stated model, listed in Section 10. Section B joins (II)–(IV) in one finite model in which the agent's actions change what it reads next. Section 8 makes the body's working state a condition of access and identifies a task domain where body history is necessary for a higher harvest. The template is not shown to be necessary for persistence: a system that can read everything it needs, or whose coupling never changes, needs neither (III) nor (IV).

A separate-register implementation can add reset, reference preparation and slow-driving actuation costs without double counting. Appendix F states that conditional bound and its full proof. These implementation charges are not a universal persistence budget.

7 A joint loop and its reference standard

The memory–self–introspection route can be joined around a body that the observer reads and changes. Records consume memory purity; a model of the instrument selects feedback and harvest levels; calibration revises that model. We first give the ledger and its identification boundary, then a finite-block operating regime and a continuing joint filter that uses its body history to revise both its instrument model and its reference decisions. A further construction lets the body determine whether those readings and updates can occur; latched recovery, when physically available, preserves later access. A labelled operation supplies orientation relative to its dispatch map, the calibration counterpart of HAFF Paper G's final cut [8]. Record purity, field preparation and timing are supplied resources in these constructions.

7.1 The model

Definition 21 (Joint model)**.** One cycle is one use; entropies are in nats and energies in units of k_BT.

(i) *World.* A persistent *body* bit $B_t = \pm 1$, flipped by a bath with probability d per cycle ($c = 1 - 2d$), and a *food* pointer $X_t = \pm 1$, fresh each cycle with $P(X_t = +1) = \frac{1}{2}$. Both have degenerate Hamiltonians and are diagonal in the pointer basis.

(ii) *Records.* Each cycle fresh pure memories collide with each carrier through pointer-controlled couplings [5, 6]: the food leaves two disjoint fragments F and G of overlap r, the body a fragment F_B and a copy G_B. Record formation costs a diagonal carrier nothing and consumes memory purity (Paper I, ledger of the collision model).

(iii) *Instrument.* Reading a fragment through the setting that matches its record gives $P(m \mid x) = (1 + \sigma g\, mx)/2$ with $0 \le g \le \sqrt{1 - r^2}$, the Helstrom gain. The *magnitude* g follows a specified bounded transition law; the unbounded Gaussian model of Section 5 is an exact comparison and requires a separate bridge to this physical instrument; the *sign* $\sigma = \pm 1$ may invert with probability η per use; the *setting* is one of D, and a

read with the wrong setting returns a fair coin. The same instrument reads food and body. (This is the gain–bias channel of Section B.2 with $b = 0$, and the setting is a discrete restriction of Paper II's direction family.) For uniform settings, the D choices are coordinate axes $\{e_1, \ldots, e_D\} \subset S^{D-1}$; exact identification removes $\ln D$ nats of setting uncertainty. This is not Paper II's continuous spherical alignment price, and the present model supplies these discrete settings rather than deriving their formation.

(iv) *Actions.* The flip of Q-RAIF Paper B, $U = e^{-i\lambda XQ}$, which with drive occupation N acts as $s\,X\rho X + (1-s)\rho$, $s = 1 - \epsilon_N$ [9], and costs $\theta_\omega(N + a\lambda^2)$ per opportunity, $\theta_\omega = \hbar\bar{\omega}/k_BT$, $a = 1$ if switched on. On the food: flip if $M = -1$, then harvest with fixed levels $E_\pm = -\ln \hat{q}(\pm)$ set from the agent's estimate, by the level protocol of Paper I. On the body: with a gate probability p_{g} known to the agent, flip if the reading is -1.

(v) *Memory and calibration.* The record register M is reset against the copy G, as in the cycle bound of Paper I, or against blank qubits. References are fragments of pointers prepared in a known state and read with the same instrument. A *self-label* is the recorded quench work of the harvest, which equals the energy of the level that the flipped food occupies and so records the food value relative to the agent's own level assignment.

With an ideal actuator, no drift, the matched setting and no body, the food cycle is Paper I's read–harvest–reset cycle; the body alone with a known instrument is the persistence task of Q-RAIF Paper B; the instrument alone is the drifting channel of Section 5. After the flip the food has $P(X' = +1) = q = (1+s\sigma g)/2$, and the harvest returns $\ln 2 - H_2(q) - d(q\|\hat{q})$, with $H_2(x) = -x\ln x - (1-x)\ln(1-x)$, by Paper I's lemma on extraction from a qubit, so $N \to \infty$ recovers the value $I(X{:}M)$ of Paper I.

7.2 The ledger

Every read leaves an outcome that the agent must eventually erase.

Proposition 22 (Reading tax)**.** *Let a classical register R with finite alphabet be updated once per cycle by a deterministic map $(R_t, Y_t) \mapsto R_{t+1}$, after which the outcome register is returned to a standard state, all registers having degenerate Hamiltonians and the update coupling them to a thermal bath with couplings that vanish at the endpoints and no other system correlated with (R_t, Y_t). Then $\beta \sum_{t\le K} W_t \ge \sum_{t\le K} H(Y_t \mid R_t) - \ln|R|$. Under these isolation assumptions, every reference read costs at least $\mathbb{E}\, H_2((1+g)/2)$ per read in the long run, whatever its preparation costs.*

Proof. As in the cycle bound of Paper I, unitarity and $D(\rho'_B\|\tau_B) \ge 0$ give $\beta W_t \ge H(R_t, Y_t) - H(R_{t+1}) = H(Y_t \mid R_t) + H(R_t) - H(R_{t+1})$; the last two terms telescope to at least $-\ln|R|$. Condition also on the sign, known input and register history. Fresh sensing noise gives $H(Y_t \mid R_t) \ge \mathbb{E}H_2((1+g_t)/2)$. A retained correlated outcome copy or newly consumed pure workspace must instead enter the reset ledger. □

The tax distinguishes a free reference input from a free cyclic processor. Under Proposition 22's isolation assumptions, an outcome stream still has a reset cost even when reference preparation is free. In a valid scalar Gaussian comparison this can supply the positive reading price w and preserve the $\nu^{2/3}$ order. For the local Bernoulli comparison, adding preparation and separate outcome erasure changes w from $\ln 2$ to $\ln 2 + H_2((1+g)/2)$; the leading comparison rises by a factor 1.21 at $g = 0.55$. The bounded physical transition and joint likelihood must still be established.

Proposition 23 (Purity available to a reset)**.** *Suppose the reset may couple M, a copy G whose marginal is retained, and n_B initially pure, independent blank qubits to a thermal bath. All registers have degenerate endpoint Hamiltonians. Then*

$$\beta W_{\text{reset}} \geq S(M \mid E_G) - n_B \ln 2.$$

Consequently, for a uniform food pointer,

$$\beta \,\text{net} \leq \chi(M{:}E_G) - H(M \mid X) + \Pi_B \leq \Pi_G + \Pi_B - H(M \mid X), \qquad \Pi_B = n_B \ln 2,$$

provided the record model gives $\chi(X{:}E_G) \leq \Pi_G$ and $M - X - E_G$ is the stated conditional product model. If the allowed reset is additionally required to consume nonnegative work, the first upper bound can also be capped by $I(X{:}M)$.

Proof. The entropy balance for reset gives the first inequality. Subtract it from the harvest bound $\beta W_{\text{harvest}} \leq I(X{:}M)$, and use $S(M \mid E_G) = H(M) - \chi(M{:}E_G)$ and $I(X{:}M) - H(M) = -H(M \mid X)$. Data processing gives $\chi(M{:}E_G) \leq \chi(X{:}E_G) \leq \Pi_G$. Pure blanks can themselves fund work during reset; only the added nonnegative-reset-work condition permits the cap by the harvest alone. □

In the uniform-pointer task, the second copy and pure blanks are resources for reset. Preparing all consumed purity from the agent's own thermal resources removes that contribution to profit. The five-memory copy of Paper I supplies $\chi(M{:}E_G) = 0.5231$ from $5 \ln 2 = 3.466$ nats of initial purity, while a blank qubit supplies $\ln 2$ of entropy capacity. The copy also serves as a readable record.

Theorem 24 (Ledger of the joint model)**.** *Use the food feedback of Definition 21, reset against the declared copies with no additional pure blanks, and account separately for every remaining processor reset and reference preparation. Let $\mathcal{C}_t$ denote the classical information retained and available to a reset, whose marginal the reset preserves. Condition each register's state on that information. With separate baths and no shared erasure accounting, the long-run average net free energy per cycle satisfies*

$$\beta\,\overline{\text{net}} \leq \mathbb{E}\big[\ln 2 - H_2(q_t)\big] - \mathbb{E}\big[S(M_t \mid E_G, \mathcal{C}_t)\big] - \mathbb{E}\big[\ell_t + w\rho_t\big] - \bar{\tau} - \theta_\omega \mathbb{E}\big[N_t + a_t \lambda_t^2\big] - U_{\text{body}}, \quad (18)$$

where $\ell_t = d(q_t \| \hat{q}_t)$ is the harvest mismatch, $w \geq \ln 2$ the reference preparation cost, $\bar{\tau}$ the cost of the remaining outcome and model-register resets, and $U_{\text{body}} \geq \mathbb{E}[S(Y_t \mid E_{G_B}, \mathcal{C}_t)] + \theta_\omega \mathbb{E}[N_B + a\lambda^2]$ the body upkeep. Without retained side information the conditional entropy reduces to $H(M_t) - \chi(M_t{:}E_G)$. Proposition 22 applies to an isolated processor among the remaining resets. To apply Theorem 11, the complete likelihood, transition law and loss

must meet its hypotheses. For constant known s and fixed sign, the local loop-gain coordinate $\vartheta = s\sigma g$ has reference information $1/(s^2(1-g^2))$; a marginal self-label has information $1/(1-\vartheta^2)$. Retaining the food reading changes the label likelihood and requires its full conditional information.

Proof. Apply Paper I's extraction lemma and conditional reset bound to each state conditioned on $\mathcal{C}_t$, then average. Book the other processor resets, reference preparation and Q-RAIF B's drive and switching work once each. The displayed Bernoulli informations follow by differentiating their likelihoods. Local quadratic harvest loss alone does not supply the global loss premise of Theorem 11. Under the separate-register assumptions the actual costs add to the displayed bound. □

Beyond leading order the calibration loss has a closed form. An agent with a belief of variance P about its gain sets its levels to $\mathbb{E}q$, and its expected harvest is $\ln 2 - H_2(\mathbb{E}[q \mid \text{records}])$: the agent earns what its belief allows, and the loss relative to a known gain is the Jensen gap $H_2(\mathbb{E}q) - \mathbb{E}H_2(q)$, which tends to $Ps^2/(2(1-s^2g^2))$ as $P \to 0$ and saturates for broad beliefs. The harvest is increasing in s, for a fixed gain belief. Its Jensen-gap mismatch penalty can also increase with s, so total harvest and mismatch penalty have different monotonicities.

7.3 What acting can reveal

Let Z_t be the hidden world state, θ the hidden parameters (instrument, actuators, world dynamics, record formation, initial law μ_θ), and let the agent's record be $(Y_1, A_1, \ldots, Y_t, A_t)$, with readings $Y_t \sim K^{\mathrm{s}}_\theta(\cdot \mid Z_t)$, actions A_t chosen by an arbitrary adaptive policy with private randomness, and transitions $Z_{t+1} \sim K^{\mathrm{w}}_\theta(\cdot \mid Z_t, A_t)$.

Definition 25 (Loop symmetry)**.** A group $\mathcal{G}$ acting on world states and parameters is a symmetry of the loop if $K^{\mathrm{s}}_{g\theta}(y \mid gz) = K^{\mathrm{s}}_\theta(y \mid z)$, $K^{\mathrm{w}}_{g\theta}(g\,\cdot \mid gz, a) = K^{\mathrm{w}}_\theta(\cdot \mid z, a)$ and $\mu_{g\theta} = g_*\mu_\theta$ for all $g \in \mathcal{G}$.

Theorem 26 (Acting cannot see the symmetry of its loop)**.** *If $\mathcal{G}$ is a loop symmetry, then for every policy the law of the agent's record is the same under $g\theta$ as under θ, and $\mathbb{E}_{g\theta}L(Z_t) = \mathbb{E}_\theta L(gZ_t)$ for every loss L. A parameter function that $\mathcal{G}$ moves has no consistent estimator from the agent's records, however it acts. If $\gamma \in \mathcal{G}$ exchanges a binary loss, $L(\gamma z) = 1 - L(z)$, then $\mathbb{E}_\theta L(Z_t) + \mathbb{E}_{\gamma\theta} L(Z_t) = 1$ for every policy and every t.*

Proof. Write the joint density of hidden variables and records as $\mu_\theta(z_0)\prod_t K^{\mathrm{s}}_\theta(y_t \mid z_t)\,\pi_t(a_t \mid \cdot)\,K^{\mathrm{w}}_\theta(z_{t+1} \mid z_t, a_t)$ and substitute $z \mapsto gz$ in every hidden variable. The policy factors do not involve z, and the others are invariant by Definition 25. Integrating out the hidden variables gives equal record laws, and the same substitution gives the loss identity. □

In the joint model, $\gamma : (B, X, \sigma) \mapsto (-B, -X, -\sigma)$ with everything else fixed is a loop symmetry for any agent that only reads and flips: the instrument depends on σx, the flip commutes with $x \mapsto -x$, the bath and the collisions are symmetric, and the gate coin is independent. The body loss $\mathbf{1}\{B = -1\}$ is exchanged. Every such agent therefore satisfies $P_\theta(B_t = -1) + P_{\gamma\theta}(B_t = -1) = 1$: if it holds its body at loss L it holds it at $1 - L$ in

the world related to it by γ, with identical records. Exact enumeration over twenty random adaptive policies of five steps gives record laws that agree to 1.2×10^{-16} in total variation; one reference read per step separates them by at least 0.30. This is the observation of the Imprint Chain that its calibration statistic depends only on ν^2 [7], extended to every policy and to the task loss. For a linear loop, $x_{t+1} = ax_t + \kappa u_t + f + w_t$, $y_t = gx_t + b + e_t$, the affine maps $x \mapsto \lambda x + \mu$ with $(g, b, \kappa, f) \mapsto (g/\lambda,\ b - g\mu/\lambda,\ \lambda\kappa,\ \lambda f + \mu(1-a))$ form a loop symmetry when the initial and process-noise laws are transformed as well; acting can identify the loop gain $g\kappa$ and never g or κ separately (Kalman likelihoods under random affine maps and dithered adaptive policies agree to 6×10^{-14}).

Proposition 27 (What the closed loop identifies)**.** *In the stationary body loop with constant gain and sign, $0 < g \le 1$, $d \in (0,1) \setminus \{1/2\}$, and the matched setting, gate coin A_t of known probability $p_{\mathrm{g}} \in (0,1)$ and flip success s,*

$$\mathbb{E}[Y_{t+1}Y_t \mid A_t = 0] = cg^2, \qquad \mathbb{E}[Y_{t+2}Y_t \mid A_t = A_{t+1} = 0] = c^2g^2,$$
$$\mathbb{E}[Y_{t+1}Y_t \mid A_t = 1] = cg^2(1-s) + cs\,\mathbb{E}Y_t.$$

Writing the three moments as K_0, K_2, K_1, and $\mu = \mathbb{E}Y_t$, one has

$$c = K_2/K_0, \qquad g^2 = K_0^2/K_2, \qquad s = (K_1 - K_0)/(c\mu - K_0).$$

Stationarity gives $\mu = cp_{\mathrm{g}}sg^2/(1 - c + cp_{\mathrm{g}}s)$, so the denominator is nonzero. Thus c, g^2 and s are determined by the agent's own records, without references, and σ is not. With a wrong setting the first and third moments vanish.

Proof. $B_{t+1} = \xi_t B_t \Phi_t$ with $\mathbb{E}\xi_t = c$ independent and $\Phi_t = -1$ if the flip is switched on and succeeds. For $A_t = 0$, $\mathbb{E}[Y_{t+1}Y_t] = \sigma gc\,\mathbb{E}[B_tY_t] = cg^2$. For $A_t = 1$, $\mathbb{E}[B_t\Phi_tY_t] = \sigma g + 2s\,\mathbb{E}[B_t\mathbf{1}\{Y_t = -1\}]$ and $\mathbb{E}[B_t\mathbf{1}\{Y_t = -1\}] = (\mathbb{E}B_t - \sigma g)/2$; multiply by σgc and use $\sigma g\,\mathbb{E}B_t = \mathbb{E}Y_t$. The lag-two identity iterates the first. □

A frame error is thus visible in the agent's own loop as a loss of loop correlation; locating the right setting still requires trying the others, which is Paper II's blind spot in binary form. What no record reveals is fixed by the following statement. Call an operation *symmetry-breaking* if its action singles out a world state in a way that the symmetry does not respect: a reference prepared in the state $+1$, a reset to $+1$, harvest levels with the higher level on -1, a bath relaxing toward a known state. Each carries a label ζ saying which state it singles out, and that label belongs to the operation's dispatch map in the sense of Paper G.

Proposition 28 (No internal standard)**.** *Let $\mathcal{G}$ act also on the labels of all symmetry-breaking operations by relabelling. If the labelled sensing, transition and initial kernels are jointly covariant, $\mathcal{G}$ remains a loop symmetry. No policy identifies an absolute jointly relabelled orientation. A system can form and retain relational standards.*

Proof. Apply the assumed joint covariance in Theorem 26. Set-theoretic closure of a kernel family alone does not imply covariance. □

The model locates an operational reference standard in a labelled kernel: an external reference, an actuator reset or level assignment, a known environmental asymmetry, or a previously interpreted record. Jointly relabelling every such kernel preserves the record law. A system can nevertheless construct relational standards and carry them between operations. Paper G places the final cut of internal selection at the dispatch map of the top executor [8]; here the corresponding calibration object is the labelled operation whose orientation the loop uses. The symmetry theorem does not derive the physical origin of that operation.

7.4 Three self-parameters, three laws

The magnitude and the setting are invariant under the sign symmetry; the sign is not.

Proposition 29 (Free loop information changes the exponent)**.** *In Theorem 11 let unlabelled reads carry $J_0 > 0$ per use at no extra cost, and let the loss be cP. Put $P_f = \frac{1}{2}\big(-\nu^2 + \sqrt{\nu^4 + 4\nu^2/J_0}\big)$. If $c \le (w/J)J_0(2P_f+\nu^2)/(P_f(P_f+\nu^2))$, which holds for $\nu \le \nu_\times = 2wJ_0^{3/2}/(Jc)\,(1+o(1))$, every policy pays at least $cP_f = c\nu/\sqrt{J_0}\,(1+O(\nu\sqrt{J_0}))$ per use, and with Gaussian reads this is attained with no references.*

Proof. $\phi(P) = \nu^2/(P(P+\nu^2))$ is convex and decreasing, so h of Theorem 11 is convex. P_f solves $\phi(P) = J_0$, and $|\phi'(P_f)| = J_0(2P_f+\nu^2)/(P_f(P_f+\nu^2))$, so under the condition $h' \le 0$ on $(0, P_f]$ and the infimum is $h(P_f) = cP_f$. As $\nu \to 0$, $P_f = \nu/\sqrt{J_0} - \nu^2/2 + O(\nu^3)$. The Kalman filter on the free reads alone has stationary variance P_f. □

The body moments establish static identifiability under Proposition 27's nondegeneracy conditions. A joint information-rate bound for the correlated record stream is a further step: the information of one overlapping product is not that rate. Proposition 29 gives an exact Gaussian realization of the linear-in-ν price when a genuine free observation channel is supplied. For self-labels the marginal Bernoulli information is $1/(1-\vartheta^2)$; retaining the earlier food read changes the conditional likelihood and its information.

Theorem 30 (Sign inversions)**.** *Suppose that sign inversions and body-bath flips are independent Bernoulli events with probabilities $\eta, d \in (0,1)$ per use. The agent knows the initial sign, has no symmetry-breaking operation, and reads only records formed in the current cycle. Put*

$$b_* = \max\{|1-2\eta|, |1-2d|\}.$$

For every read-and-flip policy,

$$\big|P(B_t = -1) - \tfrac{1}{2}\big| \le \tfrac{1}{2}b_*^t.$$

In particular the loss converges to $1/2$. If $\eta \le \min\{d, 1-d\}$, its lower bound is $\frac{1}{2}[1-(1-2\eta)^t]$.

Proof. Let a_j be the exclusive-or of the bath flip and the instrument inversion. Condition on the apparent-body trajectory, the a_j, and all independent reading and actuation randomness. This is more information than the agent receives. Conditional inversion probabilities are

$$q_1 = \frac{\eta(1-d)}{\eta(1-d) + d(1-\eta)}, \qquad q_0 = \frac{\eta d}{\eta d + (1-\eta)(1-d)}.$$

Conditional on that information, the absolute sign bias is $\prod_{j=1}^{t}|1-2q_{a_j}|$. The absolute conditional body bias has the same value, since the apparent body is then fixed. Hence

$$|\mathbb{E}B_t| \le \left(\sum_{a=0}^{1} P(a)|1-2q_a|\right)^t = \big(|1-\eta-d|+|\eta-d|\big)^t = b_*^t.$$

The a_j are independent with a law unaffected by the policy. This proves both sides of the bound. The special case follows from $b_* = 1-2\eta$ under the stated condition. □

The limit $\frac{1}{2}$ is the loss of an agent that does nothing. An inversion is the image under γ of a world kick that the loop corrected, so the records never show it. The food fares worse: after an inversion the flip pushes it the wrong way, the harvest runs at a loss, and an agent that never reads its energy store cannot tell. Reading the store is a self-label, whose standard is the agent's own level assignment. The Bayes-optimal policy, computed by backward induction over all adaptive policies for eight cycles, respects the bound in every case with $\eta \le d$ (for $g = 0.9$, $d = 0.1$, $s = 0.95$, $\eta = 0.05$: 0.307 against 0.285 at $t = 8$). Consciousness III shows that an observer can be confidently and undetectably wrong on a two-point fibre of its protocol [3]; Theorem 30 adds that acting does not change the protocol in the relevant sense, and that inversions make the error grow to chance. The hypothesis of fresh records matters. If a stored record of an earlier body state, made by the environment's fixed couplings, is read again later, the two reads compare σ at the two times, so the world's stable records detect changes of the sign though not the sign. Records of Paper F's kind [6] thus carry an agent's calibration across time, and the standard sits in the past.

Proposition 31 (Reference cost under exogenous and adaptive timing)**.** *Let a reference cost $w > 0$ and reveal the current sign exactly, and let body loss cost $V > 0$ per use. For $0 \le r < 1$, define*

$$F_r(G) = \frac{G}{2} - \frac{1-r^G}{2(1-r)}, \qquad C_r = \inf_{G\in\mathbb{N}} \frac{w + VF_r(G)}{G}.$$

Reference times fixed in advance, or randomized independently of the record, give long-run average cost at least C_{b_*}, where b_* is defined in Theorem 30. For reference times chosen from the records, the bound C_{r_*} holds with

$$r_* = \max_{a=0,1} |1-2q_a|.$$

If $\eta \to 0$ with $d \in (0,1)$ and w, V fixed, these benchmarks satisfy

$$C_{b_*} = \sqrt{2wV\eta}\,(1+o(1)), \qquad C_{r_*} = \sqrt{2wV\eta\,\min\{d/(1-d),(1-d)/d\}}\,(1+o(1)).$$

When a finite gap attains the first infimum, an ideal periodic read-and-control policy attains it; otherwise the no-reference limit gives its value $V/2$. It need not bound record-adaptive schedules.

Proof. With an exogenous gap, the mean loss at age j after a reference is at least $(1-b_*^j)/2$, by the proof of Theorem 30. For an adaptive gap one cannot use this unconditional mean

after conditioning on the absence of another reference. Instead, the same proof gives the conditional bound $(1-r_*^j)/2$ for every apparent-event history. The event that a policy has not yet requested another reference is determined by such a history and its private randomness, so this bound also holds on each adaptive gap. Summing over complete gaps and the terminal incomplete gap gives at least $KC_r - O(w)$ over K uses. Equivalently, the piecewise linear interpolation of F_r is convex, and an exogenous mixture of gap lengths cannot beat the best single length for the corresponding benchmark. Writing $r = 1 - 2a$ gives $F_r(G) = aG(G-1)/2 + O(a^2G^3)$; minimization has $G \sim \sqrt{2w/(Va)}$ and cost $\sqrt{2wVa}(1+o(1))$. For small η, $(1-b_*)/2 = \eta$ and $(1-r_*)/2 = \eta \min\{d/(1-d), (1-d)/d\}(1+O(\eta))$. Ideal reads reveal the apparent state, and ideal flips can hold it aligned with the optimal sign estimate, attaining the exogenous benchmark. □

Remark 32 (Record-adaptive references improve the benchmark)**.** Take $\eta = d = 1/10$, $w = 1/10$ and $V = 1$. Read the apparent body exactly. After accounting for the preceding commanded flip, request a reference precisely when the apparent body has changed sign. Then control the body using the last reference's sign estimate. Such events have probability $p = 2\eta(1-\eta) = 9/50$. Between them the conditional inversion probability is $q_0 = 1/82$. The stationary posterior sign bias is

$$z = p + (1-p)(1-2q_0)z = \frac{9}{10}.$$

The loss is $1/20$, and references cost $9/500$, giving total cost $17/250 = 0.068$. The exogenous-gap benchmark is $1/10$. An exact finite-state computation with sensing gain and flip success both 0.999 gives cost 0.0687253, so the separation persists away from the ideal limit. The bound for adaptive timing is $C_{r_*} = 0.0429971$ here.

Proposition 33 (Exact adaptive value at equal inversion and bath rates)**.** *In Proposition 31, take $\eta = d = 1/10$, $w = 1/10$ and $V = 1$. The minimum long-run average cost, with ideal apparent-body reads and ideal flips, over all adaptive reference policies is*

$$\rho_* = \frac{2089311}{43698622} = 0.0478118280\ldots.$$

Request a reference after an apparent innovation or after five consecutive quiet transitions since the last reference. This policy attains the value, which also bounds policies with less informative sensing or control.

Proof. An apparent innovation has probability $p = 9/50$ and conditional sign bias zero; a quiet transition has probability $z = 41/50$ and multiplies the bias by $u = 40/41$. The oracle therefore has states $k = 1, 2, \ldots$ for k quiet transitions after a reference and a zero-bias state. At state k, continued control has loss $l_k = (1-u^k)/2$; a reference resets the bias and costs w. For threshold $K = 5$, a renewal cycle has

$$D_K = \sum_{j=0}^{K-1} z^j, \qquad C_K = w + \sum_{j=1}^{K-1} z^j l_j, \qquad \rho_* = C_K/D_K.$$

Self-parameter	Sign symmetry	Without any standard	With references or labels
Setting (switches)	invariant	detected by the loop (Proposition 27); found by search	not needed
Magnitude (random walk ν)	invariant	static body moments identify gain; a valid Gaussian free channel gives price $c\nu/\sqrt{J_0}$ (Proposition 29)	$\nu^{2/3}$ (Theorem 11); marginal self-labels avoid pointer preparation; full retained history determines their information
Sign (inversion rate η)	variant	loss $\to \frac{1}{2}$ for fresh-record read-and-flip policies (Theorem 30)	positive square-root lower bound for adaptive timing; exact finite-rate optimum in Proposition 33

Table 1: What acting buys for each self-parameter of the joint model.

To verify global optimality, set $v_k = w$ for $k \geq 5$ and set $v_k = l_k - \rho_* + pw + zv_{k+1}$ backwards for $k = 4, 3, 2, 1$. Exact rational arithmetic gives

$$-\rho_* + pw + zv_1 = 0, \quad v_k < w \ (1 \leq k < 5), \quad l_5 > \rho_*.$$

Thus $v_k = \min\{w, l_k - \rho_* + pw + zv_{k+1}\}$ for every k; monotonicity of l_k proves the stopping inequality for all $k \geq 5$. The zero-bias state also stops, since $1/2 > \rho_*$. These bounded Bellman inequalities telescope over every adaptive, randomized or nonstationary policy, leaving a bounded terminal term. Dividing by the horizon proves the lower bound. Ideal readings reveal the apparent innovations, and ideal flips attain their conditional Bayes body loss, so the threshold policy is an admissible implementation of the oracle optimum. Less informative records are a garbling of this oracle. A less reliable flip can be simulated by an ideal actuator using a private coin, with its storage and reset charged to the policy. Neither restriction improves the oracle value. □

In the exogenous benchmark with $V = 1$, the small-η calculation is within 4% of $\sqrt{2w\eta}$ over the tested $\eta \leq 10^{-4}$ range. A periodic noisy-reference policy gives excess cost within a factor 2–3 of $\sqrt{2wV(1-2L_0)\eta}$, where $L_0 = 0.052$ is its no-inversion loss. Its fitted exponent is 0.41–0.43, consistent with the growing reference-block size. Proposition 33 settles the stated finite-rate ideal example. Proposition 31 supplies the general adaptive lower bound; its constant need not be sharp.

7.5 Operating and continuing regimes

A bounded two-gain block model has a supplied-resource operating margin of $0.02396\, k_B T$ per food use with a squeezed drive, body loss 0.08725 and explicit purity input. Its complete

ledger and comparison are in Appendix G.1. A separate continuing eight-state filter carries instrument and body history through every use. Under the same reference cap, adaptive timing gives body loss .10319, versus .11726 for periodic timing. Its transition/likelihood, two-seed check, 350-bit quantization certificate and resource boundary are retained in Appendix G.2. The block margin cannot be imported into that continuing protocol without its own apparatus ledger.

8 When the body controls access

The preceding loop receives a reading and can process it at every use, even when its body is at the loss state. A body that supports those operations changes the problem. Keep the bounded instrument, record couplings, field drive and body bath, but let $B = +1$ enable the main instrument, reference ports, model logic and food protocol. At $B = -1$ these ports are disconnected. The model register is held, not updated. A disabled reader cannot acquire a reference or ask a running estimator to repair it.

Definition 34 (Access-gated joint loop)**.** At each working use, read the body with likelihood $(1 + y\sigma g_G)/2$, update the model, perform the food task and write a recovery command $\ell \in \{0, 1\}$ to a latch. The working body is not actively flipped; the bath disables it with probability d. During disability the latch and model register remain unchanged. A separate standby port, if present, executes that stored command through the same field drive; it obtains no new instrument reading. The recovery probability per disabled slot is

$$r_\ell^{\mathrm{rec}} = d + \ell s(1 - 2d), \qquad 0 < d < \tfrac{1}{2}.$$

The instrument continues its gain and sign transitions in all wall-clock slots. Disabled-slot fragments and enable flags leave through output streams that the reader cannot later acquire; there is no bank of delayed readings on resumption. The reader obtains no elapsed-time counter on recovery. A resumed use is another working invocation, with its prior obtained from the event kernel below. Harvest work enters the physical ledger but is not read back as an additional self-label unless a protocol explicitly permits that read. The latch is written after all records acquired in its working invocation.

An independent enable flag copies the pointer value of B and controls the standby drive while the main processor is held. In the diagonal body model the repair leaves $+1$ fixed and sends -1 to $+1$ with probability s. A CPTP extension has Kraus operators

$$D_0 = P_+, \qquad D_1 = \sqrt{1 - s}\, P_-, \qquad D_2 = \sqrt{s}\, |+\rangle\langle -|, \qquad \sum_j D_j^\dagger D_j = I.$$

It dephases the body blocks and is nonunital. The enable flag, its preparation or reset, and the body's free-energy change enter the ledger. One explicit resource model supplies a fresh pure flag per wall-clock slot and retains the used flag in a charged output stream, consuming $\ln 2$ of flag purity per slot. Cyclic reuse requires a conditional reset and protection protocol. Table 2 collects these resource declarations. With no standby, the symmetric bath still recovers the body at rate d.

Proposition 35 (The kernel between working uses)**.** *Let* K *be the four-state gain/sign factor of* T_a *and fix a stored command with recovery rate* $r_{\mathrm{rec}} > 0$*. If* L *is the number of wall-clock slots between successive working invocations, then*

$$P(L = 1) = 1 - d, \qquad P(L = k) = d r_{\mathrm{rec}} (1 - r_{\mathrm{rec}})^{k-2} \quad (k \geq 2), \tag{19}$$

$$\mathbb{E}L = 1 + d/r_{\mathrm{rec}}, \qquad \pi_{\mathrm{w}} = \frac{r_{\mathrm{rec}}}{r_{\mathrm{rec}} + d}, \tag{20}$$

$$\mathsf{K}_\ell = (1 - d)\mathsf{K} + d r_{\mathrm{rec}} \mathsf{K}^2 [I - (1 - r_{\mathrm{rec}})\mathsf{K}]^{-1}. \tag{21}$$

π_{w} *is the long-run working fraction. A right eigenmode of* K *of eigenvalue* λ *has event eigenvalue*

$$f_{r_{\mathrm{rec}}}(\lambda) = (1 - d)\lambda + \frac{d r_{\mathrm{rec}} \lambda^2}{1 - (1 - r_{\mathrm{rec}})\lambda}.$$

The event kernel remains valid conditionally when the latch is chosen from the preceding working record.

Proof. The body either remains working for the next slot or enters a geometric recovery wait. This gives (19) and its mean. The stationary flow balance is $\pi_{\mathrm{w}} d = (1 - \pi_{\mathrm{w}}) r_{\mathrm{rec}}$; renewal averaging gives the same fraction. Instrument noise is independent of the body and recovery noise conditional on the selected latch. Sum K^L with the displayed gap probabilities. Since $\|(1 - r_{\mathrm{rec}})\mathsf{K}\|_\infty = 1 - r_{\mathrm{rec}} < 1$, the sum is the positive resolvent in (21). Applying it to an eigenvector gives $f_{r_{\mathrm{rec}}}$. No recursion is executed in disabled slots: the finite controller uses the matrix for its last stored command on the next invocation. □

The working body provides a relative calibration standard. With static sign and gain, its working readings are independent samples of $(1 + y\sigma g)/2$; their mean identifies σg for $g > 0$. With a fixed latch their Fisher information rate is $\pi_{\mathrm{w}}/(1 - g^2)$ in the signed-gain coordinate. Under a history-dependent latch, information is counted on actual acquired uses, as in Consciousness III [3]. The transformation $(B, X, \sigma) \mapsto (-B, -X, -\sigma)$ exchanges enabled and disabled blocks and hence fails to preserve this access kernel. The orientation comes from the declared body–permission coupling. Jointly relabelling that coupling, every operation and the task preserves the absolute-orientation obstruction.

The readout must be informative: $g = 0$ or wrong-setting reads give no instrument information. Continued access is a separate condition. In a hardware variant with an absorbing disabled state and conditional failure hazard at least $h > 0$ on every working use, every policy has

$$P(N \geq n) \leq (1 - h)^{n-1}, \qquad \mathbb{E}N \leq h^{-1},$$

where N is its total number of working readings. Successive conditioning proves the bound, so only finitely many readings occur almost surely. If two static parameters have positive overlapping likelihoods on finite stopping records, their stopped-record laws have positive overlap and their equal-prior Bayes error is strictly positive. Waiting longer cannot identify them. The absorbing variant removes passive recovery as well as standby.

Proposition 36 (Calibration through unobserved disability)**.** *Take known* $0 < g < 1$*, sign inversion probability* $0 < \eta < 1/2$*, one body reading on resumption and no additional current*

directional observation. Conditional on a stored recovery command, every sign estimator has average error at least

$$\min\{e_{\mathrm{rec}}, (1-g)/2\}, \qquad e_{\mathrm{rec}} = \frac{1 - f_{r_{\mathrm{rec}}}(1-2\eta)}{2} > 0.$$

Increasing r_{rec} lowers this oracle floor and increases the working fraction. For small η, $e_{\mathrm{rec}} = \eta/\pi_{\mathrm{w}} + O(\eta^2)$.

Proof. Grant the reader the true sign at its preceding working invocation. Across an unobserved gap L, its sign bias is multiplied by $(1-2\eta)^L$. Averaging gives the prior error e_{rec}. The current body reading is a binary symmetric channel of error $e = (1-g)/2$. For prior error $q \le 1/2$, its two-output Bayes error is

$$\min\{(1-q)(1-e), qe\} + \min\{(1-q)e, q(1-e)\} = \min\{q, e\}.$$

An actual history is less informative than the oracle history. For $0 < \lambda < 1$, differentiation gives

$$\frac{\partial f_{r_{\mathrm{rec}}}(\lambda)}{\partial r_{\mathrm{rec}}} = \frac{d\lambda^2(1-\lambda)}{[1-(1-r_{\mathrm{rec}})\lambda]^2} > 0.$$

The working fraction is also increasing. A paid elapsed-time counter or another current reference changes the available record and therefore changes this bound. □

Filtering at actual working invocations uses the positive four-state event kernels of Proposition 35. The certificate in Appendix H needs 150 ideal model bits and bounds asymptotic total-variation error by 1.247×10^{-7} for the stated switching rates. Its precision is fixed; identifying a continuous parameter to arbitrarily growing precision requires growing memory.

Proposition 37 (A domain where history is necessary for the food task)**.** *Fix the recovery command, a known static $0 < g < 1$, an unknown static sign with equal prior, one body reading per working invocation and no further directional sample before the food action. Within the joint food protocol, choose harvest levels from the body model before the fresh food reading M, then choose whether to flip from M. Without retained history, the maximal harvest is*

$$J_1 = \ln 2 - H_2((1+sg^2)/2).$$

Let $a_g = (1+g)/2$, $b_g = (1-g)/2$, and retain the latest three body readings. For their number k of positive results set

$$p_k = \tfrac{1}{2}\binom{3}{k}(a_g^k b_g^{3-k} + b_g^k a_g^{3-k}), \quad m_k = \frac{a_g^k b_g^{3-k} - b_g^k a_g^{3-k}}{a_g^k b_g^{3-k} + b_g^k a_g^{3-k}}.$$

The achievable harvest

$$J_3 = \sum_{k=0}^{3} p_k[\ln 2 - H_2((1+sg|m_k|)/2)]$$

is strictly larger than J_1. Thus a target harvest in $(J_1, J_3]$ requires history within these permissions. An already learned sign encoded in a program, latch or weight counts as retained history.

Proof. The fair food reading has no marginal sign information. A body-only record of posterior sign mean m allows a food polarization of at most $sg|m|$: for arbitrary flip probabilities depending on M, its magnitude is $sg|m|$ times the difference of two probabilities, whose absolute value is at most one. Choosing the appropriate orientation attains that endpoint, and matched levels attain the stated harvest. Without history $m = gY$. Three-sample Bayes gives p_k, m_k. The function $F(m) = \ln 2 - H_2((1 + sgm)/2)$ is strictly convex, with $F''(m) = (sg)^2/[1 - (sgm)^2] > 0$. Posterior means form a martingale, and two extra informative noisy samples make its conditional value nonconstant. Strict Jensen therefore gives $\mathbb{E}F(m_3) > \mathbb{E}F(m_1)$. A three-bit sliding record acquires no additional samples and remains held during disability. □

For $g = 3/5$ and $s = 49/50$, exact finite body–food–drive path enumeration gives $J_1 = 0.0635939$ and $J_3 = 0.1202672$. If levels may instead depend on the current food reading, the posterior-matched protocol family of Paper I replaces sg by g in both formulas and retains the strict history gap.

Performance is averaged over wall-clock slots. The fixed-latch harvest gain is $\pi_{\mathrm{w}}(J_3 - J_1)$ before the extra memory cost. The drive of Section G.2 changes the working fraction from $1/2$ to about 0.949083 when standby is enabled. Body readings at known $B = +1$ contain fresh instrument noise uncorrelated with the body copy. Table 2 therefore charges their accessible reset information separately, together with holding and processing. The finite-block operating margin uses a different body protocol.

Resource	Declared supply or charge	Availability and accounting
Food and body fragments	$4n \ln 2$ purity nats per wall slot if continuously allocated	Acquired and reset only on working uses; disabled outputs are not banked
Standby enable flag	$\ln 2$ supplied purity nats per wall slot in the fresh-flag implementation	Used outputs leave the reader; cyclic reuse needs a reset/protection ledger
Two drive packets	$2\theta_\omega N$ per wall slot	Unused prepared packets remain charged
Drive switching	$\theta_\omega \lambda^2[\ell(1 - \pi_{\mathrm{w}}) + \pi_{\mathrm{w}}/2]$	Recovery on disabled slots; food on working slots
Body outcomes and model	Conditional erasure, update and reliable-hold costs	No processing during disability; the body copy does not erase fresh sensing noise

Table 2: Resource declarations for access-gated comparisons. Purity is an input, not a proved total energy cost. Preparation from thermal resources and any remaining apparatus work must also be charged.

For a fully known instrument and fixed active pipeline, let v denote net value per working use after its active costs, and put $\kappa = \theta_\omega \lambda^2$. If all base supplies, including fresh flags, are

allocated equally in wall time, the two fixed latches have residual values

$$U_\ell = \pi_{\mathrm{w},\ell} v - (1 - \pi_{\mathrm{w},\ell})\kappa\ell - C_{\mathrm{base}}, \qquad U_1 - U_0 = \frac{(r_1^{\mathrm{rec}} - d)v - 2d\kappa}{2(r_1^{\mathrm{rec}} + d)}.$$

This identifies the priced gain from making future calls possible. Flag costs that depend on the latch must be added, not hidden in a common base. For an uncertain instrument, the future task value and the evidence lost while disabled must also enter the control problem. The known-parameter price threshold motivates a stronger history-dependent hypothesis.

Proposition 38 (Equal sign confidence, opposite paid repairs)**.** *There is a static, clocked version of Definition 34 in which two legally acquired histories at the same working slot have equal sign confidence and opposite optimal repair decisions. Take* $d = 1/20$, $s = 49/50$, $H = 12$, *and a uniform prior on the four static states* $(\sigma, g) \in \{\pm 1\} \times \{3/20, g_+\}$. *At slot* $t = 10$, *compare the history with positive readings at slots* 1 *and* 10 *and disability at slots* $2, \ldots, 9$, *with ten working readings of which eight are positive. Choose* g_+ *so that their sign confidences agree, with*

$$0.959375749239844 < g_+ < 0.959375749239846.$$

Both confidences are $0.876966618\ldots$, *whereas their posterior means of the unsigned gain are* $0.678161237\ldots$ *and* $0.249069192\ldots$.

Each working invocation acquires one body reading and permits the food protocol of Proposition 37, or abstention from that food task. This witness does not acquire or read back the harvest-work self-label of Definition 21: its histories contain body readings, gate outcomes and the priced clock. The harvest work still enters the physical ledger. Food switching costs $\kappa/2$ *per executed task and latched recovery costs* κ *per disabled slot. Every working invocation also pays a declared mandatory processing charge* c_{p}. *A paid clock signals the slot and horizon; base record, flag, drive and clock supplies have the same wall-time charges for both latch choices. In this witness, body-reading erasure and food reset are supplied operations, excluded from* $V(b)$ *below rather than assigned zero universal cost. Proposition 41 establishes the corresponding opposite-repair phenomenon with the terminal reading and reset charges included. At* $\kappa = 2/25$, $c_{\mathrm{p}} = 0$, *the optimal latch differences satisfy*

$$Q_1(b) - Q_0(b) \in (0.00559146, 0.00559147), \qquad Q_1(b') - Q_0(b') = -0.004272.$$

At $\kappa = 9/4000$, $c_{\mathrm{p}} = 3/25$, *their differences satisfy*

$$Q_1(b) - Q_0(b) \in (0.00587980, 0.00587982),$$
$$Q_1(b') - Q_0(b') \in (-0.00432522, -0.00432520).$$

Proof. For a gain g put $a_g = (1 + g)/2$ and $b_g = (1 - g)/2$. The posterior after n readings, k positive, is proportional to $a_g^k b_g^{n-k}$ for sign $+1$ and $b_g^k a_g^{n-k}$ for sign -1. Let $c_{n,k}(x)$ be its positive-sign probability with gains $3/20, x$. The rational function $c_{2,2}(x) - c_{10,8}(x)$ has negative and positive values at the displayed rational endpoints. Its denominator is positive, so continuity gives an exact root. Both histories can be generated by the same legal policy that holds latch zero until slot ten. Their gate paths have probabilities $d^2(1-d)^7$ and $(1-d)^9$,

independent of the static instrument; every required reading has positive likelihood. Gate and clock records therefore leave the count-based posteriors unchanged.

Write $m(b) = \mathbb{E}_b[\sigma g]$ and

$$F(x) = \ln 2 - H_2((1+x)/2), \qquad V(b) = \max\{0, F(s|m(b)|) - \kappa/2\} - c_{\mathrm{p}}, \qquad A(b) = \mathbb{E}_b V(b^Y),$$

where b^Y is the posterior after one further working reading. Only two future slots remain. If slot eleven works, its optimal continuation is common to the two current latches. It writes latch zero: a recovery switched on at slot twelve cannot create a working opportunity within this horizon. If slot eleven is disabled, the stored latch pays $\kappa\ell$ there; recovery makes slot twelve working with probability r_ℓ, otherwise the same charge is paid once more. Consequently

$$Q_1(b) - Q_0(b) = d\big[(r_1 - d)A(b) - \kappa(2 - r_1)\big], \qquad r_1 = d + s(1 - 2d). \tag{22}$$

This is the full Bellman difference for the stated action family.

The numerical intervals follow from rational enclosures, including the gain-root bracket. For $0 \le x < 1$ use

$$F(x) = \sum_{j=1}^{24} \frac{x^{2j}}{2j(2j-1)} + R_{24}(x), \qquad 0 \le R_{24}(x) \le \frac{x^{50}}{50 \cdot 49(1 - x^2)}.$$

Rational interval evaluation gives, for the first price pair, $A(b) \in (0.2236, 0.2237)$ and $A(b') = 0$. For the second it gives $A(b) \in (0.1360, 0.1361)$ and $A(b') \in (-0.0954, -0.0953)$; retaining more digits gives the stated intervals. Substitution in (22) proves the strict decisions. □

The histories distinguish how much a future callable task is worth even when their confidence about its orientation is identical. The processing charge $3/25$ is a stipulated implementation price, not a universal erasure lower bound. The first witness needs no such charge. We now use the close gains and positive drift rates of the continuing instrument, with reading and food-reset charges from its ledger.

Proposition 39 (Two moments determine terminal access value)**.** *Immediately before a final working read let $X = \sigma g_G$ have posterior law b, and put $m = \mathbb{E}_b X$ and $v = \mathbb{E}_b X^2$. Use the body reading $P(Y = y \mid X) = (1 + yX)/2$, charge its conditional entropy, and permit either the posterior-matched food protocol or abstention. For overlap r, flip success s and switching charge $\kappa/2$, the expected working-use value under these separate conditional-reset prices is*

$$\begin{gathered} A(m, v) = -H_2((1 + |m|)/2) + \sum_{y=\pm 1} \frac{1 + ym}{2}\, M(m_y), \qquad m_y = \frac{m + yv}{1 + ym}, \\ M(z) = \max\{0, S_r(0) - S_r(|z|) - H_2((1 + s|z|)/2) - \kappa/2\}. \end{gathered} \tag{23}$$

Here S_r is defined in Section G.1; a term of zero observation probability is taken by continuity. A post-record decision with two wall slots remaining has

$$Q_1(b) - Q_0(b) = d[(r_1 - d)A(m_2, v_2) - \kappa(2 - r_1)], \qquad r_1 = d + s(1 - 2d), \tag{24}$$

where m_2, v_2 are the moments of $b\mathsf{K}^2$.

Proof. Bayes' rule gives $P(Y = y) = (1+ym)/2$ and $\mathbb{E}[X \mid Y = y] = (m+yv)/(1+ym)$. The food harvest is $\ln 2 - H_2((1 + s|m_y|)/2)$, its copy-assisted reset costs $\ln 2 - S_r(0) + S_r(|m_y|)$, and switching costs $\kappa/2$. Their difference, optimized against abstention, is $M(m_y)$. The body-outcome charge is $H_2((1 + |m|)/2)$. These terms give A. For the two-slot difference repeat the branch calculation in (22); the instrument undergoes two hidden transitions before the final reading. □

Two moments suffice for this terminal task, not for the continuing joint filter. The prices are conditional-entropy limits for specified separate resets. Base packets, flags, clock, reliable holding and processor implementation have the common supplied charges of Table 2; the expression is not a total cyclic apparatus profit. Additional action-dependent costs must be included in its value before deciding to repair.

One formed model, different decisions. The body model carried between invocations is a joint belief over (G, σ), formed by the permitted record history. It has three uses within this protocol. The food controller acts only when $M(m_y) > 0$; otherwise it withholds permission to spend the switching and reset resources. The repair controller compares (24) with zero. A controller that can use a later invocation for further evidence must also forecast what today's record will make learnable. These are not three independently supplied assessments of the observer. They use different functions of the same acquired belief. A finite horizon and read quota, or a specified continuing budget, determine which functions are worth retaining and computing.

Proposition 40 (Terminal compression need not close enquiry)**.** *Use the same body likelihood as Proposition 39 and write $t = \mathbb{E}_b X^3$. Conditional on the present working read,*

$$P(Y = y) = \frac{1 + ym}{2}, \qquad m_y = \frac{m + yv}{1 + ym}, \qquad v_y = \frac{v + yt}{1 + ym}. \tag{25}$$

For a static instrument and two available working invocations, consider the body-only retained-history protocol: use the terminal food/abstention rule on each invocation, retain the body record, and do not use harvest-work labels or the food record as an extra calibration record. Its two-invocation value, before common supplied charges, is

$$A_2(b) = -H_2((1 + |m|)/2) + \sum_{y=\pm 1} \frac{1 + ym}{2}[M(m_y) + A(m_y, v_y)]. \tag{26}$$

Thus (m, v) is sufficient for the last invocation but need not determine the value of retaining a record for the next. With the four distinct supported values $\{-g_+, -g_-, g_-, g_+\}$, moments of orders zero through three determine the entire belief; fewer need not. Under positive drift, propagate that belief with the legal event or wall-slot kernel before applying the next likelihood.

Proof. Multiply the prior by $(1 + yX)/(1 + ym)$ and integrate X and X^2. This gives (25). The first invocation pays its outcome entropy and chooses $M(m_y)$; its retained outcome determines the prior of the second, whose value is $A(m_y, v_y)$. Conditioning and averaging prove (26). For the four-state support, moment recovery is an invertible Vandermonde system. To see the failure of two-moment closure without relying on a dimension count, put $a = g_-$,

$c = g_+$ and perturb any strictly positive belief in the stated order by $e(a/c, -1, 1, -a/c)$. Normalization, m and v are unchanged, whereas t changes by $2ea(a^2 - c^2) \neq 0$. For example, with $a = 0.90\sqrt{1 - 0.8^{14}}$, $c = 0.98\sqrt{1 - 0.8^{14}}$, $s = 0.979992$ and $\kappa = 0.00225388$, start from $b = (.02, .08, .10, .80)$ and take $e = .02$. Both beliefs have $(m, v) \simeq (.765001391, .892279800)$, but $\sum_y P(y)A(m_y, v_y)$ is respectively .181752936 and .180874313. The continuous entropy expressions in (23) give this strict difference. These positive-belief examples establish failure of universal compression; they are not claimed as the two histories reached from the uniform prior in Proposition 41. □

The invocation quota in this proposition is real: one body reading per invocation, no extra calibrated input and no read during a blackout. Record retention has holding, processing and eventual reset costs; the gain in (26) pays for it only if it exceeds those prices. An optional additional enquiry must be compared with foregoing that invocation or using it for the current task, including its drive, switching and record charges. The formulas are conditional task values, not a free probe or a claim of recurrent optimality. In a finite-horizon control problem the legal belief, remaining quota and available clock information suffice for the usual Bellman recursion. If the clock is absent, the disabled duration must instead be marginalized with (21); it cannot be recovered from unobserved slots. The fixed-support filter and its precision certificate implement this common model; they do not supply processor work or a cyclic reset for free.

Proposition 41 (Paid-repair decisions under positive drift)**.** *Set* $r = 0.8^7$, $h = \sqrt{1 - r^2}$, $g_- = 0.90h$, $g_+ = 0.98h$, $\delta = 3 \times 10^{-4}$, $d = 1/20$, $s = 0.979992$ *and* $\kappa = 0.00225388$. *Start from the uniform four-state prior. Take a priced clock, horizon* $H = 12$*, and a working decision at slot ten. Use the terminal protocol of Proposition 39, with no harvest-work read-back or further processing surcharge. At* $\eta = 2 \times 10^{-4}$*, a lower-confidence history optimally repairs while a higher-confidence history declines. There is also a strictly positive* η_* *in*

$$.00019999976122194292 < \eta_* < .00019999976122194293$$

at which two reachable histories have exactly equal sign confidence and opposite optimal repairs. The histories and rounded values are shown in Table 3.

Inversion rate	Slot history	Sign confidence	$Q_1 - Q_0$
0.0002	$- \times - - - - - + + ++$	.969070314	+.002191561
0.0002	$+ + - + - - - - - --$	.999973536	−.000178194
η_*	$- + + + \times \times \times + - +$	.999514604	+.000256112
η_*	$+ - - \times - + - - + -$	.999514604	−.000845366

Table 3: Working body readings are + or −; × is a disabled slot. All decisions use the same two-slot horizon, prices and action family. In the last pair confidence is exactly equal at a root in the stated interval, not merely equal after rounding.

Proof. Hold latch zero through the preceding slots. Each displayed gate path then has positive probability, independently of the instrument. All reading likelihoods and instrument

transitions are positive. Starting from the uniform prior, propagate with K between slots and multiply by $(1 + y\sigma g_G)/2$ on working readings. Thus these are reachable beliefs, not assigned posteriors.

The confidence difference of the last two histories is a continuous function of η. Outward interval evaluation gives opposite signs at the two stated rational endpoints, proving existence of an exact root. Over that whole interval the two differences in (24) lie respectively in $(.00025611, .00025612)$ and $(-.00084537, -.00084536)$. At $\eta = .0002$ the first pair lies in $(.00219156, .00219157)$ and $(-.00017820, -.00017819)$. The entropy, square-root and Bayes operations in (23) enclose these values directly; every posterior denominator is strictly positive. □

The enclosures evaluate the displayed Bayes and entropy equations directly, rather than using sampled trajectories to decide the four histories.

The first pair needs no tuned parameter and rules out an increasing confidence-threshold repair rule at the continuing instrument's own rates. The second rules out sign confidence as a sufficient decision statistic in a nearby positive-rate model. It does not assert a unique root. The restoring reader assigns more weight to the higher gain; the declining reader expects less value from a working invocation. Confidence in orientation and the value of accessible tasks are different properties of the same acquired history.

The finite horizon is material. A continuing observer must also price evidence gained on later invocations, lost evidence during disability, and the registers that carry its model. Recurrent optimal maintenance without an elapsed-time counter remains a stronger target. The two moments above explain a terminal decision without adding history as an independent reward.

9 Consequences for the other links

Paper II. Let the agent add a known push p_t, independent of the input, through an actuator with gain k, so that $y_t = \rho\, u \cdot x_t + k\rho\, u_A \cdot p_t + \xi_t$, with one in-frame read per step. If sensing and acting share one coupling ($u_A = u$), the regressions of y on the push and on the in-frame read, with the residual variance, determine u including its background, ρ, k and s; the blind spot of Paper II disappears with a single read per step when $u \cdot v \neq 0$, $k\rho \neq 0$ and sign conventions for k, ρ are fixed, and the remaining symmetry between actuator gain and input scale is fixed by the known input law. If the actuator coupling is independent of the sensing coupling, pushes determine $k\rho u_A$ and nothing about the background of u. In a simulation with $D = 6$ the shared coupling recovers ρ and k within 2% and the direction of u within 0.016; with an independent coupling the direction error is 1.3. A shared sensing/action frame supplies this constructive route. At $u \perp v$ the scale parameters remain confounded; varying the read direction repairs the degeneracy.

HAFF Paper C. The viability bound of an agent depends on it only through the product of its sensing and control influences [4], which is invariant under the sign symmetry, while viability itself is not: an inverted agent has the same influences and fails with probability $\frac{1}{2}(1 + (1 - 2\nu)(1 - 2\mu)(1 - 2q))$. Positive influences can be certified from the agent's own records; viability cannot be certified without a standard.

Several observers. Applying Theorem 26 with the sign symmetry acting on two agents at once, their relative sign is invariant and detectable from agreement, and a common inversion is not. The common error that agreement can never expose, which Foundation II measures [10], is the direction shared by the observers' standards.

10 Discussion

10.1 The memory–self–introspection route

The shared channel, loss metric and frame residual were specified in Section 1.2. The body adds a further use of the learned model: deciding which future access is worth restoring. Proposition 41 proves that sign confidence can select the wrong repair even at positive drift rates; Propositions 39–40 identify the extra retained information needed for terminal and prospective tasks. The body–permission coupling carries a relational standard. Wiring, transition laws and resource supplies remain specified inputs, and joint relabelling preserves the absolute-orientation obstruction of Theorem 26. A cyclic processor with holding and preparation costs, and a continuous self-frame with matrix-valued loop gains, remain stronger targets.

10.2 What is established and what is open

Within the stated models the results are proved, and several are sharp. The detection bound holds for every test; it is reached within a factor of about 1.3 when the drift path is known, and its order is reached by a simple residual-variance test under Brownian drift. The scalar quadratic drift law holds for every schedule and estimator; Gaussian prior/readings and controllable real precision attain its optimum, so the price of keeping up is known exactly to leading order, with its constant. The task-risk formula of Section B is exact for every joint law of the reports.

The main open problems are these. The drift law is proved for a scalar parameter and a Gaussian random walk; vector parameters, non-Gaussian increments (for which the same proof gives a weaker bound with the increment variance replaced by the inverse Fisher information of the increment law) and smoother drift (Conjecture 17) remain. Paper II conjectures that probe calibration of a D-channel frame is rate optimal for spherical Brownian/local isotropic diffusion, not for every rotation-invariant drift law [2]; the telescoping of Theorem 11 applied per background direction is a possible route, but the directions of a frame on a sphere are not independent random walks. No rule is given that learns the task risk of Proposition 45 online, and the gain proxy of Section B is not shown to be the best measure of reliability. Fisher information is the natural local quantity for these questions; nothing here shows that a physical system must compute it.

The positive-rate repair witness establishes a finite-horizon mechanism, not an optimal recurrent policy. Its extension must combine continuing maintenance, a finite processor and the cost of acquiring its decision information. A physical cycle must supply holding, processing, reference preparation and heat transport together. BRAIN's component-maintenance model [12] treats the corresponding question when learned relations span several ports; it has

Linear drift ε	10^{-2}	3×10^{-3}	10^{-3}
In-frame records, exact (asymptotic)	54 (49)	137 (129)	324 (312)
Probe records, exact (asymptotic)	58 (58)	130 (130)	271 (271)
Brownian drift σ	10^{-2}	10^{-3}	10^{-4}
In-frame records, exact (asymptotic)	17 (11)	68 (53)	282 (245)
Residual-variance test reaches errors 0.05	84	356	1565
Frozen self-model fails at $t^* = \ln 2/\sigma$	69	693	6931

Table 4: Detection times in steps for Paper II's reader ($\rho = 1$, $s = 0.5$, one record per step, errors 0.05). The rows labelled "exact (asymptotic)" are necessary times: "exact" sums the exact relative entropies, with the unwrapped angle normal of variance $2\sigma t$ under Brownian drift, and "asymptotic" is the formula of Proposition 6. The test row is the smallest horizon at which the residual-variance test of Corollary 7, with the exact χ^2 threshold, reaches both errors 0.05 (Monte Carlo).

a different repair clock and cannot inherit the single-body event kernel without an explicit mapping.

10.3 Function and experience

Reflexivity in this paper means second-order control: a system that monitors and adjusts its own monitoring. The results say what such control can detect, what it costs and what it buys in a loop. The access gate also makes the reliability of future self-knowledge depend on physical maintenance. Its working and disabled states are operational permissions, not claims about sleep, death or experience. They do not show that a calibrated self-model is, or produces, experience. The Consciousness series treats the relation between functional self-models and first-person organization with its own arguments [3].

A Worked example

A.1 Detecting drift in Paper II's two-channel reader

Take Paper II's reader with $D = 2$, $\rho = 1$, $s = 0.5$ and one record per step ($r = 1$). Then $\kappa = 36$ for in-frame records and $F_\perp(0) = 0.8$ for probes. For errors $\alpha = \beta = 0.05$, $d(0.95\|0.05) = 2.650$. Table 4 gives the smallest horizon at which the exact relative entropies of Corollary 5, summed over the records, reach d. By Proposition 6 no test with these errors can succeed earlier.

For known linear drift, the likelihood-ratio test reaches errors 0.05 at about 1.3 times the necessary horizon (4000 repetitions per point). The path-free residual-variance test needs roughly five times the Brownian lower bound; it precedes frozen-model failure at $\sigma = 10^{-3}, 10^{-4}$, but not at 10^{-2}. Its $T\sigma^{2/3}$ approaches the predicted constant 3.28. Near calibration the in-frame natural gradient is $0.0556/\theta$, whereas a probe fraction p supplies metric $0.8p$ at calibration.

ν per use	Floor (k_BT)	C (k_BT)	ρ^* (references per use)	α^*
10^{-2}	0.05511	0.05515	0.0265	1.95×10^{-3}
3×10^{-3}	0.02471	0.02471	0.0119	3.9×10^{-4}
10^{-3}	0.01188	0.01188	0.0057	9.1×10^{-5}
10^{-4}	0.002560	0.002560	0.0012	4.2×10^{-6}

Table 5: The local Gaussian comparison for a drifting gain ($g = 0.55$, $b = 0$, $w = k_BT \ln 2$). "Floor" is the Gaussian quadratic comparison $\min_P h(P)$ of Theorem 11 for the quadratic loss; C is the leading term of Corollary 12. For Paper I's divergence loss the entries are leading-order values, accurate for $\nu \lesssim 10^{-3}$. A record read through the same channel is worth $0.160\, k_BT$ (Paper I).

A.2 The local Gaussian comparison for a drifting gain

Let the gain of the gain–bias channel drift with $g \approx 0.55$, $b = 0$, so $J = G = 1.4337$, and let references cost $w = k_BT \ln 2$. Table 5 gives the local Gaussian comparison of Remark 14, its leading term C, the optimal reference rate and the optimal memory weight.

At $w = J = c = 1$ and $\nu = 10^{-2}$, average-cost value iteration on a variance grid agrees with the Gaussian floor to relative error 1.3×10^{-5}; 4000 Kalman trajectories agree within Monte Carlo error. Integer binomial reads provide a distinct comparison: at $w = 10^{-4} k_BT$, $\nu = 3 \times 10^{-3}$, the best tested grid policy costs 1.10 times the relaxed floor. For Corollary 13, $q = 10^{-4}$, $r = v = 1$, $w = 10^{-3}$ gives optimum 0.0054878. The source map identifies the full scans and fixed-seed outputs.

B A drifting instrument in a closed loop

Two finite bank models couple the estimate $(\hat{g}, \hat{b})$ to an action that changes the next world reading. Their instruments, baths and results remain separate. They specify their own assumptions and do not use (C1)–(C3).

B.1 The world and the observation map

The bank has M binary variables $z_j \in \{\pm 1\}$ with mean $x = M^{-1} \sum_j z_j$. In each epoch every variable is updated independently: it keeps its value with probability a, is replaced by a variable biased toward the agent's command u with probability c, and is replaced by a variable biased toward the environmental input d with probability r, where $a + c + r = 1$ and the biased variable is $+1$ with probability $(1 + v)/2$ for the bias $v = u$ or $v = d$. For $|u|, |d| \le 1$ this is a convex combination of the identity and two resets, hence a legal channel (Foundation III, section on legal finite feedback [11]). The probability of $+1$ is therefore $(1 + az_j + cu + rd)/2$, and

$$x_{k+1} = a\, x_k + c\, u_k + r\, d_k + \zeta_k, \qquad \mathbb{E}[\zeta_k \mid \text{past}] = 0, \quad \mathbb{E}[\zeta_k^2 \mid \text{past}] \le M^{-1}. \tag{27}$$

The count of $+1$ variables is updated exactly from two binomial draws. The simulations hold that count for efficiency and never give it to the agent. We use $(a, c, r) = (0.65, 0.25, 0.10)$.

An instrument with gain g and bias b, $|g| + |b| \le 1$, reads a uniformly random variable and returns $y = \pm 1$ with probability $(1 + gz_j + b)/2$. The index is not returned. A reference read presents an input known to be $+1$ or -1 and returns $+1$ with probability $(1 \pm g + b)/2$. Gain and bias are fixed within an epoch and drift between epochs. From n_s world reads with mean $\bar{y}$ and the current estimates the agent forms

$$\hat{x} = Q\left(\frac{\bar{y} - \hat{b}}{\max(\hat{g},\, 0.30)}\right), \tag{28}$$

where Q clips to $[-1, 1]$ and rounds to a grid of 1025 points, and it acts with $u = -h\hat{x}$, $h = 0.8$. The total number of reads per agent and epoch is 512. World loss is $x^2/4$, averaged over trajectories and over epochs after a burn-in of 200.

B.2 What reference reads measure

Proposition 42 (Reference reads)**.** *Let $p \sim \mathrm{Bin}(n, \pi_+)$ and $m \sim \mathrm{Bin}(n, \pi_-)$ be independent, with $\pi_\pm = (1 \pm g + b)/2$ and $|g| + |b| < 1$. Then $(\hat{g}, \hat{b}) = \big((p - m)/n,\ (p + m)/n - 1\big)$ is unbiased with covariance*

$$\Sigma_{\mathrm{ref}} = \frac{1}{2n}\begin{pmatrix} 1 - g^2 - b^2 & -2gb \\ -2gb & 1 - g^2 - b^2 \end{pmatrix},$$

which is the inverse of the Fisher information matrix of the $2n$ reads about (g, b). The estimator is therefore efficient.

Proof. $\mathrm{Var}\, p = n(1-(g+b)^2)/4$ and $\mathrm{Var}\, m = n(1-(b-g)^2)/4$, which gives the covariance by linearity. The pair $(p/n, m/n)$ is the maximum-likelihood estimator of the mean parameters (π_+, π_-) of a product of binomials and attains the Cramér–Rao bound $\mathrm{diag}(\pi_\pm(1 - \pi_\pm))/n$. The map to (g, b) is linear and invertible, and efficiency is preserved under it. □

The frequency estimator above is unbiased and efficient for mean-square error, but its boundary values make the plug-in expected divergence infinite at every finite sample size. To use Paper I's calibrated-work expansion, replace each estimated response probability k/n by $(k + 1/2)/(n + 1)$, or use another regularization satisfying its tail condition. The leading covariance is unchanged, and the $1/n$ work loss is then an asymptotic statement for that regularized estimator.

For that regularized estimator, Paper I computes the leading calibrated loss: acting on the estimated readout model costs $k_B T(1 + g^2/(1 - b^2))/(4n)$ per use to leading order, $0.00254\, k_B T$ for $n = 128$ at $g = 0.55$, $b = 0$. A reference read costs a world read, so n_r is a budget. The estimates are also kept in memory. A persistent memory with weight α on the new reference estimate, $\hat{\theta}_k = Q[(1 - \alpha)\hat{\theta}_{k-1} + \alpha\tilde{\theta}_k]$ for $\theta \in \{g, b\}$, trades noise against lag.

Proposition 43 (Memory under drift)**.** *Let $0 < \alpha \le 1$, $\beta = 1 - \alpha$ and $e_k = \hat{\theta}_k - \theta_k$. Suppose the reference estimate has the form $\tilde{\theta}_k = \theta_k + \xi_k$, where θ_k is determined by the information available before the reference reads of epoch k (a deterministic drift qualifies, and the true instrument state may enter without giving the observer access to it) and ξ_k has*

conditional mean zero given that information and variance at most σ^2, that $\theta_k \in [-1,1]$ and $|\theta_k - \theta_{k-1}| \le \nu$, and that Q clips to $[-1,1]$ and rounds to a grid of spacing Δ. Then

$$\|e_k\|_{L^2} \le \sqrt{\big(\beta(\|e_{k-1}\|_{L^2}+\nu)\big)^2 + \alpha^2\sigma^2} + \frac{\Delta}{2}, \qquad \|e_k\|_{L^2} \le X_\sharp + \beta^k\big(\|e_0\|_{L^2} - X_\sharp\big)_+,$$

with $X_\sharp = \dfrac{\Delta/2 + \beta^2\nu + \sqrt{\beta^2(\Delta/2+\nu)^2 + (1-\beta^2)\alpha^2\sigma^2}}{1-\beta^2}$. *If $0 < \alpha < 1$, the noise variance is exactly σ^2, and the drift is the idealised linear drift $\theta_k = \theta_0 + \nu k$ without rounding or clipping, the steady-state error has mean $-\nu(1-\alpha)/\alpha$ and variance $\alpha\sigma^2/(2-\alpha)$. Hence the persistent memory has smaller mean-square error than a memoryless estimate (weight $\alpha = 1$, error variance σ^2) exactly when*

$$\nu^2 < \frac{2\alpha^2\sigma^2}{(1-\alpha)(2-\alpha)}.$$

Proof. Put $y_k = \beta\hat\theta_{k-1} + \alpha\tilde\theta_k$, so that $y_k - \theta_k = \beta e_{k-1} - \beta(\theta_k - \theta_{k-1}) + \alpha\xi_k$. The noise ξ_k is orthogonal in L^2 to the first two terms, which are determined before it, so $\|y_k - \theta_k\|^2 \le \beta^2(\|e_{k-1}\| + \nu)^2 + \alpha^2\sigma^2$. Clipping to an interval that contains θ_k does not increase the distance to θ_k, and rounding adds at most $\Delta/2$; this is the first inequality. Write $\varphi(X)$ for its right-hand side, which is increasing with $\varphi'(X) = \beta^2(X+\nu)/\sqrt{\beta^2(X+\nu)^2 + \alpha^2\sigma^2} \le \beta$. The larger root $X_\sharp$ of $(X - \Delta/2)^2 = \beta^2(X+\nu)^2 + \alpha^2\sigma^2$ is the displayed expression, and it satisfies $X_\sharp \ge \Delta/2$, so $\varphi(X_\sharp) = X_\sharp$ and $\varphi(X_\sharp + t) \le X_\sharp + \beta t$ for $t \ge 0$. The second inequality follows by induction. For linear drift without rounding or clipping the recursion is linear, $e_k = \beta e_{k-1} - \beta\nu + \alpha\xi_k$, which gives the stated mean and variance. The mean-square comparison is $\alpha\sigma^2/(2-\alpha) + \nu^2(1-\alpha)^2/\alpha^2 < \sigma^2$. □

For Model A's $\alpha = 0.25$ and $n_r = 128$ reads per sign, the unrounded linear-drift comparison gives $\nu^* \approx 0.0161$ per epoch. It places the crossover at the scale of the observed four-to-eight-fold drift bracket; it omits the loop, gain floor and periodic drift. The Gaussian-transition optimum of Theorem 11 instead uses the stationary Kalman gain and real precision. A constant-weight estimator under linear drift has the different $\nu^{2/5}$ price of Remark 16.

The world is read through the same estimates. The next statement bounds the error of the decoded state in terms of the errors of the estimates.

Proposition 44 (Error of the decoded state)**.** *Let $g \ge g_{\min} > 0$ and let $\bar y$ be the mean of n_s readings of the bank, which are independent given the state x. Let $\hat x = Q_x\big[\mathrm{clip}\big((\bar y - \hat b)/\max(\hat g, g_{\min})\big)\big]$, where Q_x rounds to a grid of spacing Δ_x that contains ± 1. Then*

$$\|\hat x - x\|_{L^2} \le \min\left(2,\ \frac{n_s^{-1/2} + \|\hat g - g\|_{L^2} + \|\hat b - b\|_{L^2}}{g_{\min}} + \frac{\Delta_x}{2}\right).$$

Proof. Write $\bar y = gx + b + \nu$ with $\mathbb{E}[\nu \mid x] = 0$ and $\mathbb{E}[\nu^2 \mid x] \le n_s^{-1}$, and $\hat g' = \max(\hat g, g_{\min}) \ge g_{\min}$. Then $(\bar y - \hat b)/\hat g' - x = [(g - \hat g')x + (b - \hat b) + \nu]/\hat g'$. Since $g \ge g_{\min}$ one has $|g - \hat g'| \le |g - \hat g|$, and $|x| \le 1$, so the numerator has L^2 norm at most $\|g - \hat g\| + \|b - \hat b\| + n_s^{-1/2}$, and the division by $\hat g' \ge g_{\min}$ gives the bound before clipping. Clipping to $[-1,1] \ni x$ does not increase the distance to x, and rounding adds at most $\Delta_x/2$. Both $\hat x$ and x lie in $[-1,1]$, so the error is at most 2. □

Model A has $\nu_g \le 0.005357$, $\nu_b \le 0.006585$ and $g \ge 0.30$. With $\alpha = 0.25$, $n_r = 128$, $\sigma^2 = 1/(2n_r)$ and grid spacing $2/1024$, Proposition 43 gives limiting L^2 errors at most 0.0351 and 0.0376. Proposition 44, with $n_s = 256$, then bounds decoded-state error by 0.4518, compared with observed epoch-averaged error 0.135. These are expectation bounds for the stated drift, not simultaneous guarantees or bounds for the zero-gain fault.

B.3 One observer: the loop closes on the world

Model A. A single agent, a drifting instrument and a bath $d_k = 0.60 + 0.25\sin(2\pi k/177)$. Normal drift is

$$g(k) = 0.55 + 0.17\sin(2\pi k/420) + 0.03\sin(2\pi k/67),$$
$$b(k) = 0.14\sin(2\pi k/300) + 0.025\sin(2\pi k/43),$$

and a drift factor s replaces k by sk. The static instrument has $g = 0.55$, $b = 0$ and the same bath. Three policies share the interface and the budget of 512 reads. *Frozen*: 512 world reads, calibrated once from 2048 reads per sign at the start and never again. *Memoryless*: 256 world reads and 128 reads per sign each epoch, using only that epoch's references ($\alpha = 1$). *Persistent*: the same reads, $\alpha = 0.25$. Each policy pays the 4096 start-up reads. The agent does not know the drift functions.

Table 6 retains the static, normal-drift and fast-drift comparisons. A separate implementation and larger normal-drift run agreed within 1.7 combined standard errors; their full scans are identified in the source map.

Instrument	Frozen	Memoryless	Persistent
static instrument	3.426±0.036	3.589±0.009	3.520±0.008
drift ×1	5.299±0.043	3.652±0.010	3.591±0.010
drift ×4	4.642±0.040	3.654±0.010	3.622±0.009
drift ×8	4.543±0.042	3.643±0.008	3.802±0.008

Table 6: World loss $x^2/4$ ($\times 10^{-3}$, ± one standard error over 160 independent trajectories, 1200 epochs, burn-in 200).

At normal drift, persistent calibration lowers loss by 1.7% relative to memoryless calibration and 32.2% relative to the frozen policy. Freezing is best for the static instrument. The persistent-minus-memoryless difference changes from -0.032 ± 0.013 at four-fold drift to $+0.159 \pm 0.012$ at eight-fold drift ($\times 10^{-3}$). Memory therefore helps only within a drift range at this weight and read allocation.

B.4 Two observers: reliability and authority

Model B. Two agents read the same bank through different instruments (the instruments and the public event of Section 5 of Foundation II [10], here with $d_k = 0.6\,c_k$), each with the calibration of Section B.2 and the same budget of 512 reads. Each agent forms $\hat{x}_i$ by (28) and issues $u_i = -h\hat{x}_i$. The world receives the joint command $u = wu_1 + (1-w)u_2$, so $|u| \le h$ whenever $w \in [0, 1]$. The agents may also move their bias estimates toward the partner's

report, either with equal weight or weighted by $\mathrm{clip}(\hat{g}_j/0.4, 0, 1)^2$, which requires exchanging $\hat{g}$ as well as $\hat{x}$ (22 bits per sender and epoch instead of 11, so 44 for the pair instead of 22). The fixed rule is $w = \frac{1}{2}$. The convex rule is $w = s_1/(s_1 + s_2)$ with $s_i = \mathrm{clip}(\hat{g}_i/0.4, 0, 1)^2$, and $w = \frac{1}{2}$ if both vanish. In the convex rule the more reliable agent receives more of the authority, and the total never exceeds the cap h.

One-step task risk. Let the reports be $\hat{\boldsymbol{x}} = x\mathbf{1} + \epsilon$ with errors ϵ of any joint law, and let the weights $w \geq 0$, $\mathbf{1}^\top w = 1$, be determined by the information $\mathcal{F}$ available when they are chosen, which may include the reference reads of the epoch. Write $q = ch$, $z = (a-q)x + rd$, $R = \mathbb{E}[\epsilon\epsilon^\top \mid \mathcal{F}]$ (an uncentred second moment) and $b = \mathbb{E}[z\epsilon \mid \mathcal{F}]$. Weights are compared with the conditional laws of the reports and of the world held fixed.

Proposition 45 (Task risk of a weight vector)**.** *Suppose $|u| \leq 1$ and $|d| \leq 1$, so that all branch probabilities of the bank step lie in $[0, 1]$, and the action is the linear rule $u = -h\, w^\top \hat{\boldsymbol{x}}$. Then*

$$\mathbb{E}\big[\tfrac{1}{4}x'^2 \,\big|\, \mathcal{F}\big] = K_{\mathcal{F}} + \frac{1 - M^{-1}}{4}\left(q^2\, w^\top R w - 2q\, b^\top w\right),$$

where $K_{\mathcal{F}} = \frac{1}{4}\big[(1 - M^{-1})\mathbb{E}[z^2 \mid \mathcal{F}] + M^{-1}\mathbb{E}[1 - a^2 + a^2x^2 \mid \mathcal{F}]\big]$ does not depend on w.

Proof. Given x, u and d the variables move independently, so $\mathbb{E}[x' \mid x, u, d] = ax + v$ and $\mathrm{Var}(x' \mid x, u, d) = M^{-1}(1 - a^2 - v^2 - 2axv)$ with $v = cu + rd$. Hence $\mathbb{E}[x'^2 \mid x, u, d] = (1 - M^{-1})(ax + v)^2 + M^{-1}(1 - a^2 + a^2x^2)$. With $u = -h(x + w^\top\epsilon)$ one has $ax + v = z - q\, w^\top\epsilon$. Taking the conditional expectation over ϵ gives the formula. □

The proposition needs no assumption that the errors are unbiased, centred, or independent between the two agents. Quantisation and the gain floor can sit inside ϵ. If the action is saturated or non-linear the formula no longer applies, and for $q = 0$ the weights do not change the one-step risk. When the errors are unbiased given x, d and $\mathcal{F}$, $b = 0$ and R is the error covariance. The risk is a convex quadratic in w. For two agents with $w = (t, 1 - t)$, $q > 0$ and $D = R_{11} + R_{22} - 2R_{12} > 0$, its minimiser over $t \in [0, 1]$ is

$$t_* = \mathrm{clip}\left(\frac{R_{22} - R_{12} + (b_1 - b_2)/q}{D},\, 0,\, 1\right). \tag{29}$$

If $R = \mathrm{diag}(v_1, v_2) + \tau^2\mathbf{1}\mathbf{1}^\top$ and $b = 0$, the best combined error is $\tau^2 + v_1v_2/(v_1 + v_2)$: weighting removes the private part of the error and cannot touch the common part.

Proposition 46 (Estimation-optimal and task-optimal weights differ)**.** *Take $M = 20$, $x = 0.2$ (twelve variables at $+1$), $(a, c, r) = (0.65, 0.25, 0.10)$, $h = 0.8$, $d = -0.7$. Let ϵ_1 be 0 or 0.2 and ϵ_2 be -0.1 or 0.1, each with probability $\frac{1}{2}$ and independent. The weight that minimises the joint estimation error is $t = \frac{1}{3}$; the weight that minimises the next-step world loss is $t = \frac{2}{3}$.*

Proof. The estimation error is $t^2\, 0.02 + (1 - t)^2\, 0.01$, minimised at $t = \frac{1}{3}$. Here $q = 0.2$, $z = 0.02$, $R = \mathrm{diag}(0.02, 0.01)$ and $b = (0.002, 0)$, so (29) gives $t_* = \frac{2}{3}$. In every branch $|u| \leq 0.32$, so the step is legal. The values in Table 7 come from enumerating all four error branches and both binomial draws exactly, and they agree with Proposition 45. □

Allocation rule	t	Joint estimation error	Next-step world loss
Smallest joint estimation error	0.3333	0.006667	0.007525
Inverse centred variance	0.5000	0.007500	0.007501
Smallest next-step world loss	0.6667	0.010000	0.007493

Table 7: The example of Proposition 46, by exact enumeration.

Agent 1's mean error 0.1 offsets $z = 0.02$, making more authority useful despite worse estimation. Task-optimal allocation needs the effects of action as well as sensor reliability. The observable gain proxy below is not the covariance R; no online learner for the true-error quantities R, b is supplied.

The linear term $-2q\, b^\top w$ also marks where Paper I's metric applies. Paper I prices a calibration error by $\frac{1}{2}k_B T\, \delta^\top G \delta$ when the downstream protocol is matched to the posterior the agent assumes, and then a calibrated self-model is a stationary point of the loss. With a fixed linear action rule it need not be one, and a biased report can lower the world loss, as in Proposition 46.

Four cells. We cross the two interventions that can be applied separately: the update rule (equal or weighted alignment) and the authority rule (fixed or convex). One agent's instrument is damaged for 100 epochs at a random start: a bias fault raises its bias by a nominal 0.30 before the legal clip, a gain fault sets its gain to 0.05 or 0 ("zero gain": the current reading carries no information about x). Table 8 gives the increase in world loss during the window relative to a matched run without the fault. The commands in all runs are $u_i = -h\hat{x}_i$ with $\hat{x}_i$ on the 1025-point grid, so $|u_i| \le h$ exactly; the largest $|u|$ applied over 105 runs was 0.8000 and no epoch exceeded the cap. Every cell is charged the same message budget, 22 bits per sender and epoch ($\hat{x}$ and $\hat{g}$) and 44 for the pair, although the unweighted, fixed cell does not use $\hat{g}$.

Policy	Bias fault	Gain 0.05	Gain 0
Unweighted, fixed $w = \frac{1}{2}$	+0.01±0.01	+2.05±0.02	+2.53±0.02
Unweighted, convex authority	+0.03±0.01	+1.11±0.02	+1.30±0.02
Weighted, fixed $w = \frac{1}{2}$	+0.01±0.01	+1.32±0.02	+1.58±0.02
Weighted, convex authority	+0.01±0.01	+0.34±0.02	+0.31±0.02
References only, no coupling	+0.00±0.01	+1.46±0.02	+1.80±0.02

Table 8: Increase of world loss in the fault window ($\times 10^{-3}$, mean of three seed blocks of 800 trajectories ± combined standard error; the different policies use different random draws, so the comparison between rows is not a paired test).

Weighted alignment limits a failed report's influence on the other estimate; convex authority limits its influence on the world while preserving the command cap. Together they reduce the zero-gain fault increase by 87.8% relative to equal alignment and fixed authority. The interaction, -0.042 ± 0.035 ($\times 10^{-3}$), is unresolved. Fault-free losses range from 3.156 to 3.167 ($\times 10^{-3}$), without a claim of equality.

B.5 What the two models give

Both models are finite banks with a declared instrument family, a declared bath and a declared cost vector (reads, start-up references, bits of memory and message). Model A shows a self-model built from reference reads improving the agent's effect on its world, with a memory whose useful length is limited by the drift. Model B shows that reliability can be used in two separate places, in what an agent believes about its partner and in how much of the world it is allowed to move, and that the second place accounts for the larger part of the reduction when a sensor fails. The two models use different instruments and baths and are not one experiment.

C Simulation protocols for Section B

Model A. Use the parameters and policies of Section B.3 and start with $n_+ = \lfloor 0.6M \rfloor$, $M = 4096$. Each epoch reads the pre-action bank and references, updates the estimates, computes (28), then applies (27). The agent receives only read counts and its stored estimates. It retains 22 estimate bits and an 11-bit command index, pays 4096 startup references, 256 references per adaptive epoch and 4096 bank-step applications. Counters, index generation and arithmetic workspace are additional resources; no heat or work is inferred from these counts. Run lengths and errors are specified in Table 6.

Model B. The same bank and step, with the public event c_k (switching with probability 0.01 per epoch) and $d_k = 0.6\, c_k$. The instruments are

$$\begin{aligned} g_i(k) &= 0.55 + 0.17\sin(2\pi k/420 + \phi_i) + 0.03\sin(2\pi k/67 + \phi_i), \\ b_i(k) &= 0.10\sin(2\pi k/250) + 0.03\sin(2\pi k/53) \\ &\quad + 0.10\sin(2\pi k/300 + \psi_i) + 0.025\sin(2\pi k/43 + \psi_i), \end{aligned}$$

with $\phi = (0, 1.7)$ and $\psi = (0.4, 2.9)$, clipped to $|b| \le 0.98 - |g|$.

Each agent makes 512 reads per epoch, of which $2n_r$ with $n_r = 16$ are reference reads, and starts from 2048 reference reads per sign. Order within an epoch: both agents read the world and references on the same pre-action state; each updates its own calibration and forms $\hat{x}_i$; one synchronous exchange; the commands $u_i = -0.8\,\hat{x}_i$ and the joint command are formed from the values before the exchange; the world takes one joint step; the alignment update for the next epoch uses the exchanged values. Equal alignment: $\hat{b}_i \leftarrow Q(\hat{b}_i + 0.25\, e_i)$ with $e_i = \bar{y}_i - (\hat{g}_i \hat{x}_j + \hat{b}_i)$. Weighted alignment multiplies the step by $\mathrm{clip}(\hat{g}_j/0.4, 0, 1)^2$. The fault window lasts 100 epochs from a start drawn uniformly from $[800, 2400)$ for each trajectory; faulted and no-fault runs share it. Runs of 3000 epochs, burn-in 200, 800 trajectories per run, three seed blocks. The world and each agent draw from separate random streams addressed by seed, epoch and event, so a fault in one instrument does not shift the draws of the world or of the other agent; because the agents act on the world, faulted and no-fault worlds differ afterwards.

Reproducibility. The supplemental source map lists checks by manuscript table, proposition or equation label, with named scripts, archived outputs and hashes, including the

two Model A implementations and Model B. Checks cover detection, drift prices, gauge symmetry, sign inversion, body moments, finite-block margins, continuing filtering and repair. Their individual assumptions and seeds are retained with the scripts; one experiment's output is not a certificate for a different model.

D Thermodynamic implementation inputs

D.1 Thermodynamic length

Let $\lambda(t)$, $t \in [0,\tau]$, be a protocol in control-parameter space. For conjugate forces $F_i = -\partial_i H$, define the friction tensor by

$$\zeta_{ij}(\lambda) = \beta \int_0^\infty \langle \delta F_j(0)\delta F_i(t)\rangle_\lambda \, dt.$$

The thermodynamic length is $\mathcal{L} = \int_0^\tau \sqrt{\zeta_{ij}\dot\lambda^i\dot\lambda^j}\, dt$ [22]. Within this slow-driving linear-response approximation, $W_{\mathrm{ex}} \geq \mathcal{L}^2/\tau$. Constant thermodynamic speed saturates the bound along a fixed path; minimizing also over paths gives a shortest geodesic. Where the normalized correlation integrals τ_{ij} are defined, $\zeta_{ij} = k_B T\, \tau_{ij} g_{ij}$ entry by entry. A common scalar relaxation time gives $\zeta = k_B T \tau_{\mathrm{relax}} g$ only when all the relevant entries have that same time [23].

E Local tracking of a moving optimum

E.1 Tracking a moving optimum

Let $\vartheta^*(t)$ be the minimizer of $L(\cdot,t)$, differentiable in t, and let $\|\dot\vartheta^*\|_g$ be its speed in the metric g. As a monitoring function take the squared geodesic distance

$$V(t) := d_g\big(\vartheta(t),\vartheta^*(t)\big)^2. \tag{30}$$

Theorem 47 (Loop tracking bound)**.** *Suppose the trajectory of* (16) *stays in a region where $d_g(\cdot,\vartheta^*(t))^2$ is smooth and $L(\cdot,t)$ is geodesically m_L-strongly convex for some $m_L > 0$. Then*

$$\frac{dV}{dt} \;\leq\; -2\eta m_L\, V + 2\sqrt{V}\,\|\dot\vartheta^*\|_g. \tag{31}$$

Consequently: (a) V decreases whenever $\sqrt{V} > \|\dot\vartheta^*\|_g/(\eta m_L)$; (b) if $\|\dot\vartheta^*\|_g \leq \Lambda$, then

$$\limsup_{t\to\infty} V(t) \;\leq\; \frac{\Lambda^2}{(\eta m_L)^2}; \tag{32}$$

(c) if ϑ^* is constant, $V(t) \leq V(0)\, e^{-2\eta m_L t}$.

Proof. The natural gradient $g^{-1}\nabla L$ is the Riemannian gradient $\operatorname{grad} L$. At ϑ the gradient of $d_g(\cdot,\vartheta^*)^2$ is $-2\exp_\vartheta^{-1}\vartheta^*$. Strong convexity along the geodesic from ϑ to ϑ^* gives $L(\vartheta^*) \geq L(\vartheta) + \langle \operatorname{grad} L(\vartheta), \exp_\vartheta^{-1}\vartheta^*\rangle + \frac{1}{2}m_L V$, and since ϑ^* is the minimizer, $L(\vartheta) \geq L(\vartheta^*) + \frac{1}{2}m_L V$. Adding the two, $\langle \operatorname{grad} L(\vartheta), -\exp_\vartheta^{-1}\vartheta^*\rangle \geq m_L V$. The contribution of the update to dV/dt is $\langle -2\exp_\vartheta^{-1}\vartheta^*, -\eta \operatorname{grad} L\rangle \leq -2\eta m_L V$. The contribution of the moving optimum is $\langle -2\exp_{\vartheta^*}^{-1}\vartheta, \dot\vartheta^*\rangle \leq 2\sqrt{V}\|\dot\vartheta^*\|_g$ by Cauchy–Schwarz. This gives (31) and (a). For (b), $W = \sqrt{V}$ obeys $\dot W \leq -\eta m_L W + \Lambda$ wherever $V > 0$, so $\limsup W \leq \Lambda/(\eta m_L)$. For (c) set $\dot\vartheta^* = 0$. □

For the two-channel frame the loss $s^2 + \rho^2\sin^2\theta$ has $L'(0) = 0$ and $L''(0) = 2\rho^2$, so its Riemannian Hessian at calibration is $2\rho^2/g(0)$. When probes make $g(0) > 0$ (Proposition 18), continuity gives strong convexity in a neighbourhood of calibration with m_L close to $2\rho^2/g(0)$, and the theorem applies there. Without probes $g(0) = 0$ and it does not apply.

Under drift the loop cannot converge to zero mismatch; it holds the mismatch within the neighbourhood (32), which shrinks with ηm_L and grows with the drift speed. Raising η is not free: a physical implementation has to relate the adaptation rate to its energy budget (Section F), and in a stochastic loop a faster update also follows the noise, which is the trade-off Section 5 prices exactly. A second constant governs how well the metric itself is conditioned.

F Thermodynamic cost

F.1 Cost components

A physical calibration loop senses, computes, actuates and prepares references. What each costs depends on how it is realized.

Sensing and computing. Measurement, copying and logic can in principle be reversible, so reading a stream carries no positive cost by itself. If sensor and work registers irreversibly erase Shannon information at rates R_s and R_c bits per unit time, Landauer's bound gives $\dot W_{\text{sense}} \geq k_B T \ln 2\, R_s$ and $\dot W_{\text{compute}} \geq k_B T \ln 2\, R_c$.

References. A reference read requires an input in a known state. Preparing a known pointer state from the thermal state of a degenerate pointer costs at least $k_B T \ln 2$ per reference (Paper I). At R_{ref} references per unit time this is $k_B T \ln 2\, R_{\text{ref}}$, unless references are recycled.

Actuation. Moving the self-model from ϑ to a new estimate $\hat\vartheta^*$ is a finite-time transformation of whatever physical parameter stores it. In linear response, a protocol of duration τ_{recalib} and thermodynamic length $\mathcal{L}(\vartheta,\hat\vartheta^*)$ dissipates at least $\mathcal{L}^2/\tau_{\text{recalib}}$, a rate of at least $\mathcal{L}^2/\tau_{\text{recalib}}^2$ if repeated every τ_{recalib} [23].

F.2 A conditional implementation bound

Proposition 48 (Conditional implementation cost)**.** *Suppose that (i) sensing, computing, reference preparation and actuation use separate physical degrees of freedom with no shared erasure accounting; (ii) sensor and work registers erase Shannon information at rates R_s and R_c, and R_{ref} references per unit time are prepared from thermal states; and (iii) actuation*

follows a linear-response protocol of duration τ_{recalib} *and thermodynamic length* $\mathcal{L}$*, repeated every* τ_{recalib}*. Then*

$$\dot{W}_{\mathrm{loop}} \;\geq\; k_B T \ln 2 \left(R_s + R_c + R_{\mathrm{ref}}\right) + \frac{\mathcal{L}^2}{\tau_{\mathrm{recalib}}^2}. \tag{33}$$

Proof. Landauer's bound applies to the two declared erasure streams and to each reference preparation; the Sivak–Crooks bound divided by the protocol duration gives the actuation rate. Separability allows the terms to be added without double counting. Dropping separability, or using reversible sensing and computation without resets in the window, removes the corresponding terms. □

The proposition and the drift law count different things. Remark 14 counts references and the loss of acting on an imperfect self-model, in the exact Gaussian comparison. Proposition 48 adds the costs of the registers and actuators of one implementation. Together they bound what a physically realized calibration loop pays per use. Papers I and II supply no universal maintenance or processing cost that could be added to these without a concrete implementation, so no general persistence budget is claimed.

G Finite-block margins and continuing filtering

G.1 A resource-conditioned operating margin

A lower bound on drive occupation cannot by itself supply an achievable gate. We instead choose the displaced squeezed Gaussian family of Q-RAIF B and use its exact characteristic function for both error and energy. If its mean occupation is N, set

$$V_Q = \frac{1}{2N+1}, \quad \lambda^2 = \frac{\pi^2(2N+1)}{16N(N+1)}, \quad \alpha^2 = \frac{\pi^2}{16\lambda^2}, \quad s = \frac{1+\exp(-2\lambda^2 V_Q)}{2}.$$

Then $\alpha^2 + (V_Q + V_Q^{-1} - 2)/4 = N$. For the coherent comparison use $V_Q = 1$, $\lambda^2 = \pi^2/(16N)$ and the same formula for s. Each opportunity receives a fresh independent scheduled packet, costing $\theta_\omega N$ whether or not it is switched on. Actual switching adds $\theta_\omega \lambda^2$. Incoming field energy and switching work thus refer to the same state.

The gain is physical and bounded. In blocks of $L = 4096$ food uses, draw independently one of $g_- = 0.90h$ and $g_+ = 0.98h$, with equal prior probabilities, where $r = 0.8^n$ and $h = \sqrt{1-r^2}$. Keep that gain constant through the block, with matched setting and known positive sign. At the start take $m = 32$ references of the input $+1$. The count K has likelihood

$$p_g(k) = \binom{m}{k}[(1+g)/2]^k[(1-g)/2]^{m-k}; \qquad \bar{g}_k = \frac{g_- p_{g_-}(k) + g_+ p_{g_+}(k)}{p_{g_-}(k) + p_{g_+}(k)}.$$

The reference pointer preparation costs $m \ln 2$. We conservatively erase its outcomes separately for another $m \ln 2$, and erase the $(m+1)$-state count register for $\ln(m+1)$ per block. This finite-memory filter incurs no Gaussian approximation or truncation.

Write

$$S_r(x) = H_2\left(\frac{1+\sqrt{r^2+(1-r^2)x^2}}{2}\right).$$

The posterior-optimal food levels give income $\ln 2 - H_2((1+s\bar{g}_k)/2)$. The food reset against the untouched quantum copy costs $\ln 2 - S_r(0) + S_r(\bar{g}_k)$; the switching probability is exactly $1/2$. Its net, averaged over the exact law of K, is

$$A = \mathbb{E}_K[S_r(0) - S_r(\bar{g}_K) - H_2((1+s\bar{g}_K)/2)] - \theta_\omega(N+\lambda^2/2) - \frac{2m\ln 2 + \ln(m+1)}{L}.$$

The reset is the reversible conditional-entropy limit for the stated classical–quantum state, retaining the copy marginal. Its attainability uses quasistatic processing across independent replicas of the complete block protocol, conditioned on the retained count. An implementation with reset overhead below a positive margin has the same sign of net work.

Economy B adds one body read and drive opportunity per food use. The body starts each block uniform and is thermalized by a work-free swap with a fresh equilibrium bit at the block boundary. With bath flip probability $d = 0.05$ and feedback on every negative read,

$$x_{g,t+1} = 0.9[(1-s)x_{g,t} + sg], \qquad x_{g,0} = 0.$$

Given $K = k$, define $x_k = \mathbb{E}[x_{g,t} \mid k]$, $y_k = \mathbb{E}[g x_{g,t} \mid k]$, and $p_\pm = (1 \pm y_k)/2$. The conditional body-copy biases are $b_\pm = (x_k \pm \bar{g}_k)/(1 \pm y_k)$. Body erasure therefore costs

$$H_2(p_+) - S_r(x_k) + \sum_\pm p_\pm S_r(b_\pm),$$

and its drive costs $\theta_\omega[N + p_-\lambda^2]$. Subtract their exact block average from A to get B. This policy retains only K, ignores body readings as calibration data, and requires no unproved body-stream information rate.

At $n = 7$ and $\theta_\omega = 0.01$, the squeezed construction leaves $0.02396\,k_BT$ per food use after body upkeep, with body loss 0.08725; the coherent comparison leaves $-0.09847\,k_BT$. A sufficiently small irreversible overhead preserves the positive squeezed margin. This is a maintained body supplied by a record and drive stream, with a gain that changes between blocks. It is not an infinite Gaussian-gain process.

Record formation consumes $2n\ln 2$ of fresh memory purity per food use in A and $4n\ln 2$ in B, plus $mn\ln 2/L$ for reference fragments. Those memories are supplied by the environment and their purity is explicitly an input, not work produced by the cycle. Replenishing them from the agent's own thermal resources debits that input and eliminates the apparent profit. The drive additionally presupposes its preparation, phase reference and scheduled port; uncomputed apparatus overhead can be tolerated only within the displayed positive margin. The example identifies a nonempty region of positive operating margin under these resource declarations. It does not derive the resources or the reference standard from the loop.

G.2 A continuing joint filter and adaptive references

The block calculation redraws a prior and resets the body at known times. A continuing observer instead carries its model through the changes that its own history reveals. Keep the

n	θ_ω	Drive	N	A: margin	B: margin	Body loss	B: purity
3	0.01	coherent	6.220	-0.1607	-0.5798	0.1402	8.334
3	0.01	squeezed	4.220	-0.0910	-0.4899	0.1377	8.334
5	0.01	coherent	7.343	0.0533	-0.2522	0.1037	13.890
5	0.01	squeezed	4.758	0.1393	-0.1402	0.1013	13.890
7	0.01	coherent	7.865	0.1552	-0.0985	0.0896	19.446
7	0.01	squeezed	5.019	0.2492	0.0240	0.0873	19.446
5	0.05	coherent	2.522	-0.1104	-0.4709	0.1097	13.890
5	0.05	squeezed	2.454	0.0036	-0.3540	0.1032	13.890
7	0.05	coherent	2.708	-0.0196	-0.3317	0.0955	19.446
7	0.05	squeezed	2.587	0.1069	-0.1999	0.0891	19.446
10	0.10	coherent	1.587	-0.0810	-0.3884	0.0945	27.780
10	0.10	squeezed	1.919	0.0410	-0.3007	0.0840	27.780

Table 9: Achievable conditional-entropy limit in units of $k_B T$. Margins are harvest minus the listed reset, reference and drive costs after food and record resources have been supplied. Each row optimizes food-only drive within its family, then uses that same state for food and body. The B: purity column is external record-memory purity per food use, $(4n+mn/L)\ln 2$, in addition to the listed operating costs; A uses $(2n + mn/L)\ln 2$. Body loss is averaged after each step over the full block. Calibration costs 0.011684 per food use.

$n = 7$ fragments and the same single-pass squeezed drive, $N = 5.01898$, $s = 0.979992$ and $\lambda^2 = 0.225388$. Let the gain switch between $g_- = 0.90h$ and $g_+ = 0.98h$ with probability $\delta = 3 \times 10^{-4}$ per use, and the sign invert with probability $\eta = 2 \times 10^{-4}$. The body bath has $d = 0.05$. These are bounded transitions valid for arbitrarily long operation. The body, instrument and stored model continue through every reference-budget boundary.

The hidden state $Z_t = (G_t, \sigma_t, B_t)$ has eight values. Each use gives one body reading with likelihood $(1 + y\sigma g B)/2$. A requested reference gives one known-input reading with likelihood $(1 + z\sigma g)/2$. Its request can depend on the body reading and earlier records. The observer then selects a body flip by the sign of its posterior body mean, and food levels by its posterior signed gain. It receives two fresh scheduled drive packets per use, one for each task, and a maximum of $m = 32$ reference ports per $L = 4096$ uses. The ports, their timing and their input labels are supplied equally to the compared policies; an unused port creates no additional income.

For body command $a \in \{0, 1\}$, let T_a be the eight-state transition matrix. Its gain and sign factors are the symmetric switching kernels of rates δ and η. The body flips with probability d for $a = 0$, and with probability $d + s(1 - 2d)$ for $a = 1$. All entries of T_a are positive. Given the selected actions and realized records, the filter predicts with T_a and multiplies by the two likelihoods, omitting the second when no reference is requested. The request rule contributes no extra parameter-dependent likelihood factor. The fair food reading alone supplies no instrument information and is used only for its task.

Positive transition and observation kernels give a uniform finite-register certificate for this filter (Proposition 49, Appendix H). It controls quantization error even when actions and reference requests are chosen from the rounded posterior.

This filter forms its next prior from the preceding posterior and the actual transition

law. It does not request a newly assigned gain prior or reprepare the body. For the stated parameters, $a_0 = 3\times10^{-9}$, $D = 39.14572$ and $\tau = 0.999999993681$. A log grid with $\Delta = 2^{-44}$ uses 350 model bits; its asymptotic total-variation bound is 2.25×10^{-6} and its one-step excess-risk bound is 4.50×10^{-6}. These conservative uniform bounds concern quantization of a correct filter. They do not certify a wrong transition family or the arithmetic implementation. The 350-bit state is an abstract finite controller; its workspace, update throughput and erasure protocol remain to be implemented.

For a resource-matched comparison, the periodic policy uses one reference every 128 steps. The adaptive policy requests a reference whenever the posterior probability of the less likely sign exceeds 0.04, subject to the same 32-port cap. Both update the joint model from every body reading and use the same body-control rule. Starting with the uniform eight-state prior, 128 independent trajectories of 65536 steps, with the first 8192 discarded, give

	Body loss	References/use	Food mismatch
Periodic	0.11726 ± 0.00083	0.0078125	0.06726 ± 0.00048
Adaptive	0.10319 ± 0.00066	0.005704 ± 0.000020	0.05226 ± 0.00014.

Errors are one standard error across independent trajectories. Food mismatch is the posterior Jensen gap $H_2((1+s|\mathbb{E}[\sigma g \mid \mathcal{H}_t]|)/2)-\mathbb{E}[H_2((1+sg)/2) \mid \mathcal{H}_t]$ in nats. The adaptive gain mean has mean-square error 0.000353, compared with the stationary prior variance 0.001529. A separate seed and shorter horizon give body losses 0.11730 and 0.10344. Requesting too early is costly under the cap: threshold 0.01 exhausts references prematurely and gives body loss 0.19567 in that shorter run. Adaptive timing needs an allocation rule as well as a confidence statistic.

Every policy is allocated the same food and record stream, drive state and 32-port reference budget. The external record purity is at most $(4n + mn/L)\ln 2 = 19.44603$ nats per food use, and reference preparation plus conservative separate erasure is charged at the same supplied-cap rate $2m\ln 2/L$. The posterior is updated from the body's own records and the policy acts on that body, closing the inference–control loop. This establishes continuing model revision and an operational benefit at finite memory. It does not close the apparatus resource loop: conditional-copy reset, model-register processing, field preparation, phase reference and the clock still need one attainable cyclic implementation. The block margin of Table 9 cannot be imported into this continuing protocol without that ledger.

H Finite-filter certificates

H.1 Continuing eight-state filter

Proposition 49 (Finite continuing filter)**.** *Let $a_0 = \min_{a,i,j}(T_a)_{ij} > 0$, and let every observation likelihood lie in $[l_-, l_+]$ with $l_- > 0$. Suppose at most two observations precede each filter update. Set*

$$\epsilon_0 = a_0(l_-/l_+)^2, \quad A = -\ln\epsilon_0, \quad D = \max_{a,i,j,k,l} \ln \frac{(T_a)_{ik}(T_a)_{jl}}{(T_a)_{il}(T_a)_{jk}}, \qquad \tau = \tanh(D/4) < 1.$$

After the first prediction, every exact update has probability at least ϵ_0 in each state. Round each of the seven log ratios $\ln(p_i/p_8)$ to a grid of spacing Δ, then normalize their exponentials.

This uses at most

$$7\left\lceil \log_2\left(\frac{2A}{\Delta}+3\right)\right\rceil$$

bits for the model. For the exact posterior p_t^* conditioned on the same realized history and the rounded filter $\tilde{p}_t$,

$$d_H(\tilde{p}_t,p_t^*)\le \tau^t d_H(\tilde{p}_0,p_0^*)+\frac{\Delta(1-\tau^t)}{1-\tau},\qquad \|\tilde{p}_t-p_t^*\|_{\mathrm{TV}}\le \tanh\bigl(d_H(\tilde{p}_t,p_t^*)/4\bigr).$$

Here $d_H(p,q)=\max_i \ln(p_i/q_i)-\min_i \ln(p_i/q_i)$. A one-step action minimizing a loss in $[0,1]$ under $\tilde{p}_t$ has excess conditional risk at most $2\|\tilde{p}_t-p_t^*\|_{\mathrm{TV}}$. With a finite reference counter and clock, the eight-state plant and rounded controller form a finite Markov chain and possess an invariant distribution under the supplied streams.

Proof. Prediction gives each probability at least a_0. The product of at most two positive likelihoods reduces that bound by at most $(l_-/l_+)^2$. Hence the seven log ratios lie in $[-A,A]$, and rounding increases their range by at most $\Delta/2$ at each end. In the Hilbert metric, positive diagonal likelihood multiplication and normalization are isometries, and multiplication by T_a contracts by at most τ, the Birkhoff coefficient. This is the standard projective stability mechanism for finite hidden Markov filters [20]. Rounding changes the oscillation of the log ratios by at most Δ. The triangle inequality gives $e_t\le \tau e_{t-1}+\Delta$, which iterates to the displayed bound even when actions and reference requests depend on earlier rounded states. To obtain the total-variation constant, let $m\le p_i/q_i\le M$, with $M/m=e^e$. Normalization and the chord bound for $(x-1)_+$ give

$$\|p-q\|_{\mathrm{TV}}\le \frac{(M-1)(1-m)}{M-m}\le \tanh(e/4),$$

where maximizing over m at fixed M/m gives $m=e^{-e/2}$; distance zero is immediate. Comparing a minimizing action under each distribution costs at most twice that distance for a bounded loss. The model grid, budget counter, clock and eight-state plant are finite, so their joint transition has an invariant probability. This last statement need not give a unique invariant distribution or an optimal reference policy, and it concerns the declared plant and controller rather than the whole resource-supplying apparatus. □

H.2 Filter on working invocations

The hard access gate has zero likelihoods on excluded body states, so Proposition 49's eight-state certificate cannot simply be inherited. Filtering at actual working invocations avoids this issue. Each of the two event matrices is positive. For the present switching rates,

$$\min_{\ell,i,j}(\mathsf{K}_\ell)_{ij}\ge a=(1-d)\delta\eta=\frac{57}{10^9},\qquad g_{\max}<0.98.$$

One body likelihood multiplication gives a posterior lower bound $a/99$, hence a log-ratio range less than 22. Event cross-ratios are at most a^{-2}, so their common projective contraction coefficient is at most $(1-a)/(1+a)$. Round the three log ratios on a grid $\Delta=2^{-44}$. The

proof of Proposition 49, now applied once per working invocation, gives 150 ideal model bits and

$$\limsup \|\tilde{p}_n - p_n^*\|_{\mathrm{TV}} \le \frac{\Delta(1+a)}{8a} < 1.247 \times 10^{-7}.$$

For a one-step loss in $[0,1]$, regret is at most twice the total-variation distance and hence less than 2.494×10^{-7}. The latch, startup state, arithmetic, workspace and protection are separate from these model bits. In wall time the plant, held finite register and latch form a finite Markov chain; they have an invariant probability under the supplied streams. The claim neither processes new data during disability nor certifies a cyclic processor.

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